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PROJECT NUMBER	

MISSOURI DEPARTMENT OF NATURAL RESOURCES
AIR POLLUTION CONTROL PROGRAM

APPLICATION FOR AUTHORITY TO OPERATE

NOTE: Please read all instructions to complete forms properly.

FORM OP-A01 – Section A

A01.00 – GENERAL APPLICATION INFORMATION

All applications MUST be accompanied by a filling fee. The amount of the filing fee shall be as indicated in 10 CSR 10-6.065(5)(B)1.D or 10 CSR 10-6.065(6)(B)1.D.

1. INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

INSTALLATION STREET ADDRESS	COUNTY NAME OF INSTALLATION
30211 Major Avenue	Macon

CITY OF INSTALLATION STREET ADDRESS	STATE	ZIP CODE	INSTALLATION TELEPHONE NUMBER WITH AREA CODE
Macon	MO	63552	

INSTALLATION MAILING ADDRESS	INSTALLATION FAX NUMBER WITH AREA CODE
30211 Major Avenue	

CITY OF MAILING ADDRESS FOR INSTALLATION	STATE	ZIP CODE	MO SENATORIAL DISTRICT NUMBER
Macon	MO	63552	

INSTALLATION CONTACT PERSON	MO REPRESENTATIVE DISTRICT NUMBER
Ed Kacmar	

INSTALLATION CONTACT'S TITLE	EMAIL OF CONTACT PERSON FOR INSTALLATION
Plant Manager	Ed.Kacmar@POET.com

2. PARENT COMPANY NAME	MAILING ADDRESS OF PARENT COMPANY
POET Bioprocessing and Management, LLC	4615 N Lewis Ave

PARENT COMPANY CITY	STATE	ZIP CODE
Sioux Falls	SD	57104

PARENT COMPANY CONTACT PERSON	TELEPHONE NUMBER WITH AREA CODE OF PARENT COMPANY CONTACT
Shelby Wrage	(605) 965-2315

TITLE OF CONTACT PERSON FOR PARENT COMPANY	EMAIL OF CONTACT PERSON FOR PARENT COMPANY
Environmental Engineer	Shelby.Wrage@POET.com

3. TYPE OF APPLICATION

☐ PART 70 (MAJOR)
- ☐ Initial ☐ Off-Permit Change ☐ Minor Modification
- ☐ Renewal ☐ Administrative Amendment ☐ Significant Modification

☑ INTERMEDIATE STATE
- ☐ Initial ☑ Renewal ☐ Amendment

☐ BASIC STATE
- ☐ Initial ☐ Renewal ☐ Amendment

4. APPLICANT'S CERTIFICATION STATEMENT

"I certify, based on information and belief formed after reasonable inquiry, the statements and information in this document are true, accurate and complete."

SIGNATURE OF RESPONSIBLE OFFICIAL OF COMPANY	DATE
	5-12-22

TYPE OR PRINT NAME OF RESPONSIBLE OFFICIAL	TELEPHONE NUMBER WITH AREA CODE OF OFFICIAL
Scott Tuttle	660-260-6434

OFFICIAL TITLE OF RESPONSIBLE OFFICIAL	RESPONSIBLE OFFICIAL'S EMAIL
General Manager	Scott.Tuttle@POET.com

POET

MACON
30211 Major Ave.
Macon, MO 63552
Ph: (660) ~~385-2709~~ 260-6500
poet.com/macon

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CONTROL PGM

May 5, 2022

Missouri Department of Natural Resources
Air Pollution Control Program
P.O. Box 176
Jefferson City, MO 65102

RE: POET Biorefining – Macon, LLC
Intermediate Operating Permit No. OP2017-080A
Installation ID No. 121-0028
Application for Part 70 Permit Renewal

Dear MoDNR Staff:

POET Biorefining – Macon (POET) is submitting the enclosed application for a Part 70 Permit Renewal for the facility located in Macon, MO. The existing Part 70 permit expires on November 16, 2022, and therefore a renewal application is due May 15, 2022. POET requests Construction Permit No. 012020-012 be incorporated into the Part 70 Operating Permit. POET does not wish to incorporate Construction Permit No. 062020-011, and POET will request that permit be rescinded shortly.

Enclosed are the required application forms, potential to emit calculations and a $5,250 filing fee.

Please contact Shelby Wrage at (605) 965-2315 (Shelby.Wrage@POET.com) or Mike Primrose at (660) 385-8113 (Mike.Primrose@POET.com) if you have any questions or concerns regarding this application.

Regards,

Scott Tuttle
General Manager
POET Biorefining – Macon, LLC

FORM OP-A02 – APPLICATION FOR AUTHORITY TO OPERATE – SECTION A

A02.00 – APPLICATION FOR AUTHORITY TO OPERATE

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

1. LIST THE INSTALLATION'S PRINCIPAL PRODUCT(S)

PRINCIPAL PRODUCT(S)	TWO-DIGIT SIC CODE
Ethanol	28
Dried Distillers Grains with Solubles	20

2. LIST ALL OF THE INSTALLATION'S PROCESSES

PROCESSES	TWO-DIGIT SIC CODE
Ethanol Production	28
Grain Milling, DDGS Production	20
Grain Receiving/Storage	42
Ethanol Production By-Products	28

3. HAS THE INSTALLATION SUBMITTED AN EMISSION INVENTORY QUESTIONNAIRE, OR EIQ, IN THE PAST FIVE YEARS?

YES ☑ NO ☐ If No, submit one copy of a completed EIQ with this application and complete the table below.

If No, indicate the number of each EIQ form submitted with the application.

	1.1	Process Flow Diagram		2.3	VOC Process Mass-Balance Worksheet
	1.2	Summary of Emission Points		2.4	Petroleum Loading Worksheet
	2.0	Emission Point Information		2.5	Organic Liquid Storage-Fixed Roof Tank
	2.0C	Control Device Information		2.5L	General Liquid Storage Tank Information
	2.0P	Portable Plant Information		2.6	Organic Liquid Storage-Floating Roof Tank
	2.0S	Stack Information		2.7	Haul Road Fugitive Emissions Worksheet
	2.0Z	Ozone Season Information		2.8	Storage Pile Worksheet
	2.1	Fuel Combustion Worksheet		2.9	Stack Test/Continuous Emission Monitoring Worksheet
	2.2	Incinerator Worksheet		2.T	Hazardous Air Pollutant Worksheet

4. INDICATE THE NUMBER OF EACH APPLICATION FORM, LISTED BELOW, INCLUDED WITH THIS APPLICATION

1	C01.00	Insignificant Activities Required To Be Listed		D03.20	Combustion Turbines and Internal Combustion Engines
1	D01.00	Existing Plant-Wide Conditions		D03.30	Spray Booths
1	D02.00	Proposed Plant-Wide Conditions		D04.00	Alternate Operating Scenario/Voluntary Conditions
	D03.00	General Emission Unit		D05.00	Compliance Determination
2	D03.10	Indirect Heating Sources		F01.00	General Comments

FORM OP-B01 – APPLICABLE REQUIREMENTS CHECKLIST – SECTION B

B01.00 – APPLICABLE REQUIREMENTS CHECKLIST

INSTALLATION NAME	FIPS	PLANT NO.	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

ENTIRE STATE OF MISSOURI
(NOTE: ALL INSTALLATIONS MUST SUBMIT FORM OP–B01.00)

1. STATE ADMINISTRATIVE PERMIT REQUIREMENTS

APPLICABILITY			TITLE	ORGANIZATION
YES	NO	REASON		
	X	J	10 CSR 10-6.010	Ambient Air Quality Standards [1]
X		J	10 CSR 10-6.020	Definitions and Common Reference Tables [1]
X		J	10 CSR 10-6.030	Sampling Methods for Air Pollution Sources [1]
	X	J	10 CSR 10-6.040	Reference Methods [1]
X		J	10 CSR 10-6.300	Conformity of General Federal Actions to State Implementation Plans [1]
X		J	10 CSR 10-6.320	Sales Tax Exemption [2]

2. CORE PERMIT REQUIREMENTS

TITLE	ORGANIZATION
10 CSR 10-6.050	Start-Up, Shutdown, and Malfunction Conditions [1]
10 CSR 10-6.060	Construction Permits Required [1]
10 CSR 10-6.065	Operating Permits [1]
10 CSR 10-6.110	Submission of Emission Data, Emission Fees and Process Information [1]
10 CSR 10-6.130	Controlling during Episodes of High Air Pollution [1]
10 CSR 10-6.140	Restrictions of Emissions Credit for Reduced Pollutant Concentrations from the use of Dispersion Techniques [1]
10 CSR 10-6.150	Circumvention [1]
10 CSR 10-6.170	Restriction of Particulate Matter to the Ambient Air Beyond the Premises of Origin [1]
10 CSR 10-6.180	Measurement of Emissions of Air Contaminants [1]
10 CSR 10-6.210	Confidential Information [1]
10 CSR 10-6.230	Administrative Penalties [2]
10 CSR 10-6.250	Asbestos Abatement Projects-Certification, Accreditation, and Business Exemption Requirements [2]
10 CSR 10-6.280	Compliance Monitoring Usage [1]

3. STATE APPLICABLE REQUIREMENTS

APPLICABILITY			TITLE	ORGANIZATION
YES	NO	REASON		
X			10 CSR 10-6.070	New Source Performance Regulations (**NOTE**: if yes, check specific subpart on Form OP-BO2.00) [2]
X			10 CSR 10-6.075	Maximum Achievable Control Technology Regulations (**NOTE**: if yes, check specific subpart Form OP-BO3.00) [2]
	X		10 CSR 10-6.080	Emission Standards for Hazardous Air Pollutants (**NOTE**: if yes, check specific subpart Form OP-BO4.00) [2]
	X		10 CSR 10-6.090	Restriction of Emission of Fluorides From Primary Aluminum Reduction Installations [1]
	X		10 CSR 10-6.100	Alternate Emission Limits For Ozone Nonattainment Areas [2]
	X		10 CSR 10-6.120	Restriction of Emissions of Lead From Specific Lead Smelter-Refinery Installations [1]
	X		10 CSR 10-6.200	Hospital, Medical, Infectious Waste Incinerators [1]
X			10 CSR 10-6.220	Restriction of Emission of Visible Air Contaminants [1]
	X		10 CSR 10-6.240	Asbestos Abatement Projects—Registration, Notification and Performance Requirements [2]
	X		10 CSR 10-6.260	Restriction of Emission of Sulfur Compounds [1]
	X		10 CSR 10-6.270	Acid Rain Source Permits Required – If Applicable, Submit Acid Rain Permit Applications to the EPA [2]
	X		10 CSR 10-6.310	Restriction of Emissions From Municipal Solid Waste Landfills [1]
	X		10 CSR 10-6.330	Restriction of Emissions From Batch-Type Charcoal Kilns [1]
	X		10 CSR 10-6.350	Emission Limitations and Emissions Trading of Oxides of Nitrogen [1]
			10 CSR 10-6.360	Control of NOx Emissions From Electric Generating Units and Non-Electric Generating Boilers [2]
	X		10 CSR 10-6.380	Control of NOx Emissions From Portland Cement Kilns
	X		10 CSR 10-6.390	Control of NOx Emissions From Large Stationary Internal Combustion Engines
X			10 CSR 10-6.400	Restriction of Emission of Particulate Matter From Industrial Processes [1]
	X		10 CSR 10-6.405	Restriction of Particulate Matter Emissions From Fuel Burning Equipment Used for Indirect Heating
	X		10 CSR 10-6.410	Emissions Banking and Trading

[1] Federal, state and local agency enforceable regulation.
[2] State and local agency enforceable regulation.

FORM OP-B02 – APPLICABLE REQUIREMENTS CHECKLIST – SECTION B			
B02.00 – APPLICABLE REQUIREMENTS CHECKLIST			
INSTALLATION NAME POET Biorefining - Macon, LLC	FIPS 121	PLANT NUMBER 0028	YEAR SUBMITTED 2022

NEW SOURCE PERFORMANCE REGULATIONS - 10 CSR 10-6.070

(NOTE: IF CHECKED YES ON FORM OP-B01.00 FOR 10 CSR 10-6.070, PLEASE IDENTIFY SPECIFIC SUBPART. IF CHECKED NO, THIS FORM DOES NOT NEED TO BE SUBMITTED. ALL STANDARDS ARE FEDERALLY ENFORCEABLE.)

APPLICABILITY			TITLE	ORGANIZATION
YES	NO	REASON	SUBPART	(40 CFR PART 60 NEW SOURCE PERFORMANCE STANDARDS)
X			A	General Provisions
	X		B	Adoption and Submittal of State Plans for Designated Facilities
	X		C	Emission Guidelines and Compliance Times
	X		Ca	[Reserved]
	X		Cb	Emission Guidelines and Compliance Times for Municipal Waste Combusters that are Constructed on or before December 19, 1995
	X		Cc	Emission Guidelines and Compliance Times for Municipal Solid Waste Landfills
	X		Cd	Emission Guidelines and Compliance Times for Sulfuric Acid Productions Units
	X		Ce	Emission Guidelines and Compliance Times for Hospital/Medical/Infectious Waste Incinerators
	X		D	Fossil-Fuel Fired Steam Generators (construction started after 8/17/71)
	X		Da	Electric Utility Steam Generating Units(construction started after 9/18/78)
	X		Db	Industrial-Commercial-Institutional Steam Generating Units
X			Dc	Small Industrial-Commercial-Institutional Steam Generating Units
	X		E	Incinerators
	X		Ea	Municipal Waste Combustors Constructed Between 12/20/89 and 9/20/94
	X		Eb	Municipal Waste Combustors After 9/20/94
	X		Ec	Hospital/Medical/Infectious Waste Incinerators Constructed After 6-20-96
	X		F	Portland Cement Plants
	X		G	Nitric Acid Plants
	X		H	Sulfuric Acid Plants
	X		I	Hot Mix Asphalt Facilities
	X		J	Petroleum Refineries
	X		Ja	Petroleum Refineries for Which Construction, Reconstruction, or Modification Commenced After May 14, 2007
	X		K	Storage Vessels for Petroleum Liquids for Which Construction, Reconstruction, or Modification Commenced After June 11, 1973, and Prior to May 19, 1978
	X		Ka	Storage Vessels for Petroleum Liquids for Which Construction, Reconstruction, or Modification Commenced After May 18, 1978, and Prior to July 23, 1984
X			Kb	Volatile Organic Liquid Storage Vessels (Including Petroleum Liquid Storage Vessels) for Which Construction, Reconstruction, or Modification Commenced after July 23, 1984
	X		L	Secondary Lead Smelters
	X		M	Secondary Brass and Bronze Production Plants
	X		N	Primary Emissions from Basic Oxygen Process Furnaces(construction after 6/11/73
	X		Na	Secondary Emissions from Basic Oxygen Process Steelmaking Facilities (Construction started after1/20/83)
	X		O	Sewage Treatment Plants
	X		P	Primary Copper Smelters
	X		Q	Primary Zinc Smelters
	X		R	Primary Lead Smelters
	X		S	Primary Aluminum Reduction Plants
	X		T	Phosphate Fertilizer Industry; Wet-Process Phosphoric Acid Plants
	X		U	Phosphate Fertilizer Industry; Superphosphoric Acid Plants
	X		V	Phosphate Fertilizer Industry; Diammonium Phosphate Plants
	X		W	Phosphate Fertilizer Industry; Triple Superphosphate Plants
	X		X	Phosphate Fertilizer Industry; Granular Triple Superphosphate Storage Facilities
	X		Y	Coal Preparation Plants
	X		Z	Ferroalloy Production Facilities
	X		AA	Steel Plants Electric Arc Furnaces (Constructed from 11/21/74 to 8/17/83)
	X		AAa	Steel Plants Electric Arc Furnaces and Argon-oxygen Decarburization Vessels (Constructed after 8/7/83)
	X		BB	Kraft Pulp Mills
	X		CC	Glass Manufacturing Plants
	X		DD	Grain Elevators
	X		EE	Surface Coating of Metal Furniture

FORM OP-B02 – APPLICABLE REQUIREMENTS CHECKLIST – SECTION B

B02.00 – APPLICABLE REQUIREMENTS CHECKLIST

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

NEW SOURCE PERFORMANCE REGULATIONS - 10 CSR 10-6.070 (CONTINUED)

(NOTE: IF CHECKED YES ON FORM OP-B01.00 FOR 10 CSR 10-6.070, PLEASE IDENTIFY SPECIFIC SUBPART. IF CHECKED NO, THIS FORM DOES NOT NEED TO BE SUBMITTED. ALL STANDARDS ARE FEDERALLY ENFORCEABLE.)

APPLICABILITY			TITLE	ORGANIZATION
YES	NO	REASON	SUBPART	(40 CFR PART 60 NEW SOURCE PERFORMANCE STANDARDS)
	X		FF	[Reserved]
	X		GG	Stationary Gas Turbines
	X		HH	Lime Manufacturing Plants
	X		KK	Lead-Acid Battery Manufacturing
	X		LL	Metallic Mineral Processing Plants
	X		MM	Automobile and Light-Duty Truck Surface Coating Operations
	X		NN	Phosphate Rock Plants
	X		PP	Ammonium Sulfate Manufacture
	X		QQ	Graphic Arts Industry; Publication Rotogravure Printing
	X		RR	Pressure Sensitive Tape and Label Surface Coating Operations
	X		SS	Industrial Surface Coating Large Appliances
	X		TT	Metal Coil Surface Coating
	X		UU	Asphalt Processing and Asphalt Roofing Manufacture
	X		VV	Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry
X			VVa	Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry for Which Construction, Reconstruction, or Modification Commenced After November 7, 2006
	X		WW	Beverage Can Surface Coating Industry
	X		XX	Bulk Gasoline Terminals
	X		AAA	New Residential Wood Heaters
	X		BBB	Rubber Tire Manufacturing Industry
	X		CCC	[Reserved]
	X		DDD	Polymer Manufacturing Industry
	X		EEE	[Reserved]
	X		FFF	Flexible Vinyl and Urethane Coating and Printing
	X		GGG	Equipment Leaks of VOC in Petroleum Refineries
	X		GGGa	Equipment Leaks of VOC in Petroleum Refineries for which Construction, Reconstruction, or Modification Commenced After November 7, 2006
	X		HHH	Synthetic Fiber Production Facilities
	X		III	VOC Emissions from SOCMI Air Oxidation Unit Processes
	X		JJJ	Petroleum Dry Cleaners
	X		KKK	Equipment Leaks of VOC from Onshore Natural Gas Processing
	X		LLL	Onshore Natural Gas Processing-SO_2 Emissions
	X		MMM	[Reserved]
	X		NNN	VOC Emissions from SOCMI Distillation Operations
	X		OOO	Nonmetallic Mineral Processing Plants
	X		PPP	Wool Fiberglass Insulation Manufacturing Plants
	X		QQQ	VOC Emissions form Petroleum Refinery Wastewater Systems
	X		RRR	Synthetic Organic Chemical Manufacturing Reactor Processes
	X		SSS	Magnetic Tape Coating Facilities
	X		TTT	Industrial Surface Coating of Plastic Parts for Business Machines
	X		UUU	Calciners and Dryers in Mineral Industries
	X		VVV	Polymeric Coating of Supporting Substrates Facilities
	X		WWW	Landfills
	X		AAAA	Small Municipal Waste Combustion Units (started after 8/30/99, Modifications or Reconstruction after 6/6/01)
	X		BBBB	Emission Guidelines and Compliance Times for Small Municipal Waste Combustion Units Constructed On or Before August 30, 1999
	X		CCCC	Commercial and Industrial Solid Waste Incineration Units for Which Construction is Commenced After November 30, 1999 or for which Modification or Reconstruction is Commenced on or After June 1, 2001
	X		DDDD	Emission Guidelines and Compliance Times for Commercial and Industrial Solid Waste Incineration Units Constructed on or Before 11-30-99
	X		EEEE	Other Solid Waste Incineration Units for Which Construction is Commenced After December 9, 2004, or for Which Modification or Reconstruction is Commenced on or After June 16, 2006

FORM OP-B02 – APPLICABLE REQUIREMENTS CHECKLIST – SECTION B			
B02.00 – APPLICABLE REQUIREMENTS CHECKLIST			
INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

NEW SOURCE PERFORMANCE REGULATIONS - 10 CSR 10-6.070 (CONTINUED)
(NOTE: IF CHECKED YES ON FORM OP-B01.00 FOR 10 CSR 10-6.070, PLEASE IDENTIFY SPECIFIC SUBPART. IF CHECKED NO, THIS FORM DOES NOT NEED TO BE SUBMITTED. ALL STANDARDS ARE FEDERALLY ENFORCEABLE.)

APPLICABILITY			TITLE SUBPART	ORGANIZATION (40 CFR PART 60 NEW SOURCE PERFORMANCE STANDARDS)
	X		FFFF	Emission Guidelines and Compliance Times for Other Solid Waste Incineration Units that Commenced Construction On or Before December 9, 2004
	X		GGGG	[Reserved]
	X		HHHH	Emission Guidelines and Compliance Times for Coal-Fired Electric Steam Generating Units
	X		IIII	Standards of Performance for Stationary Compression Ignition Internal Combustion Engines
	X		JJJJ	Standards of Performance for Stationary Spark Ignition Internal Combustion Engines
	X		KKKK	Standards of Performance for Stationary Combustion Turbines
	X		LLLL	Standards of Performance for New Sewage Sludge Incineration Units
	X		MMMM	Emission Guidelines and Compliance Times for Existing Sewage Sludge Incineration Units

FORM OP-B03 – APPLICABLE REQUIREMENTS CHECKLIST – SECTION B

B03.00 – APPLICABLE REQUIREMENTS CHECKLIST

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

MAXIMUM ACHIEVABLE CONTROL TECHNOLOGY REGULATIONS - 10 CSR 10-6.075

(NOTE: IF CHECKED YES ON FORM OP-B01.00 FOR 10 CSR 10-6.075, PLEASE IDENTIFY THE SPECIFIC SUBPART. IF YOU CHECKED NO, THIS FORM DOES NOT NEED TO BE SUBMITTED. ALL STANDARDS ARE FEDERALLY ENFORCEABLE.)

APPLICABILITY			TITLE SUBPART	ORGANIZATION (40 CFR PART 63 NATIONAL EMISSION STANDARDS FOR HAZARDOUS AIR POLLUTANTS FOR SOURCE CATEGORIES)
YES	NO	REASON		
	X		A	General Provisions
	X		B	Requirements for Control Technology Determinations for Major Sources in Accordance with Clean Air Act Sections, Sections 112(g) and 112(j)
	X		F	Organic Hazardous Air Pollutants from the Synthetic Organic Chemical Manufacturing Industry
	X		G	Organic Hazardous Air Pollutants from the Synthetic Organic Chemical Manufacturing Industry for Process Vents, Storage Vessels, Transfer Operations, and Wastewater
	X		H	Organic Hazardous Air Pollutants for Equipment Leaks
	X		I	Organic Hazardous Air Pollutants for Certain Process Subject to the Negotiated Regulation for Equipment Leaks
	X		J	Polyvinyl Chloride Copolymers Production
	X		K	[Reserved]
	X		L	Coke Oven Batteries
	X		M	Perchloroethylene Air Emission for Dry Cleaning
	X		N	Chromium Emissions from Hard and Decorative Chromium Electroplating and from Chromium Anodizing Tanks
	X		O	Ethylene Oxide Emission for Sterilization Facilities
	X		P	[Reserved]
	X		Q	Hazardous Air Pollutants for Industrial Process Cooling Towers
	X		R	Gasoline Distribution Facilities (Bulk Gasoline Terminals and Pipeline Breakout Stations)
	X		S	Hazardous Air Pollutants from the Pulp and Paper Industry
	X		T	Halogenated Solvent Cleaning
	X		U	Group I Polymers and Resins
	X		V	[Reserved]
	X		W	Epoxy Resins Production and Non-Nylon Polyamides Production
	X		X	Hazardous Air Pollutants from Secondary Lead Smelting
	X		Y	National Emission Standards for Marine Vessel Loading and Unloading Operations
	X		Z	[Reserved]
	X		AA	Hazardous Air Pollutants from Phosphoric Acid Manufacturing Plants
	X		BB	Hazardous Air Pollutants from Phosphate Fertilizer Production Plants
	X		CC	Hazardous Air Pollutants; Petroleum Refineries
	X		DD	Off-Site Waste and Recovery Operations
	X		EE	Magnetic Tape Manufacturing Operations
	X		FF	[Reserved]
	X		GG	National Emissions Standards for Aerospace Manufacturing and Rework Facilities
	X		HH	Hazardous Air Pollutants from Oil and Natural Gas Production Facilities
	X		II	National Emission Standards for Shipbuilding & Ship Repair (Surface Coating)
	X		JJ	National Emission Standards for Wood Furniture Manufacturing Operations
	X		KK	National Emission Standard for the Printing and Publishing Industry
	X		LL	Hazardous Air Pollutants for Primary Aluminum Reduction Plants
	X		MM	Chemical Recovery Combustion Sources at Kraft, Soda, Sulfite, and Stand-Alone Semichemical Pulp Mills
	X		OO	Tanks—Level 1
	X		PP	Containers
	X		QQ	Surface Impoundments
	X		RR	Individual Drain Systems
	X		SS	Closed Vent Systems, Control Devices, Recovery Devices and Routing to a Fuel Gas System or a Process
	X		TT	Equipment Leaks—Control Level 1
	X		UU	Equipment Leaks—Control Level 2 Standards
	X		VV	Oil Water Separators and Organic-Water Separators
	X		WW	Storage Vessels (Tanks)—Control Level 2
	X		XX	Ethylene Manufacturing Process Units: Heat Exchange Systems and Waste Operations

FORM OP-B03 – APPLICABLE REQUIREMENTS CHECKLIST – SECTION B

B03.00 – APPLICABLE REQUIREMENTS CHECKLIST

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

MAXIMUM ACHIEVABLE CONTROL TECHNOLOGY REGULATIONS - 10 CSR 10-6.075

(NOTE: IF CHECKED YES ON FORM OP-B01.00 FOR 10 CSR 10-6.075, PLEASE IDENTIFY THE SPECIFIC SUBPART. IF YOU CHECKED NO, THIS FORM DOES NOT NEED TO BE SUBMITTED. ALL STANDARDS ARE FEDERALLY ENFORCEABLE.)

APPLICABILITY			TITLE SUBPART	ORGANIZATION (40 CFR PART 63 NATIONAL EMISSION STANDARDS FOR HAZARDOUS AIR POLLUTANTS FOR SOURCE CATEGORIES)
	X		YY	Hazardous Air Pollutants for Source Categories: Generic Maximum Available Control Technology Standards
	X		ZZ	[Reserved]
	X		AAA	[Reserved]
	X		BBB	[Reserved]
	X		CCC	Steel Pickling – HCl Process Facilities and Hydrochloric Acid Regeneration Plants
	X		DDD	Hazardous Air Pollutants for Mineral Wool Production
	X		EEE	Hazardous Air Pollutants from Hazardous Waste Combustors
	X		FFF	[Reserved]
	X		GGG	Pharmaceuticals Production
	X		HHH	Hazardous Air Pollutants from Natural Gas Transmission and Storage Facilities
	X		III	Hazardous Air Pollutants for Flexible Polyurethane Foam Production
	X		JJJ	Hazardous Air Pollutant Emissions: Group IV Polymers and Resins
	X		KKK	[Reserved]
	X		LLL	Hazardous Air Pollutants from the Portland Cement Manufacturing Industry
	X		MMM	Hazardous Air Pollutants for Pesticide Active Ingredient Production
	X		NNN	Hazardous Air Pollutants for Wool Fiberglass Manufacturing
	X		OOO	Manufacture of Amino/Phenolic Resins
	X		PPP	Hazardous Air Pollutant Emissions for Polyether Polyols Production
	X		QQQ	Primary Copper Smelting
	X		RRR	Secondary Aluminum Production
	X		SSS	[Reserved]
	X		TTT	Hazardous Air Pollutants for Primary Lead Smelting
	X		UUU	Petroleum Refineries:Catalytic Cracking Units, Catalytic Reforming Units, and Sulfur Recovery Units
	X		VVV	Hazardous Air Pollutants: Publicly Owned Treatment Works
	X		WWW	[Reserved]
	X		XXX	Hazardous Air Pollutants for Ferroalloys Production: Ferromanganese and Silicomanganese
	X		AAAA	Municipal Solid Waste Landfills
	X		CCCC	Manufacturing of Nutritional Yeast
	X		DDDD	Plywood and Composite Wood Products
	X		EEEE	Organic Liquids Distribution (non-gasoline)
	X		FFFF	Miscellaneous Organic Chemical Manufacturing
	X		GGGG	Solvent Extractions for Vegetable Oil Production
	X		HHHH	Wet Formed Fiberglass Mat Production
	X		IIII	Surface Coating of Automobiles and Light Duty Trucks
	X		JJJJ	Paper and Other Web Coating
	X		KKKK	Surface Coating of Metal Cans
	X		MMMM	Surface Coating of Miscellaneous Metal Parts and Products
	X		NNNN	Surface Coating of Large Appliances
	X		OOOO	Printing, Coating and Dyeing of Fabrics and Other Textiles
	X		PPPP	Surface Coating of Plastic Parts and Products
	X		QQQQ	Surface Coating of Wood Building Products
	X		RRRR	Surface Coating of Metal Furniture
	X		SSSS	Surface Coating of Metal Coil
	X		TTTT	Leather Finishing Operations
	X		UUUU	Cellulose Products Manufacturing
	X		VVVV	Boat Manufacturing
	X		WWWW	Reinforced Plastic Composites Production
	X		XXXX	Rubber Tire Manufacturing
	X		YYYY	Stationary Combustion Turbines
	X		ZZZZ	Stationary Reciprocating Internal Combustion Engines (RICE)

FORM OP-B03 – APPLICABLE REQUIREMENTS CHECKLIST – SECTION B

B03.00 – APPLICABLE REQUIREMENTS CHECKLIST

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

MAXIMUM ACHIEVABLE CONTROL TECHNOLOGY REGULATIONS - 10 CSR 10-6.075

(NOTE: IF CHECKED YES ON FORM OP-B01.00 FOR 10 CSR 10-6.075, PLEASE IDENTIFY THE SPECIFIC SUBPART. IF YOU CHECKED NO, THIS FORM DOES NOT NEED TO BE SUBMITTED. ALL STANDARDS ARE FEDERALLY ENFORCEABLE.)

APPLICABILITY			TITLE SUBPART	ORGANIZATION (40 CFR PART 63 NATIONAL EMISSION STANDARDS FOR HAZARDOUS AIR POLLUTANTS FOR SOURCE CATEGORIES)
	X		AAAAA	Lime Manufacturing Plants
	X		BBBBB	Semiconductor Manufacturing
	X		CCCCC	Coke Ovens: Pushing, Quenching and Battery Stacks
	X		DDDDD	Industrial, Commercial and Institutional Boilers and Process Heaters **(This subpart has been vacated by court action)**
	X		EEEEE	Iron and Steel Foundries
	X		FFFFF	Integrated Iron and Steel Manufacturing Facilities
	X		GGGGG	Site Remediation
	X		HHHHH	Miscellaneous Coating Manufacturing (MON)
	X		IIIII	Mercury Emissions from Mercury Cell Chlor-Alkali Plants
	X		JJJJJ	Brick and Structural Clay Products Manufacturing **(This subpart has been vacated by court action)**
	X		KKKKK	Clay Ceramics Manufacturing **(This subpart has been vacated by court action)**
	X		LLLLL	Asphalt Processing and Asphalt Roofing Manufacture
	X		MMMMM	Flexible Polyurethane Foam Fabrication Operations
	X		NNNNN	Hydrochloric Acid Production
	X		OOOOO	[Reserved]
	X		PPPPP	Engine Test Cells/Stands
	X		QQQQQ	Friction Materials Manufacturing
	X		RRRRR	Taconite Iron Ore Processing
	X		SSSSS	Refractory Products Manufacturing
	X		TTTTT	Primary Magnesium Refining
	X		UUUUU	[Reserved]
	X		VVVVV	[Reserved]
	X		WWWWW	Hospital Ethylene Oxide Sterilizers
	X		XXXXX	[Reserved]
	X		YYYYY	Area Sources: Electric Arc Furnace Steelmaking Facilities
	X		ZZZZZ	Iron and Steel Foundries Area Sources
	X		AAAAAA	[Reserved]
	X		BBBBBB	Source Category: Gasoline Distribution Bulk Terminals, Bulk Plants, and Pipeline Facilities
X			CCCCCC	Source Category: Gasoline Dispensing Facilities
	X		DDDDDD	Polyvinyl Chloride and Copolymers Production Area Sources
	X		EEEEEE	Primary Copper Smelting Area Sources
	X		FFFFFF	Secondary Copper Smelting Area Sources
	X		GGGGGG	Primary Nonferrous Metals Area Sources—Zinc, Cadmium, and Beryllium
	X		HHHHHH	Paint Stripping and Miscellaneous Surface Coating Operations at Area Sources
	X		IIIIII	[Reserved]
	X		JJJJJJ	[Reserved]
	X		KKKKKK	[Reserved]
	X		LLLLLL	Acrylic and Modacrylic Fibers Production Area Sources
	X		MMMMMM	Carbon Black Production Area Sources
	X		NNNNNN	Chemical Manufacturing Area Sources: Chromium Compounds
	X		OOOOOO	Flexible Polyurethane Foam Production and Fabrication Area Sources
	X		PPPPPP	Lead Acid Battery Manufacturing Area Sources
	X		QQQQQQ	Wood Preserving Area Sources
	X		RRRRRR	Clay Ceramics Manufacturing Area Sources
	X		SSSSSS	Glass Manufacturing Area Sources
	X		TTTTTT	Secondary Nonferrous Metals Processing Area Sources
	X		UUUUUU	[Reserved]
	X		VVVVVV	Chemical Manufacturing Area Sources
	X		WWWWWW	Area Source Standards for Plating and Polishing Operations
	X		XXXXXX	Area Source Standards for Nine Metal Fabrication and Finishing Source Categories

FORM OP-B03 – APPLICABLE REQUIREMENTS CHECKLIST – SECTION B

B03.00 – APPLICABLE REQUIREMENTS CHECKLIST

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

MAXIMUM ACHIEVABLE CONTROL TECHNOLOGY REGULATIONS - 10 CSR 10-6.075

(NOTE: IF CHECKED YES ON FORM OP-B01.00 FOR 10 CSR 10-6.075, PLEASE IDENTIFY THE SPECIFIC SUBPART. IF YOU CHECKED NO, THIS FORM DOES NOT NEED TO BE SUBMITTED. ALL STANDARDS ARE FEDERALLY ENFORCEABLE.)

APPLICABILITY			TITLE SUBPART	ORGANIZATION (40 CFR PART 63 NATIONAL EMISSION STANDARDS FOR HAZARDOUS AIR POLLUTANTS FOR SOURCE CATEGORIES)
	X		YYYYYY	Ferroalloys Production Facilities
	X		ZZZZZZ	Aluminum, Copper, and Other Nonferrous Foundries
	X		AAAAAAA	Asphalt Processing and Asphalt Roofing Manufacturing
	X		BBBBBBB	Chemical Preparations Industry
	X		CCCCCCC	Paints and Allied Products Manufacturing
	X		DDDDDDD	Area Source Standards for Prepared Feeds Manufacturing
	X		EEEEEEE	Gold Mine Ore Processing and Production Area Source Category

FORM OP-B07 – APPLICABLE REQUIREMENTS CHECKLIST – SECTION B

B07.00 – APPLICABLE REQUIREMENTS CHECKLIST

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

OUTSTATE MISSOURI AREA

(NOTE: PLEASE INCLUDE FORM OP-B07.00 ONLY IF NOT LOCATED AT FOLLOWING LOCATIONS: CITY OF ST. LOUIS, JEFFERSON, FRANKLIN, ST. CHARLES, CLAY, CASS, BUCHANAN, RAY, JACKSON, PLATTE, AND GREENE COUNTIES)

1. CORE REQUIREMENTS

TITLE	ORGANIZATION
10 CSR 10-3.030	Open Burning Restrictions [1]
10 CSR 10-3.090	Restrictions of Emission of Odors [2]

2. APPLICABLE REQUIREMENTS

APPLICABILITY			TITLE	ORGANIZATION
YES	NO	REASON		
	X		10 CSR 10-3.010	Auto Exhaust Emission Controls [1]
	X		10 CSR 10-3.160	Restriction of Emission of Fluorides From Diammonium Phosphate Fertilizer Production [1]

[1] Federal, state and local agency enforceable regulation
[2] State and local agency enforceable regulation

FORM OP-C01 – INSIGNIFICANT ACTIVITIES REQUIRED TO BE LISTED – SECTION C

C01.00 – INSIGNIFICANT ACTIVITIES

NOTE: For Part 70 applications only.

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

INSIGNIFICANT ACTIVITY	POTENTIAL ESTIMATED EMISSIONS (TONS/YR)						
EMISSION UNIT ID	PM_{10}	SO_X	NO_X	VOC	CO	LEAD	HAPS
DESCRIPTION Corrosion Inhibitor Tank				2.62E-02			2.62E-02
EMISSION UNIT ID	PM_{10}	SO_X	NO_X	VOC	CO	LEAD	HAPS
DESCRIPTION							
EMISSION UNIT ID	PM_{10}	SO_X	NO_X	VOC	CO	LEAD	HAPS
DESCRIPTION							
EMISSION UNIT ID	PM_{10}	SO_X	NO_X	VOC	CO	LEAD	HAPS
DESCRIPTION							
EMISSION UNIT ID	PM_{10}	SO_X	NO_X	VOC	CO	LEAD	HAPS
DESCRIPTION							
EMISSION UNIT ID	PM_{10}	SO_X	NO_X	VOC	CO	LEAD	HAPS
DESCRIPTION							

DUPLICATE THIS FORM AS NEEDED

FORM OP-D01 – EXISTING PLANT-WIDE CONDITIONS – SECTION D

D01.00 – EXISTING PLANT-WIDE CONDITIONS

NOTE: Include a blank form when no existing plant-wide conditions are applicable.

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

Please list in the space provided below any permit conditions which are currently applicable on a plant wide basis: (e.g., Production is limited to 10,000 units per 12 month rolling average, or a limit on the installation's hours of operation)

PERMIT NO.	APPLICABLE PERMIT CONDITION
042017-011	Emit less than 92.74 tons of total NOx in any consecutive 12-month period
042017-011	Emit less than 100 tons of total CO in any consecutive 12-month period
042017-011	Produce less than 135,000 tons of CO2 in any consecutive 12-month period
042017-011	Produce less than 58,000,000 gallons of denatured ethanol in any consecutive 12-month period
042017-011	Receive less than 3,000,000 gallons of denaturant in any consecutive 12-month period
042017-011	Produce less than 55,000,000 gallons of anhydrous ethanol in any consecutive 12-month period
042017-011	Receive less than 666,800tons of grain in any consecutive 12-month period
042017-011	Produce less than 200,000 tons of DDGS in any consecutive 12-month period
042017-011	Produce less than 80,000 tons of wet cake in any consecutive 12-month period
042017-011	Receive less than 1,600,000 gallons of chemicals (urea, enzymes, etc.) in any consecutive 12-month period
042017-011	Produce and loadout less than 20,000 tons of syrup in any consecutive 12-month period

PERMIT NO.	COMPLIANCE DEMONSTRATION METHOD	DESCRIBE METHOD AND GIVE REFERENCE
042017-011	Testing	Complete performance testing as required by permit
042017-011	Recordkeeping	Maintain records to demonstrate 12-month limits are not exceeded

DUPLICATE THIS FORM AS NEEDED

FORM OP-D02 – PROPOSED PLANT-WIDE CONDITIONS – SECTION D

D02.00 – PROPOSED PLANT-WIDE CONDITIONS

NOTE: Include a blank form when no existing plant-wide conditions are applicable.

INSTALLATION NAME	FIPS	PLANT NO.	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

Please list in the space provided below any proposed permit conditions that the installation intends to establish in this operating permit.

PROPOSED CONDITION

Incorporate conditions in Construction Permit 012020-012. No new plant-wide conditions being proposed with this application.

Please describe what methodologies you intend to use to demonstrate compliance with each of the proposed plant-wide condition(s) that are being established above: (e.g., testing, monitoring and record keeping)

PROPOSED CONDITION NO.	COMPLIANCE DEMONSTRATION METHOD	DESCRIBE METHOD AND GIVE REFERENCE

DUPLICATE THIS FORM AS NEEDED

FORM OP-D04 – ALTERNATE OPERATING SCENARIO/VOLUNTARY CONDITIONS – SECTION D

D04.00 – ALTERNATE OPERATING SCENARIO/VOLUNTARY CONDITIONS

INSTALLATION NAME	FIPS	PLANT NO.	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER (ID)	SOURCE CLASSIFICATION CODE (SCC)

1. ALTERNATE OPERATING SCENARIO (FLEXIBILITY)

ALTERNATE SCENARIO ID	SIC CODE ASSOCIATED WITH SCENARIO

DESCRIPTION

2. VOLUNTARY PERMIT CONDITIONS

POLLUTANT CONTROLLED	LIMITATION

DUPLICATE THIS FORM AS NEEDED

FORM OP-D05 – COMPLIANCE DETERMINATION METHODS – SECTION D

D05.00 – COMPLIANCE DETERMINATION

INSTALLATION NAME	FIPS	PLANT NO.	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER (ID)	SOURCE CLASSIFICATION CODE (SCC)

1. APPLICABLE REQUIREMENT

APPLICABLE REQUIREMENT	POLLUTANT(S)
No additional requirements being proposed.	

EMISSION LIMITATION OR STANDARD

Facility will determine compliance with requirements per construction permit 012020-012 and operating permit OP2017-080A

2. TESTING

DATE	TEST METHOD

SUMMARY OF RESULTS

3. MONITORING

PARAMETER MONITORED	MONITORING METHOD

MONITORING SCHEDULE

4. RECORD KEEPING

PARAMETER RECORDED	RECORD KEEPING METHOD

RECORD KEEPING SCHEDULE

5. REPORTING

REPORTING REQUIREMENT	REPORTING SCHEDULE

DUPLICATE THIS FORM AS NEEDED

D06.00 – CORE PERMIT REQUIREMENTS (NOTE: THIS IS A REQUIRED FORM FOR ALL PERMIT APPLICATIONS)

INSTALLATION NAME	FIPS	PLANT NO.	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

NOTE: The installation shall comply with each of the following emission limitations. Consult the appropriate sections in the code of federal regulations and code of state regulations for the full text of the applicable requirements.

10 CSR 10-6.050, *Start-up, Shutdown and Malfunction Conditions*

(a) In the event of a malfunction, which results in excess emissions that exceed one hour, the permittee shall submit to the director within two business days in writing the following information:
   (1) Name and location of installation.
   (2) Name and telephone number of person responsible for the installation.
   (3) Name of the person who first discovered the malfunction and precise time and date that the malfunction was discovered.
   (4) Identity of the equipment causing the excess emissions.
   (5) Time and duration of the period of excess emissions.
   (6) Cause of the excess emissions.
   (7) Air pollutants involved.
   (8) Best estimate of the magnitude of the excess emissions expressed in the units of the applicable requirement and the operating data and calculations used in estimating the magnitude.
   (9) Measures taken to mitigate the extent and duration of the excess emissions.
   (10) Measures taken to remedy the situation that caused the excess emissions and the measures taken or planned to prevent the recurrence of these situations.

(b) The permittee shall submit the paragraph (a.) information list to the director in writing at least 10 days prior to any maintenance, start-up or shutdown, which is expected to cause an excessive release of emissions that exceed one hour. If notice of the event cannot be given 10 days prior to the planned occurrence, it shall be given as soon as practicable prior to the release. If an unplanned excess release of emissions exceeding one hour occurs during maintenance, start-up or shutdown, the director shall be notified verbally as soon as practical during normal working hours and no later than the close of business of the following working day. A written notice shall follow within 10 working days.

(c) Upon receipt of a notice of excess emissions issued by an agency holding a certificate of authority under section 643.140, RSMo, the permittee may provide information showing that the excess emissions were the consequence of a malfunction, start-up or shutdown. The information, at a minimum, should be the paragraph (a.) list and shall be submitted not later than 15 days after receipt of the notice of excess emissions. Based upon information submitted by the permittee or any other pertinent information available, the director or the commission shall make a determination whether the excess emissions constitute a malfunction, start-up or shutdown and whether the nature, extent and duration of the excess emissions warrant enforcement action under section 643.080 or 643.151, RSMo.

(d) Nothing in this rule shall be construed to limit the authority of the director or commission to take appropriate action, under sections 643.080, 643.090 and 643.151, RSMo to enforce the provisions of the Air Conservation Law and the corresponding rule.

(e) Compliance with this rule does not automatically absolve the permittee of liability for the excess emissions reported.

10 CSR 10-6.060, *Construction Permits Required*

The permittee shall not commence construction, modification or major modification of any installation subject to this rule; begin operation after that construction, modification or major modification; or begin operation of any installation which has been shut down longer than five years without first obtaining a permit from the permitting authority.

10 CSR 10-6.065, *Operating Permits*

The permittee shall file for renewal of this operating permit no sooner than eighteen months, nor later than six months, prior to the expiration date of this operating permit. The permittee shall retain the most current operating permit issued to this installation on-site and shall immediately make such permit available to any Missouri Department of Natural Resources personnel upon request.

10 CSR 10-6.080, *Emission Standards for Hazardous Air Pollutants*
40 CFR Part 61 Subpart M, National Emission Standard for Asbestos

(a) The permittee shall follow the procedures and requirements of 40 CFR Part 61, Subpart M for any activities occurring at this installation which would be subject to provisions for 40 CFR Part 61, Subpart M, National Emission Standard for Asbestos.

(b) The permittee shall conduct monitoring to demonstrate compliance with registration, certification, notification and Abatement Procedures and Practices standards as specified in 40 CFR Part 61, Subpart M.

D06.00 – CORE PERMIT REQUIREMENTS (CONTINUED) (THIS IS A REQUIRED FORM FOR ALL PERMIT APPLICATIONS)

INSTALLATION NAME	FIPS	PLANT NO.	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

The installation shall comply with each of the following emission limitations. Consult the appropriate sections in the code of federal regulations and code of state regulations for the full text of the applicable requirements.

10 CSR 10-6.100, *Alternate Emission Limits*

Proposals for alternate emission limitations shall be submitted on Alternate Emission Limits Permit forms provided by the department. An installation owner or operator must obtain an Alternate Emission Limits Permit in accordance with 10 CSR 10-6.100 before alternate emission limits may become effective.

10 CSR 10-6.110, *Submission of Emission Data, Emission Fees and Process Information*

(a) The permittee shall complete and submit an Emission Inventory Questionnaire, or EIQ, in accordance with the requirements outlined in this rule.

(b) The permittee shall pay an annual emission fee per ton of regulated air pollutant emitted according to the schedule in the rule. This fee is an emission fee assessed under authority of RSMo. 643.079 to satisfy the requirements of the Federal Clean Air Act, Title V.

(c) The fees shall be due April 1 each year for emissions produced during the previous calendar year. The fees shall be payable to the Department of Natural Resources and shall be accompanied by the EIQ form or equivalent approved by the director.

10 CSR 10-6.130, *Controlling Emissions During Episodes of High Air Pollution Potential*

This rule specifies the conditions that establish an air pollution alert (yellow/red), watch or emergency and the associated procedures and emissions reduction objectives for dealing with each. The permittee shall submit an appropriate emergency plan if required by the director.

10 CSR 10-6.150, Circumvention

The permittee shall not cause or permit the installation or use of any device or any other means which, without resulting in reduction in the total amount of air contaminant emitted, conceals or dilutes an emission or air contaminant which violates a rule of the Missouri Air Conservation Commission.

10 CSR 10-6.170, Restriction of Particulate Matter to the Ambient Air Beyond the Premises of Origin

(a) The permittee shall not cause or allow to occur any handling, transporting or storing of any material; construction, repair, cleaning or demolition of a building or its appurtenances; construction or use of a road, driveway or open area; or operation of a commercial or industrial installation without applying reasonable measures as may be required to prevent, or in a manner which allows or may allow, fugitive particulate matter emissions to go beyond the premises of origin in quantities that the particulate matter may be found on surfaces beyond the property line or origin. The nature or origin of the particulate matter shall be determined to a reasonable degree of certainty by a technique proven to be accurate and approved by the director.

(b) The permittee shall not cause nor allow to occur any fugitive particulate matter emissions to remain visible in the ambient air beyond the property line of origin.

(c) Should it be determined that noncompliance has occurred, the director may require reasonable control measures as may be necessary.

10 CSR 10-6.180, *Measurement of Emissions of Air Contaminants*

(a) The director may require any person responsible for the source of emission of air contaminants to make or have made tests to determine the quantity or nature, or both, of emission of air contaminants from the source. The director may specify testing methods to be used in accordance with good professional practice. The director may observe the testing. All tests shall be performed by qualified personnel.

(b) The director may conduct tests of emissions of air contaminants from any source. Upon request of the director, the person responsible for the source to be tested shall provide necessary ports in stacks or ducts and other safe and proper sampling and testing facilities, exclusive of instruments and sensing devices as may be necessary for proper determination of the emission of air contaminants.

(c) The director shall be given a copy of the test results in writing and signed by the person responsible for the tests.

10 CSR 10-6.250, *Asbestos Abatement Projects – Certification, Accreditation, and Business Exemption Requirements*

The permittee shall conduct all asbestos abatement projects within the procedures established for certification and accreditation by 10 CSR 10-6.250. This rule requires individuals who work in asbestos abatement projects to be certified by the department's Air Pollution Control Program. This rule requires training providers who offer training for asbestos abatement occupations to be accredited by the department's Air Pollution Control Program. This rule requires persons who hold exemption status from certain requirements of this rule to allow the department to monitor training provided to employees. Each individual who works in asbestos abatement projects must first obtain certification for the appropriate occupation from the department. Each person who offers training for asbestos abatement occupations must first obtain accreditation from the department. Certain business entities that meet the requirements for state-approved exemption status must allow the department to monitor training classes provided to employees who perform asbestos abatement.

D06.00 – CORE PERMIT REQUIREMENTS (CONTINUED) (THIS IS A REQUIRED FORM FOR ALL PERMIT APPLICATIONS)

INSTALLATION NAME	FIPS	PLANT NO.	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

The installation shall comply with each of the following emission limitations. Consult the appropriate sections in the code of federal regulations and code of state regulations for the full text of the applicable requirements.

Yes	No	Regulation *(Please check the appropriate response regarding applicability)*
	✓	**10 CSR 10-2.070** (Kansas City Metropolitan Area)
✓		**10 CSR 10-3.090** (Outstate Area)
	✓	**10 CSR 10-4.070** (Greene County)

Restriction of Emission of Odors

No person may cause, permit or allow the emission of odorous matter in concentrations and frequencies or for durations that odor can be perceived when one volume of odorous air is diluted with seven volumes of odor-free air for two separate trials not less than 15 minutes apart within the period of 1 hour.

This requirement is not federally enforceable.

10 CSR 10-5.160, *(Not Applicable if not in St. Louis Metropolitan Area)* ***Restriction of Emission of Odors***

No person shall emit odorous matter as to cause an objectionable odor on or adjacent to:

(a) Residential, recreational, institutional, retail sales, hotel or educational premises.
(b) Industrial premises when air containing odorous matter is diluted with 20 or more volumes of odor-free air; or
(c) Premises other than those in paragraphs (1)A.1. and (2) of the rule when air containing odorous matter is diluted with four or more volumes of odor-free air.

The previously mentioned requirement shall apply only to objectionable odors. An odor will be deemed objectionable when 30 percent or more of a sample of the people exposed to it believe it to be objectionable in usual places of occupancy; the sample size to be at least 20 people or 75 percent of those exposed if fewer than 20 people are exposed.

This requirement is not federally enforceable.

10 CSR 10-6.280, *Compliance Monitoring Usage*

(a) The permittee is not prohibited from using the following in addition to any specified compliance methods for the purpose of submission of compliance certificates:
  (1) Monitoring methods outlined in 40 CFR Part 64.
  (2) Monitoring method(s) approved for the permittee pursuant to 10 CSR 10-6.065, "Operating Permits", and incorporated into an operating permit.
  (3) Any other monitoring methods approved by the director.
(b) Any credible evidence may be used for the purpose of establishing whether a permittee has violated or is in violation of any such plan or other applicable requirement. Information from the use of the following methods is presumptively credible evidence of whether a violation has occurred by a permittee:
  (1) Monitoring methods outlined in 40 CFR Part 64.
  (2) A monitoring method approved for the permittee pursuant to 10 CSR 10-6.065, "Operating Permits", and incorporated into an operating permit.
  (3) Compliance test methods specified in the rule cited as the authority for the emission limitations.
(c) The following testing, monitoring or information gathering methods are presumptively credible testing, monitoring, or information gathering methods:
  (1) Applicable monitoring or testing methods, cited in:
    - 10 CSR 10-6.030, "Sampling Methods for Air Pollution Sources".;
    - 10 CSR 10-6.040, "Reference Methods".
    - 10 CSR 10-6.070, "New Source Performance Standards".
    - 10 CSR 10-6.080, "Emission Standards for Hazardous Air Pollutants".
  (2) Other testing, monitoring, or information gathering methods, if approved by the director, that produce information comparable to that produced by any method listed above.

D06.00 – CORE PERMIT REQUIREMENTS (CONTINUED) (THIS IS A REQUIRED FORM FOR ALL PERMIT APPLICATIONS)

INSTALLATION NAME	FIPS	PLANT NO.	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

The installation shall comply with each of the following emission limitations. Consult the appropriate sections in the code of federal regulations and code of state regulations for the full text of the applicable requirements.

10 CSR 10-5.040, *(Delete if not in St. Louis Metropolitan Area)* ***Use of Fuel in Hand-Fired Equipment Prohibited***
It shall be unlawful to operate any hand-fired fuel-burning equipment in the St. Louis, Missouri metropolitan area. This regulation shall apply to all fuel-burning equipment including, but not limited to, furnaces, heating and cooking stoves and hot water furnaces. It shall not apply to wood-burning fireplaces and wood-burning stoves in dwellings, nor to fires used for recreational purpose, nor to fires used solely for the preparation of food by barbecuing. Hand-fired fuel-burning equipment is any stove, furnace, or other fuel-burning device in which fuel is manually introduced directly into the combustion chamber.

Yes	No	Regulation *(Please check the appropriate response regarding applicability)*
	✓	**10 CSR 10-2.100** (Kansas City Metropolitan Area)
✓		**10 CSR 10-3.030** (Outstate Area)
	✓	**10 CSR 10-4.090** (Greene County)
	✓	**10 CSR 10-5.070** (St. Louis Metropolitan Area)

Open Burning Restrictions
(a) The permittee shall not conduct, cause, permit or allow a salvage operation, the disposal of trade wastes or burning of refuse by open burning.
(b) Exception - Open burning of trade waste or vegetation may be permitted only when it can be shown that open burning is the only feasible method of disposal or an emergency exists which requires open burning.
(c) Any person intending to engage in open burning shall file a request to do so with the director. The request shall include the following:
  (1) The name, address and telephone number of the person submitting the application; The type of business or activity involved; A description of the proposed equipment and operating practices, the type, quantity and composition of trade wastes and expected composition and amount of air contaminants to be released to the atmosphere where known.
  (2) The schedule of burning operations.
  (3) The exact location where open burning will be used to dispose of the trade wastes.;
  (4) Reasons why no method other than open burning is feasible.
  (5) Evidence that the proposed open burning has been approved by the fire control authority which has jurisdiction.
(d) Upon approval of the open burning permit application by the director, the person may proceed with the operation under the terms of the open burning permit. Be aware that such approval shall not exempt the installation from the provisions of any other law, ordinance or regulation.
(e) The permittee shall maintain files with letters from the director approving the open burning operation and previous DNR inspection reports.

St. Louis City Ordinance 64749, Sec 17, *(Not Applicable if not in City Limits of St. Louis City)* **Open Burning Restrictions**
(a) No person shall cause, suffer, allow or permit the open burning of refuse.
(b) No person shall conduct, cause or permit the conduct of a salvage operation by open burning.
(c) No person shall conduct, cause or permit the disposal of trade waste by open burning.
(d) No person shall cause or permit the open burning of leaves, trees or the byproducts therefrom, grass, or other vegetation.
(e) It shall be prima-facie evidence that the person who owns or controls property on which open burning occurs, has caused or permitted said open burning.

10 CSR 10-5.240, *(Not Applicable if not in St. Louis Metropolitan Area)* ***Additional Air Quality Control Measures May be Required When Sources Are Clustered in a Small Land Area***
The Air Conservation Commission may prescribe more restrictive air quality control requirements that are more restrictive and more extensive than provided in regulations of general application for:
  (a) Areas in which there are one or more existing sources and/or proposed new sources of particulate matter in any circular area with a diameter of two miles (including sources outside metropolitan area) from which the sum of particulate emissions allowed from these sources by regulations of general application are or would be greater than 2000 tons per year or 500 pounds per hour.
  (b) Areas in which there are one or more existing sources and/or proposed new sources of sulfur dioxide in any circular area with a diameter of two miles from which the sum of sulfur dioxide emissions from these sources allowed by regulations of general application are or would be greater than 1000 tons for any consecutive three months or 1000 pounds per hour."

D06.00 – CORE PERMIT REQUIREMENTS (CONTINUED) (THIS IS A REQUIRED FORM FOR ALL PERMIT APPLICATIONS)

INSTALLATION NAME	FIPS	PLANT NO.	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

The installation shall comply with each of the following emission limitations. Consult the appropriate sections in the code of federal regulations and code of state regulations for the full text of the applicable requirements.

Title VI – 40 CFR Part 82, *Protection of Stratospheric Ozone*

(a) The permittee shall comply with the standards for labeling of products using ozone-depleting substances pursuant to 40 CFR Part 82, Subpart E:
   (1) All containers in which a class I or class II substance is stored or transported, all products containing a class I substance, and all products directly manufactured with a class I substance must bear the required warning statement if it is being introduced into interstate commerce pursuant to §82.106.
   (2) The placement of the required warning statement must comply with the requirements pursuant to §82.108.
   (3) The form of the label bearing the required warning statement must comply with the requirements pursuant to §82.110.
   (4) No person may modify, remove, or interfere with the required warning statement except as described in §82.112.

(b) The permittee shall comply with the standards for recycling and emissions reduction pursuant to 40 CFR part 82, Subpart F, except as provided for motor vehicle air conditioners, or MVACs, in Subpart B:
   (1) Persons opening appliances for maintenance, service, repair, or disposal must comply with the required practices pursuant to §82.156.
   (2) Equipment used during the maintenance, service, repair, or disposal of appliances must comply with the standards for recycling and recovery equipment pursuant to §82.158.
   (3) Persons performing maintenance, service, repair, or disposal of appliances must be certified by an approved technician certification program pursuant to §82.161.
   (4) Persons disposing of small appliances, MVACs, and MVAC-like appliances must comply with record keeping requirements pursuant to §82.166. ("MVAC-like" appliance as defined at §82.152).
   (5) Persons owning commercial or industrial process refrigeration equipment must comply with the leak repair requirements pursuant to §82.156.
   (6) Owners/operators of appliances normally containing 50 or more pounds of refrigerant must keep records of refrigerant purchased and added to such appliances pursuant to §82.166.

(c) If the permittee manufactures, transforms, imports, or exports a class I or class II substance, the permittee is subject to all the requirements as specified in 40 CFR part 82, Subpart A, Production and Consumption Controls.

(d) If the permittee performs a service on motor (fleet) vehicles when this service involves ozone-depleting substance refrigerant (or regulated substitute substance) in the motor vehicle air conditioner, the permittee is subject to all the applicable requirements as specified in 40 CFR part 82, Subpart B, Servicing of Motor Vehicle Air conditioners. The term "motor vehicle" as used in Subpart B does not include a vehicle in which final assembly of the vehicle has not been completed. The term "MVAC" as used in Subpart B does not include the air-tight sealed refrigeration system used as refrigerated cargo, or system used on passenger buses using HCFC-22 refrigerant.

The permittee shall be allowed to switch from any ozone-depleting substance to any alternative that is listed in the Significant New Alternatives Program (SNAP) promulgated pursuant to 40 CFR part 82, Subpart G, Significant New Alternatives Policy Program.
Federal Only - 40 CFR part 82

FORM OP-E01 – COMPLIANCE PLAN/STATUS – SECTION E

E01.00 – COMPLIANCE PLAN/STATUS

INSTALLATION NAME	FIPS	PLANT NO.	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

Completion of this form of the operating permit forms package is mandatory for all sources. Complete this form once for each application.

1. COMPLIANCE STATUS WITH ALL APPLICABLE REQUIREMENTS EFFECTIVE AT THE TIME OF THE ISSUANCE OF THIS PERMIT.

WILL YOUR INSTALLATION BE IN COMPLIANCE WITH ALL APPLICABLE REQUIREMENTS AT THE TIME OF THE PERMIT ISSUANCE AND CONTINUE TO COMPLY WITH THESE REQUIREMENTS FOR THE DURATION OF THE PERMIT?

YES ☑ NO ☐ (IF NO, COMPLETE A COMPLIANCE PLAN AS DESCRIBED IN THE INSTRUCTIONS ON FORM OP–F01.00)

2. COMPLIANCE STATUS WITH ALL APPLICABLE REQUIREMENTS EFFECTIVE DURING THE PERMIT TERM.

WILL YOUR INSTALLATION BE IN COMPLIANCE WITH ALL APPLICABLE REQUIREMENTS TAKING EFFECT DURING THE TERM OF THE PERMIT?

YES ☑ NO ☐ (IF NO, COMPLETE A COMPLIANCE PLAN AS DESCRIBED IN THE INSTRUCTIONS ON FORM OP–F01.00)

3. COMPLIANCE STATUS WITH ENHANCED MONITORING AND COMPLIANCE CERTIFICATION.

IS THE INSTALLATION IDENTIFIED IN THIS APPLICATION IN COMPLIANCE WITH ALL APPLICABLE ENHANCED MONITORING AND COMPLIANCE CERTIFICATION REQUIREMENTS?

YES ☑ NO ☐ (IF NO, COMPLETE A COMPLIANCE PLAN AS DESCRIBED IN THE INSTRUCTIONS ON FORM OP–F01.00)

4. SCHEDULE OF SUBMISSION OF COMPLIANCE CERTIFICATION DURING THE PERMIT TERM.

FREQUENCY OF SUBMITTALS	BEGINNING DATE
Annual	04/01/2023

5. CERTIFICATION STATEMENT FOR PART 70 MINOR PERMIT MODIFICATIONS.

I hereby certify that this request for a permit modification meets the criteria described in 10 CSR 10-6.065(5)(e)5.b.(i) for minor permit modifications, and request that the minor permit modification procedures be followed.

SIGNATURE OF RESPONSIBLE OFFICIAL OF COMPANY	DATE
	5-22-22

6. CERTIFICATION OF COMPLIANCE WITH ALL APPLICABLE REQUIREMENTS.

Except for requirements identified in the above statement for which compliance is not achieved, I hereby certify that, based on information and belief formed after reasonable inquiry, the air contaminant source identified in this application is in compliance with all applicable requirements.

SIGNATURE OF RESPONSIBLE OFFICIAL OF COMPANY	DATE
	5.12.22

TYPE OR PRINT NAME OF RESPONSIBLE OFFICIAL	OFFICIAL TITLE OF RESPONSIBLE OFFICIAL
Scott Tuttle	General Manager

FORM OP-F01 – GENERAL COMMENTS – SECTION F

F01.00 – GENERAL COMMENTS

INSTALLATION NAME	FIPS	PLANT NO.	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

1. GENERAL INFORMATION

This application is for the Part 70 permit renewal and to incorporate construction permit 012020-012 into the Part 70 permit.

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
FS008		30200530

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

DDGS Flat Storage Fugitives

DESCRIPTION OF EMISSION UNIT

DDGS Flat Storage Fugitives

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
NA	NA

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	22.8 tph

STACK NO.	TEMPERATURE	FLOW RATE
NA	°F	Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
NA		%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
FS006		30200530

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Wetcake Production

DESCRIPTION OF EMISSION UNIT

Wetcake Production

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
NA	NA

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	60 tph

STACK NO.	TEMPERATURE	FLOW RATE
NA	°F	Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
NA		%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
FS005		30200530

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Tank Farm Equipment Fugitives

DESCRIPTION OF EMISSION UNIT

Tank Farm Equipment Fugitives

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
NA	NA

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	NA

STACK NO.	TEMPERATURE	FLOW RATE
NA	°F	Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
NA		%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
FS003		30200530

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Cooling Tower

DESCRIPTION OF EMISSION UNIT

Cooling Tower

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
NA	NA

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	18,500 gpm

STACK NO.	TEMPERATURE	FLOW RATE
NA	°F	Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
NA		%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
FS002		30200530

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Fugitive VOC Equipment Leaks

DESCRIPTION OF EMISSION UNIT

Fugitive VOC Equipment Leaks

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
NA	NA

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	NA

STACK NO.	TEMPERATURE	FLOW RATE
NA	°F	Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
NA		%	%
ADDITIONAL CONTROL DEVICE TYPE	**POLLUTANT(S) CONTROLLED**	**CONTROL EFFICIENCY**	**CAPTURE EFFICIENCY**
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
FS001		30200530

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Grain Receiving and DDGS Handling Fugitives

DESCRIPTION OF EMISSION UNIT

Grain Receiving and DDGS Handling Fugitives

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
NA	NA

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	135 ton DDGS/hour, 6300 ton grain/day

STACK NO.	TEMPERATURE	FLOW RATE
NA	°F	Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
NA		%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-38		30200530

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Grain Receiving #2

DESCRIPTION OF EMISSION UNIT

Grain Receiving Pit #2 and Corn Conveyor

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	560 tons grain/hour

STACK NO.	TEMPERATURE	FLOW RATE
EP-38	Ambient °F	35,000 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Baghouse	PM, PM10, PM2.5	99 %	80 %
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM/PM10/PM2.5	042017-011	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-37		30200817

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Hammermill #4

DESCRIPTION OF EMISSION UNIT

Hammermill #4

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	22.5 tons grain/hour

STACK NO.	TEMPERATURE	FLOW RATE
SV035	Ambient °F	12,000 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Baghouse	PM, PM10, PM2.5	99 %	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM/PM10/PM2.5	032003-008C and 102007-014	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-36		30200817

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Hammermill #3

DESCRIPTION OF EMISSION UNIT

Hammermill #3

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	22.5 tons grain/hour

STACK NO.	TEMPERATURE	FLOW RATE
SV035	Ambient °F	12,000 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Baghouse	PM, PM10, PM2.5	99 %	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM/PM10/PM2.5	032003-008C and 102007-014	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-35		30200817

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Hammermill #2

DESCRIPTION OF EMISSION UNIT

Hammermill #2

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	22.5 tons grain/hour

STACK NO.	TEMPERATURE	FLOW RATE
SV035	Ambient °F	12,000 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Baghouse	PM, PM10, PM2.5	99 %	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM/PM10/PM2.5	032003-008C and 102007-014	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-34		30200817

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Hammermill #1

DESCRIPTION OF EMISSION UNIT

Hammermill #1

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	22.5 tons grain/hour

STACK NO.	TEMPERATURE	FLOW RATE
SV034	Ambient °F	12,000 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Baghouse	PM, PM10, PM2.5	99 %	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM/PM10/PM2.5	032003-008C and 102007-014	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-32		30205030

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Ethanol Storage Tank #3

DESCRIPTION OF EMISSION UNIT

Ethanol Storage Tank #3

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2000	1,000,000 gallons

STACK NO.	TEMPERATURE	FLOW RATE
EP-32	Ambient °F	Ambient Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Internal Floating Roof	VOC and HAPs	%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
VOC, HAPs	NSPS Kb	
VOC, HAPs	042017-011	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-31		40899999

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Flare for Truck Loadout

DESCRIPTION OF EMISSION UNIT

Flare for Truck Loadout

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
John Zink	

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2002	0.0567 MMBtu, 60,000 gph

STACK NO.	TEMPERATURE	FLOW RATE
EP-31	1800 °F	380 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Flare (C019)	HAPs, VOC, CO, NOx	98 %	99.2 %
ADDITIONAL CONTROL DEVICE TYPE	**POLLUTANT(S) CONTROLLED**	**CONTROL EFFICIENCY**	**CAPTURE EFFICIENCY**
Flare (C019)	SOx	98 %	99.2 %

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
VOC, HAPs	042017-011	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-28		30200540

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

DDGS Storage Silo #1

DESCRIPTION OF EMISSION UNIT

DDGS Storage Silo #1

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
Landig/SMI	

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2002	22.83 tph DDGS

STACK NO.	TEMPERATURE	FLOW RATE
SV018	°F	3,500 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Baghouse (C010)	PM/PM10/PM2.5	99 %	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM10	042017-011 and 032003-008C	0.151

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-27		30200930

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

DDGS Fluid Bed Cooler

DESCRIPTION OF EMISSION UNIT

DDGS Fluid Bed Cooler

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
Barr-Rosin	

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2002	22.83 tph DDGS

STACK NO.	TEMPERATURE	FLOW RATE
SV027	120 °F	26,500 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Baghouse (C009)	PM/PM10/PM2.5	99 %	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM10	042017-011 and 032003-008C	0.909

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-22 through EP-25		30201002

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Centrifuges #1 through #4 Bypass Stack

DESCRIPTION OF EMISSION UNIT

Centrifuges #1 through #4 Bypass Stack

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	200 gpm (each centrifuge)

STACK NO.	TEMPERATURE	FLOW RATE
SV005	180 °F	275 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%
ADDITIONAL CONTROL DEVICE TYPE	**POLLUTANT(S) CONTROLLED**	**CONTROL EFFICIENCY**	**CAPTURE EFFICIENCY**
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
	Routes to Distillation Scrubber (C005)	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-18		40899999

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Ethanol Loadout to Rail - Unflared

DESCRIPTION OF EMISSION UNIT

Ethanol Loadout to Rail - Unflared

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2002	60,000 gph

STACK NO.	TEMPERATURE	FLOW RATE
	°F	Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
None		%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-17A and EP-17B		30200530

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Unpaved and Paved Road Fugitives

DESCRIPTION OF EMISSION UNIT

Unpaved and Paved Road Fugitives

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
NA	NA

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	NA

STACK NO.	TEMPERATURE	FLOW RATE
NA	°F	Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
NA		%	%

ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-15		30205030

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Ethanol Storage Tank #2

DESCRIPTION OF EMISSION UNIT

Ethanol Storage Tank #2

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2000	180,000 gallons

STACK NO.	TEMPERATURE	FLOW RATE
EP-15	Ambient °F	Ambient Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Internal Floating Roof	VOC and HAPs	%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
VOC, HAPs	NSPS Kb	
VOC, HAPs	042017-011	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-14		30205030

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Ethanol Storage Tank #1

DESCRIPTION OF EMISSION UNIT

Ethanol Storage Tank #1

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2000	180,000 gallons

STACK NO.	TEMPERATURE	FLOW RATE
EP-14	Ambient °F	Ambient Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Internal Floating Roof	VOC and HAPs	%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
VOC, HAPs	NSPS Kb	
VOC, HAPs	042017-011	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-13		30205030

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Denaturant Storage Tank #2

DESCRIPTION OF EMISSION UNIT

Denaturant Storage Tank #2

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2000	18,000 gallons

STACK NO.	TEMPERATURE	FLOW RATE
EP-13	Ambient °F	Ambient Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Internal Floating Roof	VOC and HAPs	%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
VOC, HAPs	NSPS Kb	
VOC, HAPs	042017-011	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-12		30205030

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Denaturant Storage Tank #1

DESCRIPTION OF EMISSION UNIT

Denaturant Storage Tank #1

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2000	49,000 gallons

STACK NO.	TEMPERATURE	FLOW RATE
EP-12	Ambient °F	Ambient Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Internal Floating Roof	VOC and HAPs	%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
VOC, HAPs	NSPS Kb	
VOC, HAPs	042017-011	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-11		30205030

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

190 Proof Ethanol Storage Tank

DESCRIPTION OF EMISSION UNIT

190 Proof Ethanol Storage Tank

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2000	67,000 gallons

STACK NO.	TEMPERATURE	FLOW RATE
Ep-11	Ambient °F	Ambient Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Internal Floating Roof	VOC and HAPs	%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
VOC, HAPs	NSPS Kb	
VOC, HAPs	042017-011	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-09		30201002

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Grain Dryer

DESCRIPTION OF EMISSION UNIT

Grain Dryer

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
	31 MMBTU, 7.61 tph grain

STACK NO.	TEMPERATURE	FLOW RATE
SV009	°F	Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
None		%	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM	10 CSR 10-6.220 & 10-6.400	15.97 lb/hr

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-07		30200930

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

DDGS Silo Bypass

DESCRIPTION OF EMISSION UNIT

DDGS Silo Bypass

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2002	22.83 tph DDGS

STACK NO.	TEMPERATURE	FLOW RATE
SV007	100 °F	3,500 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Baghouse (C006)	PM/PM10/PM2.5	99 %	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM10	042017-011 and 032003-008C	0.151

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-06		30200930

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

RTO Stack

DESCRIPTION OF EMISSION UNIT

DDGS Dryers #1 and #2

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
ICM, Barr-Rosin, and CECO RTO	

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
1999, 2002, and 2009	68.5 MMBtu/hr, 22.83 DDGS tph

STACK NO.	TEMPERATURE	FLOW RATE
SV006	359 °F	52,800 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
RTO	PM/PM10/PM2.5	90 %	100 %
ADDITIONAL CONTROL DEVICE TYPE	**POLLUTANT(S) CONTROLLED**	**CONTROL EFFICIENCY**	**CAPTURE EFFICIENCY**
RTO	VOC, HAPs, CO	98, 95, 90 %	100 %

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM	10 CSR 10-6.400	33.34
PM10	042017-011A	4.83
PM2.5	042017-011A	4.33

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-05		30201014

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Distillation Bypass Stack (C004)

DESCRIPTION OF EMISSION UNIT

Distillation Bypass Stack (C004)

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
Custom	

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
1999	38,000 gallons of beer

STACK NO.	TEMPERATURE	FLOW RATE
SV005	90 °F	5,800 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Scrubber	VOC	95 %	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Scrubber	HAPs	50 %	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
VOC	042017-011A	
Acetaldehyde	042017-011A	
Total HAPs	042017-011A	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-04		30201014

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Fermentation Scrubber(s) Bypass Stack (C003 or C025)

DESCRIPTION OF EMISSION UNIT

Fermentation Scrubber(s) Bypass Stack (C003 or C025)

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
Custom	

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
1999 and 2021	38,000 gallons of beer

STACK NO.	TEMPERATURE	FLOW RATE
SV004	90 °F	5,800 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Scrubber(s)	VOC	95 %	%
ADDITIONAL CONTROL DEVICE TYPE	**POLLUTANT(S) CONTROLLED**	**CONTROL EFFICIENCY**	**CAPTURE EFFICIENCY**
Scrubber(s)	HAPs	50 %	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
VOC	042017-011A	
Acetaldehyde	042017-011A	
Total HAPs	042017-011A	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-02		

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

DDGS Loading

DESCRIPTION OF EMISSION UNIT

DDGS Loading

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
1999	135 tph DDGS

STACK NO.	TEMPERATURE	FLOW RATE
SV002	Ambient °F	8,000 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Baghouse (C002)	PM/PM10/PM2.5	99 %	%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM/PM10/PM2.5	012020-012	

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.00 – GENERAL EMISSION UNITS

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon, LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-01		30200530

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Grain Receiving #1

DESCRIPTION OF EMISSION UNIT

Grain Receiving, Transfer, Storage, Conveyors, Grain Scalper and Surge Bin

MANUFACTURER	MODEL NUMBER/SERIAL NUMBER
Roskamp-Champion	

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2012	490 tons grain/hour

STACK NO.	TEMPERATURE	FLOW RATE
EP-01	Ambient °F	25,000 Ft^3/min

2. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
Baghouse	PM, PM10, PM2.5	99 %	80 %
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY	CAPTURE EFFICIENCY
		%	%

3. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (CSR#, CFR#, PERMIT NO., ETC.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM/PM10/PM2.5	042017-011 and 032003-008C	2.11 lbs/hr

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.10 – BOILERS, FURNACES AND OTHER INDIRECT HEATING SOURCES

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-30		10200602

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Boiler #2

DESCRIPTION OF EMISSION UNIT

Boiler #2

MANUFACTURER	MODEL NO./SERIAL NO.
Johnston	PFTS1800

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
2002	72.6 MM BTU/hr

2. FUEL DATA

PRIMARY FUEL TYPE AND GRADE	MAXIMUM SULFUR CONTENT
Natural Gas	%
SECONDARY FUEL TYPE AND GRADE	MAXIMUM SULFUR CONTENT
	%

3. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY
		%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY
		%

4. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (e.g, CSR#, CFR#, PERMIT NO.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM10	042017-011A	0.079 lb/hr

DUPLICATE THIS FORM AS NEEDED

FORM OP-D03 – EMISSION UNIT INFORMATION – SECTION D

D03.10 – BOILERS, FURNACES AND OTHER INDIRECT HEATING SOURCES

INSTALLATION NAME	FIPS	PLANT NUMBER	YEAR SUBMITTED
POET Biorefining - Macon LLC	121	0028	2022

EMISSION UNIT ID	EIQ REFERENCE NUMBER	SOURCE CLASSIFICATION CODE
EP-8		10200602

1. EMISSION UNIT DESCRIPTION

INSTALLATION'S NAME FOR THIS EMISSION UNIT

Boiler #1

DESCRIPTION OF EMISSION UNIT

Boiler #1

MANUFACTURER	MODEL NO./SERIAL NO.
Johnston	PFTS1500

CONSTRUCTION DATE	MAXIMUM HOURLY DESIGN RATE
1999	60.5 MM BTU/hr

2. FUEL DATA

PRIMARY FUEL TYPE AND GRADE	MAXIMUM SULFUR CONTENT
Natural Gas	%
SECONDARY FUEL TYPE AND GRADE	MAXIMUM SULFUR CONTENT
	%

3. ASSOCIATED AIR POLLUTION CONTROL EQUIPMENT

CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY
		%
ADDITIONAL CONTROL DEVICE TYPE	POLLUTANT(S) CONTROLLED	CONTROL EFFICIENCY
		%

4. APPLICABLE REQUIREMENTS

POLLUTANT	APPLICABLE REQUIREMENT AUTHORITY (e.g, CSR#, CFR#, PERMIT NO.)	EMISSION LIMIT OR STANDARD (INCLUDING UNITS)
PM10	042017-011A	0.06 lb/hr

DUPLICATE THIS FORM AS NEEDED

POET Biorefining - Macon
Macon, Missouri
Potential Emission Summary - Proposed

Controlled Emissions		TSP		PM10		PM2.5		NOx		SOx		VOC		CO		Acetaldehyde		Acrolein		Total HAP	
Emission Point/Stack Vent	Description	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	ton/yr	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy
EP-1	Grain Receiving #1	0.64	2.82	0.64	2.82	0.64	2.82														
EP-38	Grain Receiving #2	0.90	3.94	0.90	3.94	0.90	3.94														
EP-34	Hammermill #1	0.31	1.35	0.31	1.35	0.31	1.35														
EP-35	Hammermill #2	0.31	1.35	0.31	1.35	0.31	1.35														
EP-36	Hammermill #3	0.31	1.35	0.31	1.35	0.31	1.35														
EP-37	Hammermill #4	0.31	1.35	0.31	1.35	0.31	1.35														
EP-4	Fermentation Bypass Stack: Fermentation (11 Fermenters @ 270,000 gallons each), 1 Beer Well (394,000 gallons)											21.50	5.38			3.00	0.750	0.0121	0.0030	3.064	0.766
EP-4	Distillation Bypass Stack: Distillation (Evaporator, Side Stripper, Beer Stripper, Rectifier, Surge Tank Side Stripper, Product Rundown Tank, 5 Molecular Sieves)											0.93	0.23			0.14	0.034	0.0118	0.0030	0.151	0.038
EP-22, EP-23, EP-24, EP-25	Centrifuge Bypass Stacks: Emissions from (4) Centrifuges during RTO bypass											2.78	0.70			0.235	0.0587	0.0043	0.0011	0.248	0.062
EP-6	RTO	4.83	21.16	4.83	21.16	4.33	18.97	13.46	58.96	0.091	0.40	3.64	15.94	11.00	48.18	0.1422	0.623	0.0778	0.3407	0.658	2.880
EP-9	Grain Dryer	15.63	3.88	4.08	1.01	0.89	0.22	3.04	0.75	0.018	0.0045	0.17	0.04	2.55	0.63					0.0574	0.0142
EP-27	DDG Fluid Bed Cooler	0.91	3.98	0.91	3.98	0.91	3.98					2.95	12.94			0.165	0.610	0.0114	0.0499	0.205	0.898
EP-28	DDGS Storage Silo #1	0.090	0.39	0.090	0.39	0.090	0.39					0.17	0.73			0.0155	0.0683	0.00721	0.00721	0.024	0.107
EP-7	DDGS Silo Bypass	0.090	0.39	0.090	0.39	0.090	0.39					0.17	0.73			0.0155	0.0683	0.00721	0.00721	0.024	0.107
EP-2	DDGS Loading	0.21	0.90	0.21	0.90	0.21	0.90					0.99	0.731			0.0922	0.0683	0.00974	0.00721	0.145	0.107
EP-10	Ethanol Loadout											36.51	19.28			0.0865	0.0808			0.223	0.165
EP-31	Flare for Truck Loadout							0.22	0.97					0.56	2.44						
EP-8	Boiler #1	0.45	1.97	0.45	1.97	0.45	1.97	2.97	12.99	0.036	0.16	0.33	1.43	4.98	21.82					0.112	0.490
EP-30	Boiler #2	0.54	2.37	0.54	2.37	0.54	2.37	3.56	15.59	0.043	0.19	0.39	1.71	5.98	26.19					0.134	0.588
EP-11	190 Proof Ethanol Storage Tank											0.085	0.37			0.00035	0.0015			0.000	0.002
EP-12	Denaturant Storage Tank #1											0.35	1.53							0.005	0.022
EP-13	Denaturant Storage Tank #2											0.29	1.25							0.004	0.018
EP-14	Ethanol Storage Tank #1											0.028	0.12			0.00011	0.00048			0.000	0.001
EP-15	Ethanol Storage Tank #2											0.028	0.12			0.00011	0.00048			0.000	0.001
EP-32	Ethanol Storage Tank #3											0.05	0.24			0.00022	0.0010			0.000	0.001
FS001	Grain Receiving Fugitives	3.54	3.00	1.16	0.98	0.20	0.17														
FS001	DDGS Loadout Fugitives	0.089	0.088	0.022	0.016	0.022	0.016					0.99	0.73			0.092	0.068	0.0097	0.0072	0.145	0.107
Total (Title V and PSD Applicable) Potential Emissions		29.16	50.28	15.16	45.34	10.50	41.54	23.25	89.26	0.19	0.75	72.33	64.19	25.07	99.25	4.00	2.63	0.151	0.43	5.20	6.37

Controlled Emissions - Fugitive Sources Not included in Title V or PSD applicability Included for HAP major threshold		TSP		PM10		PM2.5		NOx		SOx		VOC		CO		Acetaldehyde		Acrolein		Total HAP	
Fugitive Source	Description	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	ton/yr	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy
EP-17A, EP-17B	Unpaved Roads	0.76	2.37	0.21	0.67	0.02	0.07														
EP-17A, EP-17B	Paved Roads	14.14	2.66	2.83	0.54	0.69	0.13														
FS003	DDGS Flat Storage	0.02	0.088	0.0037	0.0160	0.0037	0.0160					0.17	0.73			0.016	0.068	0.00165	0.00721	0.0245	0.1072
FS002	Equipment Leaks											0.83	3.65			0.0268	0.117	0.00087	0.00381	0.0278	0.1223
FS003	Cooling Towers	0.69	3.04	0.49	2.13	0.29	1.28														
FS005	Tank Farm Equipment Fugitives											0.93	4.09			0.0027	0.0117			0.007	0.031
FS006	Wet Cake Production											3.26	8.46							0.94	2.44
	Total	15.6	8.2	3.5	3.3	1.0	1.5		0.0		0.0	5.2	16.9		0.0	0.045	0.187		0.01102	1.0	2.7
Facility Wide Potential Emissions			58.43		48.69		43.04		89.26		0.75		81.12		99.25		2.83		0.438		9.08

POET Biorefining - Macon
Macon, Missouri
Facility Parameters

Receiving	Current	Proposed	Difference		
Annual Grain Receiving	20,599,251	20,599,251	-	bu/yr	
Annual Grain Receiving[1]	666,800	666,800	-	ton/yr	Permit Limit
Grain Density	56	56	-	lb/bushel	
Gallons Ethanol Produced per Bushel of Corn	2.67	2.67	-	gal/bu	

Production (Based on on Worst Case Emissions)	Current	Proposed	Difference		
Total Gallons Anhydrous Ethanol Produced per Year	55,000,000	55,000,000	-	gallons/yr	Permit Limit
E-85 Operations					
Gallons E-85 Produced:	1,000,000	1,000,000	-	gallons/year	
Denaturant Throughput[3]:	300,000	300,000	-	gallons/year	
Gallons Anhydrous Ethanol Loaded out in E-85 Service[2]	700,000	700,000	-	gallons/year	
Denatured Ethanol Operation[5]:					
Gallons Denatured Ethanol Produced:	57,000,000	57,000,000	-	gallons/year	
Denaturant Throughput[4]:	2,700,000	2,700,000	-	gallons/year	
Gallons Anhydrous Ethanol Loaded out in Denatured Service	54,300,000	54,300,000			
Maximum Denaturant Throughput	3,000,000	3,000,000			Permit Limit
Combined Maximum Loadout (Undenatured Ethanol or Denatured Ethanol and E85)	58,000,000	58,000,000			
Anhydrous Ethanol Storage	4,000,000	4,000,000		gallons/year	

DDGS Production	Current	Proposed	Difference		
Hourly DDGS Production:	22.83	22.83	-	ton/hour	
Annual DDGS Produced:	200,000	200,000	-	ton/year	Permit Limit
Percent Grain Throughput that becomes DDGS	28.0%	28.0%			

Other Production Data	Current	Proposed	Difference		
Pounds of Grain Oil Produced per Bushel of Grain	1	1	-	lb/bushel	
Annual Corn Oil Production	10,300	10,300	-	tons/year	
Liquid Carbon Dioxide	35,000	35,000	-	tons/year	
CO2 Max trucks calc. based on 800 kg/m3 being the minimum density of liquid CO2 shipped by truck	0.0033	0.0033	-	ton/gallon CO2	
Number of Hammermills	4	4	-	tons/year	
Wet Cake Production	80,000	312,500	232,500	tons/year	
Syrup Shipment	20,000	20,000	-	tons/year	Permit Limit
CO2 Shipment	135,000	135,000	-	tons/year	Permit Limit
Chemical Shipment	1,600,000	1,600,000	-	gallons/year	Permit Limit

Truck Data	Current	Proposed	Difference	Average Weight of Truck (tons)	Quantity Transported per Truck		Maximum Number of Trucks Hourly	Maximum Daily Number of Trucks Daily	Paved miles traveled per truck	Unpaved miles traveled per truck
DDGS Haul Out	8000	8000	0 truck/yr	27.5	25	tons	8	60	0.50	0.10
Ethanol Haul Out	7125	7125	0 truck/yr	27.5	8,000	gallons	5	45	0.40	
Denaturant Delivery	375	375	0 truck/yr	27.5	8,000	gallons	5	4	0.40	
Grain Delivery - Route 1	1706	1706	0 truck/yr	27.5	25	tons	30	240	0.80	0.30
Grain Delivery - Route 2	15359	15359	0 truck/yr	27.5	25	tons	30	240	0.80	0.30
Corn Oil Haul Out	412	412	0 truck/yr	27.5	25	tons	2	4	0.50	0.30
Liquid CO2 Haul Out	5400	5400	0 truck/yr	27.5	25	tons	3	22	0.70	0.10
Wetcake Haul Out[6]	3200	12500	9,300 truck/yr	27.5	25	tons	1	80	0.60	
Syrup Haul Out	800	800	0 truck/yr	27.5	25	tons	1	20	0.50	
Chemical delivery	200	200	0 truck/yr	27.5	8,000	gallons	1	2	0.40	0.30

Cooling Tower Data

Drift Loss	0.005%	
Total Dissolved Solids	1500	milligrams/liter

[1] Drift loss as provided by the manufacturer.

[2] From the current operating permit, the total dissolved solids (TDS) concentration in the circulated cooling water shall not exceed an average TDS concentration of 1,500 ppm for any consecutive 12-month period.

[1] Assumes all ethanol loaded out is undenatured for worst case grain receiving.
[2] Assumed maximum production of E-85.
[3] E-85 can be blended anywhere between 70% to 83% undenatured ethanol, depending on atmospheric conditions. Assume denaturant is 30% of E-85 product.
[4] Assume denaturant is 4.7% of denatured alcohol product. Denaturant is typically between 1.8 and 2.2%.
[5] Worst case emissions occur when the maximum E85 product is loaded out.
[6] Spent grain may only be loaded out as DDGS or wetcake.

POET Biorefining - Macon
Macon, Missouri
List of Emission Sources

							Control Equipment		
Emission Point/ Stack Vent	Emission Unit	Control Equipment ID	Description	Current Capacity	Units	Pollutants	Description	Capacity	Units
SV001	EP-1	C001	Grain Receiving #1	490	ton/hr	TSP, PM10, PM2.5	Fabric Filter	25,000	dscfm
			Grain Transfer	490	ton/hr	TSP, PM10, PM2.5			
			Grain Storage	490	ton/hr	TSP, PM10, PM2.5			
			Grain Conveyors (#1 and #2)	490	ton/hr	TSP, PM10, PM2.5			
			Grain Scalper	90	ton/hr	TSP, PM10, PM2.5			
			Surge Bin	90	ton/hr	TSP, PM10, PM2.5			
SV038	EP-38	C022	Grain Receiving #2	560	ton/hr	TSP, PM10, PM2.5	Fabric Filter	35000	dscfm
SV034	EP-34	C013	Hammermill #1	22.5	ton/hr	TSP, PM10, PM2.5	Fabric Filter	12000	dscfm
SV035	EP-35	C014	Hammermill #2	22.5	ton/hr	TSP, PM10, PM2.5	Fabric Filter	12000	dscfm
SV036	EP-36	C015	Hammermill #3	22.5	ton/hr	TSP, PM10, PM2.5	Fabric Filter	12000	dscfm
SV037	EP-37	C016	Hammermill #4	22.5	ton/hr	TSP, PM10, PM2.5	Fabric Filter	12000	dscfm
SV004	EP-4	C003/C025	Fermentation Bypass Stack: Fermentation (11 Fermenters @ 270,000 gallons each), 1 Beer Well (394,000 gallons)			VOC, HAP	Wet Scrubbers, RTO	500	Bypass hrs/yr
SV006	EP-4	C003/C025 & C005	Fermenters #1 and #2 270,000 gallons each	38,000	gal/hr (beerfeed rate)	VOC, HAP	Wet Scrubbers, RTO		
			Fermenters #3 and #4 270,000 gallons each						
			Fermenters #5 and #6 270,000 gallons each						
			Fermenters #7 270,000 gallons						
			Fermenters #8 270,000 gallons						
			Fermenters #9, #10, and #11 270,000 gallons each						
			Beer Well 394,000 gallons						
SV005	EP-5	C004	Distillation Bypass Stack: Distillation (Evaporator, Side Stripper, Beer Stripper, Rectifier, Surge Tank Side Stripper, Product Rundown Tank, 6 Molecular Sieves)			VOC, HAP	Wet Scrubber	500	Bypass hrs/yr
SV006	EP-5	C004 & C005	Beer Stripper	38,000	gal/hr (beerfeed rate)	VOC, HAP	Wet Scrubber, RTO		
			Rectifier Column						
			Side Stripper						
			One (1) Set of Six (6) Molecular Sieves						
			One (1) Set of Evaporators						
SV005	EP-22		Centrifuge Bypass Stacks: Emissions from (4) Centrifuges during RTO bypass	200	gal/min (each centrifuge)	VOC, HAP		500	Bypass hrs/yr
	EP-23								
	EP-24								
	EP-25								
SV006	EP-6	C005	DDGS Dryer #1	22.83	ton/hr	TSP, PM10, PM2.5, NOx, SOx, CO, VOC, HAP	Multiclone, RTO		
				68.5	MMBTU/hr				
SV006	EP-6	C005	DDGS Dryer #2	22.83	ton/hr	TSP, PM10, PM2.5, NOx, SOx, CO, VOC, HAP	Multiclone, RTO		
				68.5	MMBTU/hr				
SV006	EP-6	C005	RTO	18	MMBTU/hr	TSP, PM10, PM2.5, NOx, SOx, CO, VOC, HAP		8760	hrs/yr
SV009	EP-9		Grain Dryer	31	MMBTU/hr	TSP, PM10, PM2.5, NOx, SOx, CO, VOC, HAP			
SV027	EP-27	C009	DDG Fluid Bed Cooler	22.83	ton/hr	TSP, PM10, PM2.5, VOC, HAP	Fabric Filter	26,500	dscfm
SV002	EP-2	C002	DDGS Loading	135.00	ton/hr	TSP, PM10, PM2.5, VOC, HAP	Fabric Filter	8,000	dscfm

Emission Point/ Stack Vent	Emission Unit	Control Equipment ID	Description	Current Capacity	Units	Pollutants	Description	Capacity	Units
SV018	EP-28	C010	DDGS Storage Silo #1	22.83 200,000	ton/hr ton/yr	TSP, PM10, PM2.5, VOC, HAP	Fabric Filter	3,500	dscfm
SV007	EP-7	C006	DDGS Silo Bypass	22.8 200,000	ton/hr ton/yr	TSP, PM10, PM2.5, VOC, HAP	Fabric Filter	3,500	dscfm
EP-31	EP-31		Flare for Truck Loadout	0.0567	MMBtu/Hr	NOx, SOx, VOC, CO	Flare		
EP-18	EP-18		Ethanol Loadout	60,000	gal/hr	VOC, HAP	Flare		
SV008	EP-8		Boiler #1	60.5	MMBTU/hr	TSP, PM10, PM2.5, NOx, SOx, VOC, CO	Low Nox Burner		
SV019	EP-30		Boiler #2	72.6	MMBTU/hr	TSP, PM10, PM2.5, NOx, SOx, VOC, CO	Low Nox Burner		
EP-11	EP-11		190 Proof Ethanol Storage Tank	67,000	gallons	VOC, HAP	Internal Floating Roof		
EP-12	EP-12		Denaturant Storage Tank #1	49,000	gallons	VOC, HAP	Internal Floating Roof		
EP-13	EP-13		Denaturant Storage Tank #2	18,000	gallons	VOC, HAP	Internal Floating Roof		
EP-14	EP-14		Ethanol Storage Tank #1	180,000	gallons	VOC, HAP	Internal Floating Roof		
EP-15	EP-15		Ethanol Storage Tank #2	180,000	gallons	VOC, HAP	Internal Floating Roof		
EP-29	EP-32		Ethanol Storage Tank #3	1,000,000	gallons	VOC, HAP	Internal Floating Roof		

Fugitive Source			Description			Pollutants	Control Equipment
FS001			Grain Receiving Fugitives	6300	ton/day	TSP, PM10, PM2.5	None
FS001			DDGS Loadout Fugitives	135.00	ton/hr	TSP, PM10, PM2.5, VOC, HAP	None
FS008			DDGS Flat Storage	22.8 200,000	ton/hr ton/yr	TSP, PM10, PM2.5, VOC, HAP	None
FS002			Equipment Leaks	NA	NA	VOC, HAP	Leak Detection and Repair Program
FS003			Cooling Towers	18,500	GPM	TSP, PM10, PM2.5	High Efficiency Mist Eliminators
EP-17A, EP-17B			Unpaved Roads	NA	NA	TSP, PM10, PM2.5	None
EP-17A, EP-17B			Paved Roads	NA	NA	TSP, PM10, PM2.5	None
FS005			Tank Farm Equipment Fugitives	NA	NA	VOC, HAP	Leak Detection and Repair Program
FS006			Wet Cake Production	60	ton/hr	TSP, PM10, PM2.5,	None

POET Biorefining - Macon
Macon, Missouri
Current PTE Summary (As Stated in Current Permit and Statement of Basis, Issued February 19, 2019)

Limited Potential Emissions Summary

Controlled Emissions		PM	PM10	PM2.5	NOx	SOx	VOC	CO	Acetaldehyde	Acrolein	Total HAP
Emission Point	Description	tpy	tpy	tpy	tpy	tpy	tpy	tpy	tpy	tpy	tpy
EP-1	Grain Receiving #1 & #2 & DDGS Loadout	2.82	2.82	2.82							
EP-38	Grain Receiving #2	3.94	3.94	3.94							
EP-4	Fermentation Bypass Stack: Fermentation (11 Fermenters @ 270,000 gallons each), 1 Beer Well (394,000 gallons)						5.38		0.75	0.003	0.8
EP-4	Distillation Bypass Stack: Distillation (Evaporator, Side Stripper, Beer Stripper, Rectifier, Surge Tank Side Stripper, Product Rundown Tank, 6 Molecular Sieves)						0.23		0.03	0.003	0.04
EP-22, EP-23, EP-24, EP-25	Centrifuge Bypass Stacks: Emissions from (4) Centrifuges during RTO bypass						0.70		0.06	0.0	0.1
EP-6	RTO	21.16	21.16	18.97	58.96	0.40	15.94	48.18	0.62	0.34	2.88
EP-7	DDGS Silo Bypass	0.39	0.39	0.39			0.73				
EP-2	DDGS Loading	0.90	0.90	0.90			0.73		0.07	0.01	0.11
EP-8	Boiler #1	1.97	1.97	1.97	12.99	0.16	1.43	21.82			0.49
EP-9	Grain Dryer	3.88	1.01	0.22	0.75	0.00	0.04	0.63			0.01
EP-11	190 Proof Ethanol Storage Tank						0.37		0.00		0.0016
EP-12	Denaturant Storage Tank #1						1.53				0.02
EP-13	Denaturant Storage Tank #2						1.25				0.02
EP-14	Ethanol Storage Tank #1						0.12		0.0		0.00
EP-15	Ethanol Storage Tank #2						0.12		0.0		0.00
EP-18	Ethanol Loadout						19.28		0.08		0.16
EP-27	DDG Fluid Bed Cooler	3.98	3.98	3.98			12.94		0.81	0.05	0.90
EP-28	DDGS Storage Silo #1	0.39	0.39	0.39			0.73		0.07	0.01	0.11
EP-32	Ethanol Storage Tank #3						0.24		0.00		0.0
EP-30	Boiler #2	2.37	2.37	2.37	15.59	0.19	1.71	26.19			0.59
EP-31	Flare for Truck Loadout				0.97			2.44			
EP-34	Hammermill #1	1.35	1.35	1.35							
EP-35	Hammermill #2	1.35	1.35	1.35							
EP-36	Hammermill #3	1.35	1.35	1.35							
EP-37	Hammermill #4	1.35	1.35	1.35							
	Total Point Sources	47.21	44.35	41.36	89.26	0.75	63.46	99.26			

Controlled Emissions - Not Included in Title V or PSD applicability		PM	PM10	PM2.5	NOx	SOx	VOC	CO	Acetaldehyde	Acrolein	Total HAP
Fugitive Source	Description	tpy	tpy	tpy	tpy	tpy	tpy	tpy	tpy	tpy	tpy
FS001	Grain Receiving Fugitives	3.00	0.98	0.17							
FS001	DDGS Loadout Fugitives	0.07	0.02	0.02			0.73		0.07	0.01	0.11
FS002	Equipment Leaks						3.65		0.12	0.00	0.122
FS003	Cooling Towers	3.04	2.13	1.28							
EP-17A, EP-17B	Unpaved Roads	2.37	0.67	0.07							
EP-17A, EP-17B	Paved Roads	2.68	0.54	0.13							
FS008	DDGS Flat Storage	0.07	0.02	0.02			0.73		0.07	0.01	0.11
FS005	Tank Farm Equipment Fugitives						4.09		0.01		0.031
FS006	Wet Cake Production						8.46				2.44
Total Fugitive Sources		**11.2**	**4.3**	**1.7**	**0.0**	**0.0**	**17.7**	**0.0**			

Total Potential Emissions		**58.43**	**48.69**	**43.04**	**89.26**	**0.75**	**81.12**	**99.26**	**2.76**	**0.43**	**8.97**

POET Biorefining - Macon
Macon, Missouri
Potential HAP Emission Summary - Proposed

		Facility Wide Total			EP-8			EP-30		
		HAP			Boiler #1			Boiler #2		
HAP Name	CAS #	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy
Acetaldehyde	75-07-0	4.05	18.71	2.63	-	-	-	-	-	-
Acrolein	107-02-8	0.14	7.07	0.438	-	-	-	-	-	-
Arsenic	7440-38-2	0.00	0.00	0.00	-	-	-	-	-	-
Benzene	71-43-2	0.01	0.02	0.12	0.0001	0.0005	0.0005	0.0001	0.0007	0.0007
Beryllium	7440-41-7	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cadmium	7440-43-7	0.00	0.00	0.00	0.0001	0.0003	0.0003	0.0001	0.0003	0.0003
Carbon Disulfide	75-15-0	0.00	0.00	0.00	-	-	-	-	-	-
Chromium	7440-47-3	0.00	0.00	0.00	0.0001	0.0004	0.0004	0.0001	0.0004	0.0004
Cobalt	7440-48-4	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	98-82-8	0.00	0.00	0.00	-	-	-	-	-	-
Ethylbenzene	100-41-4	0.00	0.00	0.00	-	-	-	-	-	-
Formaldehyde	50-00-0	0.09	1.68	0.19	0.0044	0.0195	0.0195	0.0053	0.0234	0.0234
Hexane	110-54-3	0.58	2.44	3.55	0.107	0.468	0.468	0.128	0.561	0.561
Lead	NA	0.00	0.00	0.00	0.0000	0.0001	0.0001	0.0000	0.0001	0.0001
Manganese	7439-96-5	0.00	0.00	0.00	0.0000	0.0001	0.0001	0.0000	0.0001	0.0001
Mercury	7439-97-6	0.00	0.00	0.00	0.0000	0.0001	0.0001	0.0000	0.0001	0.0001
Methanol	67-56-1	1.33	25.95	3.74	-	-	-	-	-	-
Naphthalene	91-20-3	0.00	0.00	0.00	0.0000	0.0002	0.0002	0.0000	0.0002	0.0002
Nickel	7440-02-0	0.00	0.00	0.00	0.0001	0.0005	0.0005	0.0001	0.0007	0.0007
Propionaldehyde	123-38-6	0.00	0.00	0.00	-	-	-	-	-	-
Selenium	7782-49-2	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Toluene	108-88-3	0.00	0.01	0.04	0.0002	0.0009	0.0009	0.0002	0.0011	0.0011
Xylenes	1330-20-7	0.00	0.00	0.00	-	-	-	-	-	-
	Total HAP			**10.93**	**0.111921**	**0.490213**	**0.490213**	**0.134305**	**0.588256**	**0.588256**

POET Biorefining - Macon
Macon, Missouri
Potential HAP Emission Summary

		EP-2			FS001			FS008			EP-4		
		DDGS Loading			DDGS Loadout Fugitives			DDGS Flat Storage			Fermentation Bypass Stack: Fermentation (11 Fermenters @ 270,000 gallons each), 1 Beer Well (394,000 gallons)		
HAP Name	CAS #	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy
Acetaldehyde	75-07-0	0.0922	0.4037	0.0683	0.092	0.404	0.068	0.0156	0.0683	0.0683	3.000	0.750	0.750
Acrolein	107-02-8	0.0097	0.0426	0.0072	0.010	0.043	0.007	0.0016	0.0072	0.0072	0.012	0.0030	0.0030
Arsenic	7440-38-2	-	-	-	-	-	-	-	-	-	-	-	-
Benzene	71-43-2	-	-	-	-	-	-	-	-	-	-	-	-
Beryllium	7440-41-7	-	-	-	-	-	-	-	-	-	-	-	-
Cadmium	7440-43-7	-	-	-	-	-	-	-	-	-	-	-	-
Carbon Disulfide	75-15-0	-	-	-	-	-	-	-	-	-	-	-	-
Chromium	7440-47-3	-	-	-	-	-	-	-	-	-	-	-	-
Cobalt	7440-48-4	-	-	-	-	-	-	-	-	-	-	-	-
Cumene	98-82-8	-	-	-	-	-	-	-	-	-	-	-	-
Ethylbenzene	100-41-4	-	-	-	-	-	-	-	-	-	-	-	-
Formaldehyde	50-00-0	0.0312	0.1365	0.0087	0.012	0.051	0.009	0.0020	0.0087	0.0087	0.0017	0.0004	0.0004
Hexane	110-54-3	-	-	-	-	-	-	-	-	-	-	-	-
Lead	NA	-	-	-	-	-	-	-	-	-	-	-	-
Manganese	7439-96-5	-	-	-	-	-	-	-	-	-	-	-	-
Mercury	7439-97-6	-	-	-	-	-	-	-	-	-	-	-	-
Methanol	67-56-1	0.0117	0.0512	0.0231	0.031	0.136	0.023	0.0053	0.0231	0.0231	0.0502	0.0125	0.0125
Naphthalene	91-20-3	-	-	-	-	-	-	-	-	-	-	-	-
Nickel	7440-02-0	-	-	-	-	-	-	-	-	-	-	-	-
Propionaldehyde	123-38-6	-	-	-	-	-	-	-	-	-	-	-	-
Selenium	7782-49-2	-	-	-	-	-	-	-	-	-	-	-	-
Toluene	108-88-3	-	-	-	-	-	-	-	-	-	-	-	-
Xylenes	1330-20-7	-	-	-	-	-	-	-	-	-	-	-	-
	Total HAP	**0.144736**	**0.633942**	**0.107212**	**0.144736**	**0.633942**	**0.107212**	**0.024478**	**0.107212**	**0.107212**	**3.063921**	**0.765980**	**0.765980**

POET Biorefining - Macon
Macon, Missouri
Potential HAP Emission Summary

		EP-6			EP-6 - Combustion HAP			EP-27			EP-28			EP-7			EP-4	
		RTO			RTO			DDG Fluid Bed Cooler			DDGS Storage Silo #1			DDGS Silo Bypass			Distillation Bypass Stack (Evaporator, Side Stripper, I Rectifier, Surge Tank Sid Product Rundown Tank, (Sieves)	
HAP Name	CAS #	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy
Acetaldehyde	75-07-0	0.142	12.46	0.623	-	-	-	0.184952	0.810088	0.8101	0.015587	0.0683	0.068269	0.015587	0.0683	0.068269	0.1357	0.0678
Acrolein	107-02-8	0.078	6.81	0.341	-	-	-	0.0114	0.0499	0.0499	0.0016	0.0072	0.0072	0.0016	0.0072	0.0072	0.0118	0.0059
Arsenic	7440-38-2	-	-	-	0.0000	0.0001	0.0001	-	-	-	-	-	-	-	-	-	-	-
Benzene	71-43-2	-	-	-	0.0000	0.0014	0.0002	-	-	-	-	-	-	-	-	-	-	-
Beryllium	7440-41-7	-	-	-	0.0000	0.0000	0.0000	-	-	-	-	-	-	-	-	-	-	-
Cadmium	7440-43-7	-	-	-	0.0002	0.0007	0.0007	-	-	-	-	-	-	-	-	-	-	-
Carbon Disulfide	75-15-0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Chromium	7440-47-3	-	-	-	0.0002	0.0009	0.0009	-	-	-	-	-	-	-	-	-	-	-
Cobalt	7440-48-4	-	-	-	0.0000	0.0001	0.0001	-	-	-	-	-	-	-	-	-	-	-
Cumene	98-82-8	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ethylbenzene	100-41-4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Formaldehyde	50-00-0	0.016	1.36	0.068	-	-	-	0.0048	0.0200	0.0200	0.0053	0.0231	0.0231	0.0020	0.0087	0.0087	0.000658	0.0003
Hexane	110-54-3	-	-	-	0.1563	1.198	0.884	-	-	-	-	-	-	-	-	-	-	-
Lead	NA	-	-	-	0.0001	0.0003	0.0003	-	-	-	-	-	-	-	-	-	-	-
Manganese	7439-96-5	-	-	-	0.0001	0.0003	0.0003	-	-	-	-	-	-	-	-	-	-	-
Mercury	7439-97-6	-	-	-	0.0000	0.0002	0.0002	-	-	-	-	-	-	-	-	-	-	-
Methanol	67-56-1	0.264	23.17	1.158	-	-	-	0.0042	0.0183	0.0183	0.0020	0.0087	0.0087	0.0053	0.0231	0.0231	0.00315149	0.0016
Naphthalene	91-20-3	-	-	-	0.0001	0.0004	0.0002	-	-	-	-	-	-	-	-	-	-	-
Nickel	7440-02-0	-	-	-	0.0003	0.0014	0.0014	-	-	-	-	-	-	-	-	-	-	-
Propionaldehyde	123-38-6	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Selenium	7782-49-2	-	-	-	0.0000	0.0000	0.0000	-	-	-	-	-	-	-	-	-	-	-
Toluene	108-88-3	-	-	-	0.0003	0.0023	0.0013	-	-	-	-	-	-	-	-	-	-	-
Xylenes	1330-20-7	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Total HAP	**0.500000**	**43.800000**	**2.190000**	**0.157588**	**1.206160**	**0.690236**	**0.205098**	**0.898329**	**0.898329**	**0.024478**	**0.107212**	**0.107212**	**0.024478**	**0.107212**	**0.107212**	**0.151305**	**0.075653**

POET Biorefining - Macon
Macon, Missouri
Potential HAP Emission Summary

		Distillation Beer Stripper, Rectifier Stripper, Molecular	EP-11		
			190 Proof Ethanol Storage Tank		
HAP Name	CAS #	Lim tpy	lb/hr	Unc tpy	Lim tpy
Acetaldehyde	75-07-0	0.0339	0.0004	0.0015	0.0015
Acrolein	107-02-8	0.0030	-	-	-
Arsenic	7440-38-2	-	-	-	-
Benzene	71-43-2	-	-	-	-
Beryllium	7440-41-7	-	-	-	-
Cadmium	7440-43-7	-	-	-	-
Carbon Disulfide	75-15-0	-	-	-	-
Chromium	7440-47-3	-	-	-	-
Cobalt	7440-48-4	-	-	-	-
Cumene	98-82-8	-	-	-	-
Ethylbenzene	100-41-4	-	-	-	-
Formaldehyde	50-00-0	0.0002	-	-	-
Hexane	110-54-3	-	-	-	-
Lead	NA	-	-	-	-
Manganese	7439-96-5	-	-	-	-
Mercury	7439-97-6	-	-	-	-
Methanol	67-56-1	0.0008	0.0000	0.0001	0.0001
Naphthalene	91-20-3	-	-	-	-
Nickel	7440-02-0	-	-	-	-
Propionaldehyde	123-38-6	-	-	-	-
Selenium	7782-49-2	-	-	-	-
Toluene	108-88-3	-	-	-	-
Xylenes	1330-20-7	-	-	-	-
	Total HAP	**0.037826**	**0.000377**	**0.001649**	**0.001649**

POET Biorefining - Macon
Macon, Missouri
Potential HAP Emission Summary

		EP-12			EP-13			EP-14			EP-15			EP-32	
		Denaturant Storage Tank #1			Denaturant Storage Tank #2			Ethanol Storage Tank #1			Ethanol Storage Tank #2			Ethanol Storage Ta	
HAP Name	CAS #	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy
Acetaldehyde	75-07-0	-	-	-	-	-	-	0.000109	0.000477	0.000477	0.0001	0.0005	0.0005	0.0002	0.0010
Acrolein	107-02-8	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Arsenic	7440-38-2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzene	71-43-2	0.0003	0.0013	0.0013	0.000246	0.001079	0.001079	-	-	-	-	-	-	-	-
Beryllum	7440-41-7	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Cadmium	7440-43-7	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Carbon Disulfide	75-15-0	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Chromium	7440-47-3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Cobalt	7440-48-4	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Cumene	98-82-8	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ethylbenzene	100-41-4	0.0000	0.0000	0.0000	0.000001	0.000006	0.000006	-	-	-	-	-	-	-	-
Formaldehyde	50-00-0	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Hexane	110-54-3	0.0046	0.0203	0.0203	0.003783	0.016570	0.016570	-	-	-	-	-	-	-	-
Lead	NA	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Manganese	7439-96-5	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Mercury	7439-97-6	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Methanol	67-56-1	-	-	-	-	-	-	0.000008	0.000036	0.000036	0.0000	0.0000	0.0000	0.0000	0.0001
Naphthalene	91-20-3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Nickel	7440-02-0	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Propionaldehyde	123-38-6	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Selenium	7782-49-2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Toluene	108-88-3	0.0001	0.0005	0.0005	0.000085	0.000371	0.000371	-	-	-	-	-	-	-	-
Xylenes	1330-20-7	0.0000	0.0000	0.0000	0.000009	0.000040	0.000040	-	-	-	-	-	-	-	-
	Total HAP	**0.0050**	**0.0221**	**0.0221**	**0.00412**	**0.01807**	**0.01807**	**0.00012**	**0.00051**	**0.00051**	**0.00012**	**0.00051**	**0.00051**	**0.00024**	**0.00106**

POET Biorefining - Macon
Macon, Missouri
Potential HAP Emission Summary

			EP-22, EP-23, EP-24, EP-25			FS002			FS005			EP-18			EP-9		
		k #3	Centrifuge Bypass Stacks: Emissions from (4) Centrifuges during RTO bypass			Equipment Leaks			Tank Farm Equipment Fugitives			Ethanol Loadout			Grain Dryer		
HAP Name	CAS #	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy	lb/hr	Unc tpy	Lim tpy
Acetaldehyde	75-07-0	0.0010	0.235	1.028	0.0587	0.0268	2.1215	0.1172	0.0027	0.372	0.0117	0.086	0.082	0.0808	-	-	-
Acrolein	107-02-8	-	0.0043	0.0187	0.001	0.0009	0.0690	0.0038	-	-	-	-	-	-	-	-	-
Arsenic	7440-38-2	-	-	-	-	-	-	-	-	-	-	-	-	-	6.08E-06	1.51E-06	1.51E-06
Benzene	71-43-2	-	-	-	-	-	-	-	0.0002	0.0081	0.0011	0.008	0.005	0.115	6.38E-05	1.58E-05	1.58E-05
Beryllium	7440-41-7	-	-	-	-	-	-	-	-	-	-	-	-	-	3.85E-07	9.06E-08	9.06E-08
Cadmium	7440-43-7	-	-	-	-	-	-	-	-	-	-	-	-	-	3.34E-05	8.30E-06	8.30E-06
Carbon Disulfide	75-15-0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Chromium	7440-47-3	-	-	-	-	-	-	-	-	-	-	-	-	-	4.25E-05	1.06E-05	1.06E-05
Cobalt	7440-48-4	-	-	-	-	-	-	-	-	-	-	-	-	-	2.55E-06	6.34E-07	6.34E-07
Cumene	98-82-8	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ethylbenzene	100-41-4	-	-	-	-	-	-	-	0.0000	0.0000	0.0000	0.0001	0.0001	0.001	-	-	-
Formaldehyde	50-00-0	-	0.0029	0.0127	0.001	0.0002	0.0172	0.0010	-	-	-	-	-	-	2.28E-03	5.66E-04	5.66E-04
Hexane	110-54-3	-	-	-	-	-	-	-	0.0038	0.0937	0.0155	0.119	0.071	1.772	5.47E-02	1.36E-02	1.36E-02
Lead	NA	-	-	-	-	-	-	-	-	-	-	-	-	-	1.52E-05	3.77E-06	3.77E-06
Manganese	7439-96-5	-	-	-	-	-	-	-	-	-	-	-	-	-	1.15E-05	2.87E-06	2.87E-06
Mercury	7439-97-6	-	-	-	-	-	-	-	-	-	-	-	-	-	7.90E-06	1.96E-06	1.96E-06
Methanol	67-56-1	0.0001	0.0058	0.0254	0.001	0.0001	0.0052	0.0003	0.0002	0.0279	0.0009	0.006	0.008	0.005	-	-	-
Naphthalene	91-20-3	-	-	-	-	-	-	-	-	-	-	-	-	-	1.85E-05	4.60E-06	4.60E-06
Nickel	7440-02-0	-	-	-	-	-	-	-	-	-	-	-	-	-	6.38E-05	1.58E-05	1.58E-05
Propionaldehyde	123-38-6	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Selenium	7782-49-2	-	-	-	-	-	-	-	-	-	-	-	-	-	7.60E-07	1.89E-07	1.89E-07
Toluene	108-88-3	-	-	-	-	-	-	-	0.0001	0.002	0.0004	0.003	0.002	0.040	1.03E-04	2.57E-05	2.57E-05
Xylenes	1330-20-7	-	-	-	-	-	-	-	0.0000	0.0002	0.0000	0.0003	0.0002	0.004	-	-	-
	Total HAP	**0.00106**	**0.25**	**1.08**	**0.062**	**0.0279**	**2.2130**	**0.1223**	**0.0070**	**0.5020**	**0.0306**	**0.2228**	**0.1656**	**2.0184**	**0.05736**	**0.01424**	**0.01424**

POET Biorefining - Macon
Macon, Missouri
Potential HAP Emission Summary

		FS006		
		Wet Cake Production		
HAP Name	CAS #	lb/hr	Unc tpy	Lim tpy
Acetaldehyde	75-07-0	-	-	-
Acrolein	107-02-8	-	-	-
Arsenic	7440-38-2	-	-	-
Benzene	71-43-2	-	-	-
Beryllium	7440-41-7	-	-	-
Cadmium	7440-43-7	-	-	-
Carbon Disulfide	75-15-0	-	-	-
Chromium	7440-47-3	-	-	-
Cobalt	7440-48-4	-	-	-
Cumene	98-82-8	-	-	-
Ethylbenzene	100-41-4	-	-	-
Formaldehyde	50-00-0	-	-	-
Hexane	110-54-3	-	-	-
Lead	NA	-	-	-
Manganese	7439-96-5	-	-	-
Mercury	7439-97-6	-	-	-
Methanol	67-56-1	0.938	2.443	2.443
Naphthalene	91-20-3	-	-	-
Nickel	7440-02-0	-	-	-
Propionaldehyde	123-38-6	-	-	-
Selenium	7782-49-2	-	-	-
Toluene	108-88-3	-	-	-
Xylenes	1330-20-7	-	-	-
	Total HAP	**0.94**	**2.44**	**2.44**

POET Biorefining - Macon
Macon, Missouri
Unpaved Road Traffic - Fugitive Emissions

All trucks are assumed to have an empty weight of 15 tons and a full weight of 40 tons. Actual ethanol and denaturant trucks have a capacity of 8,000 gallons and DDGS trucks have a capacity of 25 tons.

Assume for the limited potential to emit calculations that 100% of the ethanol and DDGS are trucked off-site. Also assume that 100% of the denaturant and grain is recieved by truck. Actual operations will result in some of the ethanol and DGGS being shipped off-site by rail.

Conversion Factors:

1 ton = 2000 lb

Equation from AP-42 Section 13.2.2 Unpaved Roads

$E=k(s/12)^a(W/3)^b$

$E_{ext} = E[(365-P)/365]$

Formula Variables

Description	Variable	Value
Value used for lb/ton PM2.5	k	0.15
Value used for lb/ton PM10	k	1.5
Value used for lb/ton PM	k	4.9
Value used for lb/ton PM2.5	a	0.9
Value used for lb/ton PM10	a	0.9
Value used for lb/ton PM	a	0.7
Value used for lb/ton PM2.5	b	0.45
Value used for lb/ton PM10	b	0.45
Value used for lb/ton PM	b	0.45
Surface material silt content (%)	sL	8.00
Surface material moisture content	M	5.00
Mean vehicle weight (tons)[1]	W	27.50
Number of Days in a year with at least 0.254 mm of precipitation	P	105

Vehicle Type[2]	Average Weight of Truck (tons)[1]	Quantity Transported per Truck		Maximum Number of Trucks (Trucks/yr)	Limited Number of Trucks (Trucks/yr)	Maximum Number of Trucks (Trucks/hr)[2]	Maximum Daily Number of Trucks (Trucks/day)	Unpaved Miles Traveled per Truck	Limited VMT/yr
DDGS Haul Out	27.5	25	tons	8,000	8,000	8	60	0.10	800
Ethanol Haul Out	27.5	8,000	gallons	7,125	7,125	5	45	0.00	0
Denaturant Delivery	27.5	8,000	gallons	375	375	5	4	0.00	0
Grain Delivery - Route 1	27.5	25	tons	1,706	1,706	30	240	0.30	512
Grain Delivery - Route 2	27.5	25	tons	15,359	15,359	30	240	0.30	4,608
Corn Oil Haul Out	27.5	25	tons	412	412	2	4	0.30	124
Liquid CO2 Haul Out	27.5	25	tons	5,400	5,400	3	22	0.10	540
Wetcake Haul Out6	27.5	25	tons	12,500	12,500	1	60	0.00	NA
Syrup Haul Out	27.5	25	gallons	800	800	1	20	0.00	NA
Chemical delivery	27.5	8,000	tons	200	200	1	2	0.30	60
Total	27.5					85	697		6,643

Fugitive dust control plan requires application of dust suppressant.
Control Efficiency of dust suppresant = 90%

Unpaved Haul Road Emissions	$E_{(p/VMT)}$	E_{ext}	Uncontrolled (Lb/hr)	Uncontrolled (TPY)	Precipitation Reduced (TPY)
Emissions PM2.5	0.28	0.20	0.21	0.94	0.67
Emissions PM10	2.82	2.01	2.14	9.37	6.68
Emissions PM	10.00	7.12	7.58	33.21	23.68

Unpaved Haul Road Emissions	Controlled Lb/hour	Dust Suppressant Only Controlled (TPY)	Dust Suppressant and Precipitation Reduction (TPY)
Emissions PM2.5	0.021	0.09	0.067
Emissions PM10	0.21	0.94	0.67
Emissions PM	0.76	3.32	2.37

Notes:

[1] Mean vehicle weight (W) accounts for the vehicles weight (load or no load depending on the vehicle) while traveling only on the portion of the road in which the vehicle can travel fast enough to create fugitive emissions.

[2] Denaturant Delivery and Ethanol Haul Out can not occur simultaneously.

POET Biorefining - Macon
Macon, Missouri
EP-FS-02 Road Traffic Emission Calculations

All trucks are assumed to have an empty weight of 15 tons and a full wieght of 40 tons. Actual ethanol and denaturant trucks have a capacity of 8,000 gallons and DDGS trucks have a capacity of 25 tons.

Assume for the potential to emit calculations that 100% of the ethanol and DDGS are trucked off-site. Also assume that 100% of the denaturant ang grain is recieved by truck. Actual operations will result in some of the ethanol and DDGS being shipped off-site by rail.
Road Path 1 is used for delivery of grain delivery to the new pit = 0.77 miles total
Road Path 2 is used for ethanol, DDGS, wetcake, CO2 and syrup shipment, and chemical delivery = 0.35 miles
All roads are paved and swept as needed to reduce fugitive emissions

Equation from AP-42 Chapter 13.2.1.3

Equation (1):
$E_{ext}=[k(sL)^{0.91}x(W)^{1.02}](1-P/4N)$

Description	Variable	Value
Value used for PM	k	0.011
Value used for PM10	k	0.0022
Value used for PM2.5	k	0.00054
Surface material silt content (g/m²)	sL	0.70
Maximum Instantaneous Control Efficiency	No Control	0%
Road Cleaning Frequency	N/A	N/A
Mean vehicle weight (tons)[1]	W	27.50
Precipitation Frequency	P	105
Number of days in averaging period	N	365
Constant		0.91
Constant		1.02

DDGS Haul Out by Truck	100%
Ethanol Haul Out by Truck	100%
Denaturant Delivery by Truck	100%
Grain Delivery by Truck	100%
Corn Oil Haul Out by Truck	100%
Grain Delivery by Truck	100%

Road Path 1 Emission Calculations

Vehicle Type[3]	Average Weight of Truck (tons)[1]	Quantity Transported per Truck		Maximum Number of Trucks (Trucks/yr)	Limited Number of Trucks (Trucks/yr)	Maximum Number of Trucks (Trucks/hr)[2]	Maximum Daily Number of Trucks (Trucks/day)	Paved Miles Traveled per Truck	Limited VMT/yr	Limited VMT/hr	Limited VMT/day
DDGS Haul Out	27.5	25	tons	8,000	8,000	8	60	0.50	4,000	4	30
Ethanol Haul Out	27.5	8000	gallons	7,125	7,125	5	45	0.40	2,850	2	18
Denaturant Delivery	27.5	8000	gallons	338	338	5	4	0.40	135	2	2
Grain Delivery - Route 1	27.5	25	tons	1,706	1,706	30	240	0.80	1,365	24	192
Grain Delivery - Route 2	27.5	25	tons	15,359	15,359	30	240	0.80	12,287	24	192
Corn Oil Haul Out	27.5	25	tons	412	412	2	4	0.50	206	1	2
Liquid CO2 Haul Out	27.5	25	tons	5,400	5,400	3	22	0.70	3,780	2	15
Wetcake Haul Out[3]	27.5	25	tons	12,500	12,500	1	50	0.50	6,250	1	39
Syrup Haul Out[3]	27.5	25	gallons	800	800	1	20	0.50	400	1	10
Chemical delivery	27.5	8000	tons	200	200	1	2	0.40	80	0	1
Total	**27.50**			51,839	51,839	48	697	5.50	24,703	61	492

Paved Haul Road Emissions	$E_{ext(PM)}$	(TPY)	Maximum lb/hour	Maximum lb/day
Emissions PM	0.234		14.14	114.91
Annual PM Emissions	0.217	2.56		
Emissions PM10	0.047		2.83	11.29
Annual PM10 Emissions	0.043	0.54		
Emissions PM2.5	0.011		0.69	2.77
Annual PM2.5 Emissions (25% winter)	0.011	0.13		

Notes:
[1] Mean vehicle weight (W) accounts for the vehicles weight (load or no load depending on the vehicle)
Assumed all maximum loads are 40 tons total (15 tons truck weight + 25 tons load weight), which is conservative.
[2] Denaturant Delivery and Ethanol Haul Out can not occur simultaneously.
[3] The majority of DDGS go out via rail, so assuming all DDGS are loaded out by truck to calculate fugitive emissions from roads is conservative. The amount of distillers grain produced as wetcake is small and is not included in annual fugitive road emissions due to the conservative assumption that all DDGS goes out by truck. Any wetcake produced would reduce the amount of DDGS loaded out to some degree.
Syrup loadout is not included for the same reasons that wetcake is not included.

POET Biorefining - Macon
Macon, Missouri
EP-01 Grain Receiving Pit

POET Macon proposes to replace the Grains Receiving 1 baghouse (C01) with a new baghouse. POET will retain the C001 designation for the replacement baghouse.

The following calculations are for the emissions of total suspended particulate (TSP), particulate matter less than 10 micron (PM10) and particulate matter less than 2.5 micron (PM2.5) from point sources at the proposed ethanol facility. For the purposes of these calculations, TSP, PM10, and PM2.5 were assumed to be equal.

The controlled potential TSP/PM10/PM2.5 emissions were calculated by multiplying the design grain loading from the fabric filter by the design flow rate for the source and converting to a pound per hour emission rate. The equation for the calculations is:

Ec lb/hr = G gr/dscf x 1 lb / 7000 gr x Q dscfm x 60 minutes / hour

Where:
E = Emission rate of TSP/PM10/PM2.5 in lb/hr
G = Outlet Grain Loading in grain per dry standard cubic foot of air
Q = Flow Rate in dry standard cubic feet per minute

The uncontrolled potential emission were calculated by multiplying the AP-42 emission factor by the annual throughput in tons/year. The equation for the uncontrolled emission rate is:

Eu ton/yr = Ef * Q
Where:
Q = Annual Throughput

Qgrain =		666,800	tons/year
		490	tons/hour
Conversion Factors:			
	1 ton =	2000	lb
	1 lb =	7000	gr
	1 year =	8760	hours
	1 hour =	60	minutes
Assumptions:			
Maximum controlled TSP/PM10 emission rate[1] =		0.003	gr/dscf
Control Efficiency[2] =		99%	
Emission Factor PM Grain Receiving[3] =		0.18	lb/ton
Emission Factor PM10 Grain Receiving[3] =		0.059	lb/ton
Emission Factor PM2.5 Grain Receiving[3] =		0.010	lb/ton

Given:
All point source emissions from these emission units will be controlled by a fabric filter.
Exhaust Flow Rate = 25,000 dscfm

	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Limited Uncontrolled[5]	Controlled[4]	Controlled[4]
	lbs/hr	Unc. TPY	TPY	lbs/hr	TPY
PM	64.29	281.57	60.01	0.64	2.82
PM10	64.29	281.57	19.67	0.64	2.82
PM2.5	64.29	281.57	3.33	0.64	2.82

[1] This is assumed loading. Stack tests have indicated that this is a conservative assumption
[2] Assumed control efficiency
[3] EF taken from AP-42 Table 9.9.1-1 for straight truck receiving. Conservatively assumed all receiving via truck.
[4] Controlled is based on maximum flow rate
[5] Limited is based on annual throughput

POET Biorefining - Macon
Macon, Missouri
EP-38 Grain Receiving #2

The following calculations are for the emissions of total suspended particulate (TSP), particulate matter less than 10 micron (PM10) and particulate matter less than 2.5 micron (PM2.5) from point sources at the proposed ethanol facility. For the purposes of these calculations, TSP, PM10, and PM2.5 were assumed to be equal.

The controlled potential TSP/PM10/PM2.5 emissions were calculated by multiplying the design grain loading from the fabric filter by the design flow rate for the source and converting to a pound per hour emission rate. The equation for the calculations is:

Ec lb/hr = G gr/dscf x 1 lb / 7000 gr x Q dscfm x 60 minutes / hour

Where:
E = Emission rate of TSP/PM10/PM2.5 in lb/hr
G = Outlet Grain Loading in grain per dry standard cubic foot of air
Q = Flow Rate in dry standard cubic feet per minute

The uncontrolled potential emission were calculated by multiplying the AP-42 emission factor by the annual throughput in tons/year. The equation for the uncontrolled emission rate is:

Eu ton/yr = Ef * Q

Where:
Q = Annual Throughput

Qgrain =	666,800	tons/year
Qgrain =	560	tons/hour

Conversion Factors:

1 ton =	2000	lb
1 lb =	7000	gr
1 year =	8760	hours
1 hour =	60	minutes

Assumptions:

Maximum controlled TSP/PM10 emission rate[1] =	0.003	gr/dscf
Control Efficiency[2] =	99%	
Emission Factor PM Grain Receiving[3] =	0.18	lb/ton
Emission Factor PM10 Grain Receiving[3] =	0.059	lb/ton
Emission Factor PM2.5 Grain Receiving[3] =	0.010	lb/ton

Given:
All point source emissions from these emission units will be controlled by a fabric filter.
Exhaust Flow Rate = 35,000 dscfm

	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Limited Uncontrolled[5]	Controlled[4]	Controlled[4]
	lbs/hr	Unc. TPY	TPY	lbs/hr	TPY
PM	90.00	394.20	60.01	0.90	3.94
PM10	90.00	394.20	19.67	0.90	3.94
PM2.5	90.00	394.20	3.33	0.90	3.94

[1] This is assumed loading. Stack tests have indicated that this is a conservative assumption
[2] Assumed control efficiency
[3] EF taken from AP-42 Table 9.9.1-1 for straight truck receiving. Conservatively assumed all receiving via truck. DDGS loadout factors taken from AP-42 Table 9.9.1-2 for animal feed shipping.
[4] Controlled is based on maximum flow rate
[5] Limited is based on annual throughput

POET Biorefining - Macon
Macon, Missouri
FS008 Uncaptured Emissions From DDGS Storage - Fugitive Emissions

Some DDGS is stored in a pile inside the building (flat storage). While DDGS is loaded into flat storage, the building will capture 80% of the particulate matter emitted. Therefore, DDG flat storage will be a source of fugitive particulate emissions. DDGS flat storage is also a source of VOC and HAP emissions. It is assumed that the building and baghouse provide no control of the VOC & HAPs emitted during the loadout operations.

The uncontrolled potential emission for particulates will be calculated by multiplying the appropriate USEPA AP-42 emission factor by the maximum loading rate for the process and assuming that all TSP/PM10/PM2.5 is emitted.

The controlled potential emissions for particulates will be calculated by multiplying the appropriate USEPA AP-42 emission factor by the maximum loading rate for the process, assuming that 80% of the TSP/PM10/PM2.5 is captured by the building.

The equation for calculating the uncontrolled potential fugitive emissions is:

Eu lb/hr = Ef lb/ton x W tons

The equation for calculating the controlled potential fugitive emissions is:

Ec lb/hr = (Ef lb/ton x W tons) x [(100-Ce) / 100]

Where:

Ec = Emission rate in lb/hr
Ef = Emission factor in lb/ton loaded
W = Tons loaded per unit time
Ce = Capture efficiency of shed and aspiration system

Given:

Maximum Hourly Flat Storage Loading Rate =	22.83	ton/hour
Maximum Daily Loading Rate =	547.9	tons/day
Maximum Annual Loading Rate =	200,000	ton/year
PM Emission Factor DDGS Loadout[1] =	0.0033	lb/ton
PM10 Emission Factor DDGS Loadout[1] =	0.0008	lb/ton
Capture Efficiency of DDGS Storage building	80%	
PM2.5 =	100%	of PM10
1 ton =	2000	lbs
1 year =	8760	hours

DDGS Flat Storage Potential to Emit - PM/PM10/PM2.5			
	Hourly Emission Rate	Daily Emission Rate	Annual Emission Rate
	lb/hr	lb/day	ton/year
Uncontrolled TSP	0.08	1.81	0.33
Uncontrolled PM10	0.02	0.44	0.08
Uncontrolled PM2.5	0.02	0.44	0.08
	Hourly Emission Rate	Daily Emission Rate	Annual Emission Rate
	lb/hr	lb/day	ton/year
Controlled TSP	0.015	0.4	0.066
Controlled PM10	0.0037	0.1	0.0160
Controlled PM2.5	0.0037	0.1	0.0160

[1] Emission factors taken from AP-42 Table 9.9.1-2

POET Biorefining - Macon
Macon, Missouri
FS008 Uncaptured Emissions From DDGS Storage - Fugitive Emissions

DDGS loadout is a source of VOC and HAP emissions. The following calculations are for the fugitive emissions of VOC and HAPs. The potential to emit is based on stack testing at other POET facilities. It is assumed that the building does not capture any of the VOC & HAPs emitted.

POET - Gowrie Stack Test 11-9-17 - Grain Receiving/DDGS Loadout Baghouse

Test Results	Run 1	Run 2	Run 3	Average
	lb/hr	lb/hr	lb/hr	lb/hr
VOC	0.25	0.26	0.25	0.25
Acetaldehyde	0.023	0.024	0.024	0.024
Acrolein[1]	0.015	0.015	0.015	0.015
Methanol	0.008	0.008	0.008	0.008
Formaldehyde	0.003	0.003	0.003	0.003
Process Rate	Run1	Run 2	Run 3	Average
	ton/hr	ton/hr	ton/hr	ton/hr
DDGS Loadout	104	104	104	104

Emission Factors (DDGS Loadout - Captured Emissions)[2]	
	lb/ton
VOC	2.44E-03
Acetaldehyde	2.28E-04
Acrolein	1.44E-04
Methanol	7.69E-05
Formaldehyde	2.88E-05

Safety Factor = 3
Acrolein Adjustment Factor[3] = 0.5

DDGS Flat Storage Potential to Emit - VOC & HAPs			
	Hourly PTE	Annual PTE (based on 8760 hrs/yr)	Annual Limited PTE[4]
	lbs/hr	TPY	TPY
VOC	0.17	0.73	0.731
Acetaldehyde	0.016	0.068	0.0683
Acrolein	0.0016	0.007	0.0072
Methanol	0.005	0.023	0.0231
Formaldehyde	0.0020	0.009	0.0087
Total HAPs	0.024	0.11	0.107

[1]On 1/13/2019, Montrose Environmental Group, Inc. completed further data analysis on Acrolein for Gowrie's 2017 DDGS Loadout Baghouse Stack Test. The reprocessed data indicates < 0.102 ppmv (0.015 lb/hr) for all 3 runs.

[2]Emission factors are determined based on stack tests completed at POET - Gowrie.

[3]Acrolein forms in the DDGS dryers from fat and fat breakdown products, with the majority being emitted through the RTO stack. A small amount may be present in between the voids of the DDGS solids, which is then emitted through the fluid bed cooler. Acrolein levels in DDGS handling emissions measured during performance tests have historically been less than the detection value. POET is working with stack testers to lower detection levels.

[4]Limited is based on annual throughput.

Methodology
Emission Factor (lb/ton) = average stack test result (lb/hr) / average tested throughput (ton/hr)
Hourly PTE (lbs/hr) = Emission factor (lb/ton) x proposed throughput (ton/hr) x safety factor
Annual PTE (@ 8760 hrs/yr)(tons/yr) = Hourly PTE (lb/hr) x 8760 hours/yr / 2000 lbs/ton
Annual Limited PTE (tons/yr) = Emission factor (lb/ton) x proposed throughput (ton/yr) x safety factor / 2000 (lbs/ton)

POET Biorefining - Macon
Macon, Missouri
EP-2 DDGS Loading

Dried and cooled distillers grain (DDGS) can be diverted to a silo for storage or to the flat storage. Flat storage is located in the same building where grain is received, the Grains and DDGS Flat Storage Building. A reversing conveyor allows DDGS from the silo to divert to either a truck or to rail, or DDGS on flat storage can be diverted to either a truck or to rail. DDGS cannot be diverted from both the silo and flat storage to a single delivery point. DDGS can not be loaded out to a rail and truck at the same time.

DDGS handling emissions were previously aspirated to C001/SV001. Following permit issuance, DDGS handling emissions will be aspirated to a dedicated fabric filter (C002) and exhausted to atmosphere through SV002. Fugitive emissions from the DDGS loadout are accounted for through FS001 Fugitive DDG. Uncontrolled emissions and controlled emissions are calculated below.

Max DDGS Handling Rate =	135.00	tons/hr	short term throughput
	200,000	ton/year	total facility annual throughput limit of DDGS
Conversion 1 ton =	2000	lbs	
Conversion 1 year =	8,760	hours	
Conversion - 1 hr =	60	minutes	
Conversion - 1 lb =	7,000	grains	

Assumptions:
Given:
All point source emissions from these emission units will be controlled by a fabric filter.

Maximum controlled TSP/PM10/PM2.5 emission rate[1] =	0.003	gr/dscf
Exhaust Flow Rate =	8,000	dscfm
Control Efficiency[2] =	99.0%	

	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Controlled[3]	Controlled
	lb/hr	Unc. TPY	lb/hr	TPY
PM	20.57	90.10	0.206	0.901
PM_{10}	20.57	90.10	0.206	0.901
$PM_{2.5}$	20.57	90.10	0.206	0.901

[1] This is assumed loading. Stack tests have indicated that this is a conservative assumption.
[2] Assumed control efficiency to calculate uncontrolled emissions
[3] Controlled is based on maximum flow rate
[4] Limited is based on annual throughput

DDGS handling is a source of VOC and HAP emissions. The following calculations are for the emissions of VOC and HAPs. The baghouse does not provide control of VOC or HAPs, and therefore the emissions are exhausted as uncontrolled.

POET - Gowrie Stack Test 11-9-17 - Grain Receiving/DDGS Loadout Baghouse

Test Results	Run 1	Run 2	Run 3	Average
	lb/hr	lb/hr	lb/hr	lb/hr
VOC	0.25	0.26	0.25	0.25
Acetaldehyde	0.023	0.024	0.024	0.024
Acrolein[1]	0.015	0.015	0.015	0.015
Methanol	0.008	0.008	0.008	0.008
Formaldehyde	0.003	0.003	0.003	0.003
Process Rate	Run1	Run 2	Run 3	Average
	ton/hr	ton/hr	ton/hr	ton/hr
DDGS Loadout	104	104	104	104

Emission Factors (DDGS Loadout - Captured Emissions)[2]	
	lb/ton
VOC	2.44E-03
Acetaldehyde	2.28E-04
Acrolein	1.44E-04
Methanol	7.69E-05
Formaldehyde	2.88E-05

Safety Factor =	3
Acrolein Adjustment Factor[3] =	0.5

	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Limited Uncontrolled[4]	Controlled	Controlled (based on 8760 hrs/yr)	Limited Controlled[4]
	lbs/hr	Unc. TPY	TPY	lbs/hr	lbs/hr	TPY
VOC	0.987	4.32	0.731	0.987	4.32	0.731
Acetaldehyde	0.0922	0.404	0.0683	0.0922	0.404	0.0683
Acrolein	0.0097	0.043	0.0072	0.010	0.043	0.0072
Formaldehyde	0.0312	0.136	0.0087	0.0312	0.136	0.0087
Methanol	0.0117	0.051	0.0231	0.0117	0.051	0.0231
Total HAPs	0.145	0.634	0.107	0.145	0.634	0.107

[1]On 1/13/2019, Montrose Environmental Group, Inc. completed further data analysis on Acrolein for Gowrie's 2017 DDGS Loadout Baghouse Stack Test. The reprocessed data indicates < 0.102 ppmv (0.015 lb/hr) for all 3 runs.

[2]Emission factors are determined based on stack tests completed at POET - Gowrie.

[3]Acrolein forms in the DDGS dryers from fat and fat breakdown products, with the majority being emitted through the RTO stack. A small amount may be present in between the voids of the DDGS solids, which is then emitted through the fluid bed cooler. Acrolein levels in DDGS handling emissions measured during performance tests have historically been less than the detection value. POET is working with stack testers to lower detection levels.

[4]Limited is based on annual throughput.

Methodology

Emission Factor (lb/ton) = average stack test result (lb/hr) / average tested throughput (ton/hr)

Uncontrolled PTE (lbs/hr) = Emission factor (lb/ton) x proposed throughput (ton/hr) x safety factor

Uncontrolled PTE (tons/yr) = Uncontrolled PTE (lb/hr) x 8760 hours/yr / 2000 lbs/ton

Limited Uncontrolled PTE (tons/yr) = Emission factor (lb/ton) x annual throughput (tons/yr) / 2000 lbs * safety factor

Controlled PTE (lbs/hr) = Uncontrolled PTE (lbs/hr)

Controlled PTE (tons/yr) = Uncontrolled PTE (tons/yr)

Limited Controlled PTE (tons/yr) = Limited uncontrolled PTE (tons/yr)

POET Biorefining - Macon
Macon, Missouri
FS-01 Fugitive Grain Handling Source Emission Calculations

The grain unloading operations will be located inside a dump shed and will be aspirated to a fabric filter. The shed enclosure will provide control of the TSP and PM10 and PM2.5 emissions. The loading shed and aspiration system will capture 95% of the particulate matter emitted during loading operations. Therefore, grain unloading will be a source of fugitive particulate emissions.

The uncontrolled potential emission will be calculated by multiplying the appropriate USEPA AP-42 emission factor by the maximum dump rate for the process and assuming that all TSP/PM10/PM2.5 is emitted.

The controlled potential emissions will be calculated by multiplying the appropriate USEPA AP-42 emission factor by the maximum dump rate for the process, assuming that 95% of the TSP/PM10/PM2.5 is captured by the aspiration system with the remainder being emitted.

The equation for calculating the uncontrolled potential fugitive emissions is:

Eu lb/hr = Ef lb/ton x W tons

The equation for calculating the controlled potential fugitive emissions is:

Ec lb/hr = (Ef lb/ton x W tons) x [(100-CE) / 100]

Where:
Eu = Uncontrolled Potential Emissions
Ef = AP-42 Emission Factor for Grain Receiving
W = Tons of grain received
Ce = Capture efficiency of shed and aspiration system

Given:

Maximum Hourly Truck Dump Rate[2] =	393.8	ton/hour
Limited Daily Truck Dump Rate	6,300	tons/day
Maximum Annual Truck Dump Rate =	666,800	ton/year
Hopper Truck TSP Emission Factor[1] =	0.035	lb/ton
Hopper Truck PM10 Emission Factor[1]=	0.0078	lb/ton
Hopper Truck PM2.5 Emission Factor[1]=	0.0013	lb/ton
Straight Truck TSP Emission Factor[1] =	0.18	lb/ton
Straight Truck PM10 Emission Factor [1]=	0.059	lb/ton
Straight Truck PM2.5 Emission Factor [1]=	0.01	lb/ton
Shed Capture Efficiency =	95%	
1 ton =	2000	lbs

	Hourly Emission Rate	Daily Emission Rate	Annual Emission Rate
	lb/hr	lb/day	ton/year
Uncontrolled TSP	70.9	1134.0	60.0
Uncontrolled PM10	23.2	371.7	19.7
Uncontrolled PM2.5	3.9	63.0	3.3

	Hourly Emission Rate	Daily Emission Rate	Annual Emission Rate
	lb/hr	lb/day	ton/year
Fugitive TSP	3.5	56.7	3.0
Fugitive PM10	1.2	18.6	1.0
Fugitive PM2.5	0.2	3.2	0.2

[1] Emission factors taken from AP-42 Table 9.9.1-1
[2] Based on daily truck dump limit and 16 hours per day

POET Biorefining - Macon
Macon, Missouri
EP-FS-01 Fugitive DDG Source Emission Calculations

The loading of DDG solids into trucks or containers will be conducted inside a shed. A portion of the loading operations will be aspirated to a fabric filter. Due to physical constraints of the loading operations, truck or container loadout may not be conducted simultaneously with rail loadout. During rail, truck, or container loadout, the shed and aspiration system will capture 80% of the particulate matter emitted. Therefore, DDG loading will be a source of fugitive particulate emissions.

The emissions from the loading operations into trucks are higher than railcar loading. Therefore, the potential emission calculations will be based on truck loading for all DDG.
The uncontrolled potential emission will be calculated by multiplying the appropriate USEPA AP-42 emission factor by the maximum loading rate for the process and assuming that all TSP/PM10/PM2.5 is emitted.
The controlled potential emissions will be calculated by multiplying the appropriate USEPA AP-42 emission factor by the maximum loading rate for the process, assuming that 80% of the TSP/PM10/PM2.5 is captured by the shed aspiration system with the remainder being emitted.

The equation for calculating the uncontrolled potential fugitive emissions is:

Eu lb/hr = Ef lb/ton x W tons

The equation for calculating the controlled potential fugitive emissions is:

Ec lb/hr = (Ef lb/ton x W tons) x [(100-Ce) / 100]

Where:

Ec = Emission rate in lb/hr
Ef = Emission factor in lb/ton loaded
W = Tons loaded per unit time
Ce = Capture efficiency of shed and aspiration system

Given:

Maximum Hourly Truck/Container Loading Rate =	135.00	ton/hour
Maximum Daily Loading Rate =	3240	tons/day
Maximum Annual Loading Rate =	200,000	ton/year
TSP Emission Factor DDGS Loadout[1] =	0.0033	lb/ton
PM10 Emission Factor DDGS Loadout[1] =	0.0008	lb/ton
Capture Efficiency of Rail Shed and Aspiration System =	80%	
Capture Efficiency of Truck/Container Shed =	80%	
PM2.5 =	100%	of PM10
1 ton =	2000	lbs
1 year =	8760	hours

Total DDGS Loadout Potential to Emit - PM/PM10/PM2.5			
	Hourly Emission Rate	Daily Emission Rate	Annual Emission Rate
	lb/hr	lb/day	ton/year
Uncontrolled TSP	0.45	10.7	0.33
Uncontrolled PM10	0.11	2.6	0.08
Uncontrolled PM2.5	0.11	2.6	0.08
	Hourly Emission Rate	Daily Emission Rate	Annual Emission Rate
	lb/hr	lb/day	ton/year
Controlled TSP	0.09	2.1	0.07
Controlled PM10	0.02	0.5	0.02
Controlled PM2.5	0.02	0.5	0.02

[1] Emission factors taken from AP-42 Table 9.9.1-1

POET Biorefining - Macon
Macon, Missouri
EP-FS-01 Fugitive DDG Source Emission Calculations

DDGS loadout is a source of VOC and HAP emissions. The following calculations are for the fugitive emissions of VOC and HAPs. The potential to emit is based on stack testing at other POET facilities. It is assumed that the building does not capture any of the VOC & HAPs emitted.

POET - Gowrie Stack Test 11-9-17 - Grain Receiving/DDGS Loadout Baghouse

Test Results	Run 1	Run 2	Run 3	Average
	lb/hr	lb/hr	lb/hr	lb/hr
VOC	0.25	0.26	0.25	0.25
Acetaldehyde	0.023	0.024	0.024	0.024
Acrolein[1]	0.015	0.015	0.015	0.015
Methanol	0.008	0.008	0.008	0.008
Formaldehyde	0.003	0.003	0.003	0.003
Process Rate	Run1	Run 2	Run 3	Average
	ton/hr	ton/hr	ton/hr	ton/hr
DDGS Loadout	104	104	104	104

Emission Factors (DDGS Loadout - Captured Emissions)[2]	
	lb/ton
VOC	2.44E-03
Acetaldehyde	2.28E-04
Acrolein	1.44E-04
Methanol	7.69E-05
Formaldehyde	2.88E-05

Safety Factor = 3
Acrolein Adjustment Factor[3] = 0.5

Total DDGS Loadout Potential to Emit - VOC & HAPs			
	Hourly PTE	Annual PTE (based on 8760 hrs/yr)	Annual Limited PTE[4]
	lbs/hr	TPY	TPY
VOC	0.987	4.32	0.731
Acetaldehyde	0.092	0.404	0.068
Acrolein	0.010	0.043	0.0072
Methanol	0.031	0.136	0.023
Formaldehyde	0.012	0.051	0.0087
Total HAPs	0.145	0.634	0.107

[1]On 1/13/2019, Montrose Environmental Group, Inc. completed further data analysis on Acrolein for Gowrie's 2017 DDGS Loadout Baghouse Stack Test. The reprocessed data indicates < 0.102 ppmv (0.015 lb/hr) for all 3 runs.

[2]Emission factors are determined based on stack tests completed at POET - Gowrie.

[3]Acrolein forms in the DDGS dryers from fat and fat breakdown products, with the majority being emitted through the RTO stack. A small amount may be present in between the voids of the DDGS solids, which is then emitted through the fluid bed cooler. Acrolein levels in DDGS handling emissions measured during performance tests have historically been less than the detection value. POET is working with stack testers to lower detection levels.

[4]Limited is based on annual throughput.

[5]GOW stack test includes multiple drop points (conveyor, elevator) in DDGS loadout system. Therefore, this emission factor is applied to truck and container (2 drop points) collectively.

Methodology
Emission Factor (lb/ton) = average stack test result (lb/hr) / average tested throughput (ton/hr)
Hourly PTE (lbs/hr) = Emission factor (lb/ton) x proposed throughput (ton/hr) x safety factor
Annual PTE (@ 8760 hrs/yr)(tons/yr) = Hourly PTE (lb/hr) x 8760 hours/yr / 2000 lbs/ton
Annual Limited PTE (tons/yr) = Emission factor (lb/ton) x proposed throughput (ton/yr) x safety factor / 2000 (lbs/ton)

POET Biorefining - Macon
Macon, Missouri
EP-9 Grain Dryer

Combustion Emissions

Primary Fuel Type: **Natural Gas**		
Firing Capacity:	31	MMBtu/hr
Heat Value:	1,020	Btu/cf
Fuel Burning Capacity:	30,392	cf/hr
Grain Capacity of Drier:	2,500	bu/hr
Fuel Burning Capacity/Bushel:	7,473	Btu/bu
Annual Capacity[1]	2,059,925	bu/yr
Annual Capacity[1]	66,680	ton/yr
Fuel Use Limit:	15,394	MMBtu/yr
Fuel Use Limit:	15.1	Mcf/yr
Limited Capacity	**496.6**	**hr/year**

1 - Approximately 10% of the grain is dried to reduce the moisture content

Pollutant	Emission Factor[1] (lb/MMcf)	Emission Rate (lb/hr)	Maximum Uncontrolled Emissions (tons/yr)
PM	7.6	0.23	0.06
PM_{10}	7.6	0.23	0.06
$PM_{2.5}$	7.6	0.23	0.06
SOx	0.6	0.02	0.00
NOx	100.0	3.04	0.75
VOC	5.5	0.17	0.04
CO	84.0	2.55	0.63

	Emission Factor [1]	Maximum Uncontrolled Emissions	
HAP Pollutant	(lb/MMcf)	(lb/hr)	(ton/yr)
Benzene	0.0021	6.38E-05	1.58E-05
Formaldehyde	0.075	2.28E-03	5.66E-04
Hexane	1.8	5.47E-02	1.36E-02
Naphthalene	0.00061	1.85E-05	4.60E-06
Toluene	0.0034	1.03E-04	2.57E-05
Arsenic	0.0002	6.08E-06	1.51E-06
Berylium	0.000012	3.65E-07	9.06E-08
Cadmium	0.0011	3.34E-05	8.30E-06
Chromium	0.0014	4.25E-05	1.06E-05
Cobalt	0.000084	2.55E-06	6.34E-07
Lead	0.0005	1.52E-05	3.77E-06
Manganese	0.00038	1.15E-05	2.87E-06
Mercury	0.00026	7.90E-06	1.96E-06
Nickel	0.0021	6.38E-05	1.58E-05
Selenium	0.000025	7.60E-07	1.89E-07
		0.05736	**0.01424**

1 - Emission factor is from AP-42, 5th Edition, Section 1.4, 7/98

Process Emissions

Pollutant	Emission Factor[2] (lb/ton)	Emission Rate (lb/hr)	Maximum Uncontrolled Emissions (tons/yr)
PM	0.22	15.4	3.82
PM_{10}	0.055	3.85	0.96
$PM_{2.5}$	0.00935	0.65	0.16

2 - Emission factor is from AP-42, Table 9.9.1-1., 3/03

Total Emissions

Pollutant	Emission Rate (lb/hr)	Maximum Uncontrolled Emissions (tons/yr)
PM	15.63	3.88
PM_{10}	4.08	1.01
$PM_{2.5}$	0.89	0.22
SOx	0.02	0.00
NOx	3.04	0.75
VOC	0.17	0.04
CO	2.55	0.63

POET Biorefining - Macon
Macon, Missouri
EP-34 - Hammermill #1

The following calculations are for the emissions of total suspended particulate (TSP), particulate matter less than 10 micron (PM10), and particulate matter less than 2.5 micron (PM2.5) from point sources at the proposed ethanol facility. For the purposes of these calculations, TSP, PM10, and PM2.5 are assumed to be equal.

The controlled potential TSP/PM10/PM2.5 emissions were calculated by multiplying the design grain loading from the fabric filter by the design flow rate for the source and converting to a pound per hour emission rate. The equation for the calculations is:

Ec lb/hr = G gr/dscf x 1 lb / 7000 gr x Q dscfm x 60 minutes / hour

Where:
E = Emission rate of TSP/PM10 in lb/hr
G = Outlet Grain Loading in grain per dry standard cubic foot of air
Q = Flow Rate in dry standard cubic feet per minute

Where:
Qgrain = 666,800 tons/year
Number of Hammermills 4

Conversion Factors:

1 ton =	2000	lb
1 lb =	7000	gr
1 year =	8760	hours
1 hour =	60	minutes

Assumptions:
Maximum controlled TSP/PM10/PM2.5 emission rate[1] = 0.003 gr/dscf
Control Efficiency[2] = 99%

Given:
Capacity = 22.5 ton/hr
All point source emissions from these emission units will be controlled by a fabric filter.
Exhaust Flow Rate = 12,000 dscfm

	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Limited Uncontrolled[4]	Controlled[3]	Controlled[3]
	lbs/hr	Unc. TPY	TPY	lbs/hr	TPY
PM	30.86	135.15	114.31	0.31	1.35
PM10	30.86	135.15	114.31	0.31	1.35
PM2.5	30.86	135.15	114.31	0.31	1.35

[1] This is assumed loading. Stack tests have indicated that this is a conservative assumption
[2] Assumed control efficiency
[3] Controlled is based on maximum flow rate
[4] Limited is based on annual throughput

POET Biorefining - Macon
Macon, Missouri
EP-35 - Hammermill #2

The following calculations are for the emissions of total suspended particulate (TSP), particulate matter less than 10 micron (PM10), and particulate matter less than 2.5 micron (PM2.5) from point sources at the proposed ethanol facility. For the purposes of these calculations, TSP, PM10, and PM2.5 are assumed to be equal.

The controlled potential TSP/PM10/PM2.5 emissions were calculated by multiplying the design grain loading from the fabric filter by the design flow rate for the source and converting to a pound per hour emission rate. The equation for the calculations is:

Ec lb/hr = G gr/dscf x 1 lb / 7000 gr x Q dscfm x 60 minutes / hour

Where:
E = Emission rate of TSP/PM10 in lb/hr
G = Outlet Grain Loading in grain per dry standard cubic foot of air
Q = Flow Rate in dry standard cubic feet per minute

Where:
Qgrain = 666,800 tons/year
Number of Hammermills 4

Conversion Factors:

1 ton =	2000	lb
1 lb =	7000	gr
1 year =	8760	hours
1 hour =	60	minutes

Assumptions:
Maximum controlled TSP/PM10/PM2.5 emission rate[1] = 0.003 gr/dscf
Control Efficiency[2] = 99%

Given:
Capacity = 22.5 ton/hr
All point source emissions from these emission units will be controlled by a fabric filter.
Exhaust Flow Rate = 12,000 dscfm

	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Limited Uncontrolled[4]	Controlled[3]	Controlled[3]
	lbs/hr	Unc. TPY	TPY	lbs/hr	TPY
PM	30.86	135.15	114.31	0.31	1.35
PM10	30.86	135.15	114.31	0.31	1.35
PM2.5	30.86	135.15	114.31	0.31	1.35

[1] This is assumed loading. Stack tests have indicated that this is a conservative assumption
[2] Assumed control efficiency
[3] Controlled is based on maximum flow rate
[4] Limited is based on annual throughput

POET Biorefining - Macon
Macon, Missouri
EP-36 - Hammermill #3

The following calculations are for the emissions of total suspended particulate (TSP), particulate matter less than 10 micron (PM10), and particulate matter less than 2.5 micron (PM2.5) from point sources at the proposed ethanol facility. For the purposes of these calculations, TSP, PM10, and PM2.5 are assumed to be equal.

The controlled potential TSP/PM10/PM2.5 emissions were calculated by multiplying the design grain loading from the fabric filter by the design flow rate for the source and converting to a pound per hour emission rate. The equation for the calculations is:

Ec lb/hr = G gr/dscf x 1 lb / 7000 gr x Q dscfm x 60 minutes / hour

Where:
E = Emission rate of TSP/PM10 in lb/hr
G = Outlet Grain Loading in grain per dry standard cubic foot of air
Q = Flow Rate in dry standard cubic feet per minute

Where:
Qgrain = 666,800 tons/year
Number of Hammermills 4

Conversion Factors:

1 ton =	2000	lb
1 lb =	7000	gr
1 year =	8760	hours
1 hour =	60	minutes

Assumptions:
Maximum controlled TSP/PM10/PM2.5 emission rate[1] = 0.003 gr/dscf
Control Efficiency[2] = 99%

Given:
Capacity = 22.5 ton/hr
All point source emissions from these emission units will be controlled by a fabric filter.
Exhaust Flow Rate = 12,000 dscfm

	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Limited Uncontrolled[4]	Controlled[3]	Controlled[3]
	lbs/hr	Unc. TPY	TPY	lbs/hr	TPY
PM	30.86	135.15	114.31	0.31	1.35
PM10	30.86	135.15	114.31	0.31	1.35
PM2.5	30.86	135.15	114.31	0.31	1.35

[1] This is assumed loading. Stack tests have indicated that this is a conservative assumption
[2] Assumed control efficiency
[3] Controlled is based on maximum flow rate
[4] Limited is based on annual throughput

POET Biorefining - Macon
Macon, Missouri
EP-37 - Hammermill #4

The following calculations are for the emissions of total suspended particulate (TSP), particulate matter less than 10 micron (PM10), and particulate matter less than 2.5 micron (PM2.5) from point sources at the proposed ethanol facility. For the purposes of these calculations, TSP, PM10, and PM2.5 are assumed to be equal.

The controlled potential TSP/PM10/PM2.5 emissions were calculated by multiplying the design grain loading from the fabric filter by the design flow rate for the source and converting to a pound per hour emission rate. The equation for the calculations is:

Ec lb/hr = G gr/dscf x 1 lb / 7000 gr x Q dscfm x 60 minutes / hour

Where:
E = Emission rate of TSP/PM10 in lb/hr
G = Outlet Grain Loading in grain per dry standard cubic foot of air
Q = Flow Rate in dry standard cubic feet per minute

Where:
Qgrain = 666,800 tons/year
Number of Hammermills 4

Conversion Factors:

1 ton =	2000	lb
1 lb =	7000	gr
1 year =	8760	hours
1 hour =	60	minutes

Assumptions:
Maximum controlled TSP/PM10/PM2.5 emission rate[1] = 0.003 gr/dscf
Control Efficiency[2] = 99%

Given:
Capacity = 22.5 ton/hr
All point source emissions from these emission units will be controlled by a fabric filter.
Exhaust Flow Rate = 12,000 dscfm

	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Limited Uncontrolled[4]	Controlled[3]	Controlled[3]
	lbs/hr	Unc. TPY	TPY	lbs/hr	TPY
PM	30.86	135.15	114.31	0.31	1.35
PM10	30.86	135.15	114.31	0.31	1.35
PM2.5	30.86	135.15	114.31	0.31	1.35

[1] This is assumed loading. Stack tests have indicated that this is a conservative assumption
[2] Assumed control efficiency
[3] Controlled is based on maximum flow rate
[4] Limited is based on annual throughput

POET Biorefining - Macon
Macon, Missouri
EP-7 Boiler Calculations

The facility has two natural gas fired steam boilers. Both boilers are not subject to the provisions of the United Stated Environmental Protection Agency's (USEPA's) New Source Performance Standard (NSPS), Subpart Db, Standard for Institutional, Commercial, and Industrial Steam Generating Units.

The boilers will be equipped with low NOx burners.

The potential emissions for the boiler is calculated by multiplying an emission factor by the rated capacity of the boiler. AP-42 emission factors from Section 1.4 were converted to lb/MMBtu assuming a heating value of 1020 Btu/ft3 for natural gas. The emission factors are:

AP-42 Table 1.4-3 & Table 1.4-4

HAP Compound	Emission Factor lb/MMSCF
Benzene	0.0021
Formaldehyde	0.075
Hexane	1.8
Naphthalene	0.00061
Toluene	0.0034
Arsenic	0.0002
Berylium	0.000012
Cadmium	0.0011
Chromium	0.0014
Cobalt	0.000084
Lead	0.00046
Manganese	0.00038
Mercury	0.00026
Nickel	0.0021
Selenium	0.000024

Conversion Factors:	Assumed for natural gas	
1 lb/MMBtu =	1020	Btu/scf
1 ton =	2000	lb
1 year =	8760	hours

	Emission Factor lb/MMbtu	Source
TSP	0.0075	AP-42 Section 1.4 * 1020 Btu/scf
PM10	0.0075	AP-42 Section 1.4 * 1020 Btu/scf
PM2.5	0.0075	AP-42 Section 1.4 * 1020 Btu/scf
NOx	0.0490	AP-42 Section 1.4 * 1020 Btu/scf
SOx	0.0006	AP-42 Section 1.4 * 1020 Btu/scf
VOC	0.0054	AP-42 Section 1.4 * 1020 Btu/scf
CO	0.0824	AP-42 Section 1.4 * 1020 Btu/scf

Boiler 1

HAPs		Size	60.5	MMBtu/hr
HAP Name	CAS #	lb/hr	Unc tpy	Lim tpy
Benzene	71-43-2	0.00012	0.00055	0.00055
Berylium	7440-41-7	0.00000	0.00000	0.00000
Cadmium	7440-43-7	0.00007	0.00029	0.00029
Chromium	7440-47-3	0.00008	0.00036	0.00036
Cobalt	7440-48-4	0.00000	0.00002	0.00002
Formaldehyde	50-00-0	0.00445	0.01948	0.01948
Hexane	110-54-3	0.10676	0.46763	0.46763
Lead	NA	0.00003	0.00012	0.00012
Manganese	7439-96-5	0.00002	0.00010	0.00010
Mercury	7439-97-6	0.00002	0.00007	0.00007
Naphthalene	91-20-3	0.00004	0.00016	0.00016
Nickel	7440-02-0	0.00012	0.00055	0.00055
Selenium	7782-49-2	0.00000	0.00001	0.00001
Toluene	108-88-3	0.00020	0.00088	0.00088
	Total HAP	0.11192	0.49021	0.49021

	TSP		PM10		PM2.5		NOx		SOx		VOC		CO	
	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	ton/yr	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy
Boiler 1	0.45	1.97	0.45	1.97	0.45	1.97	2.97	12.99	0.04	0.16	0.33	1.43	4.98	21.82
Uncontrolled Boiler 1	0.45	1.97	0.45	1.97	0.45	1.97	2.97	12.99	0.04	0.16	0.33	1.43	4.98	21.82

POET Biorefining - Macon
Macon, Missouri
EP-30 Boiler Calculations

The facility has two natural gas fired steam boilers. Both boilers are not subject to the provisions of the United Stated Environmental Protection Agency's (USEPA's) New Source Performance Standard (NSPS), Subpart Db, Standard for Institutional, Commercial, and Industrial Steam Generating Units.

The boilers will be equipped with low NOx burners.

The potential emissions for the boiler is calculated by multiplying an emission factor by the rated capacity of the boiler. AP-42 emission factors from Section 1.4 were converted to lb/MMBtu assuming a heating value of 1020 Btu/ft3 for natural gas. The emission factors are:

AP-42 Table 1.4-3 & Table 1.4-4

HAP Compound	Emission Factor lb/MMSCF
Benzene	0.0021
Formaldehyde	0.075
Hexane	1.8
Naphthalene	0.00061
Toluene	0.0034
Arsenic	0.0002
Berylium	0.000012
Cadmium	0.0011
Chromium	0.0014
Cobalt	0.000084
Lead	0.00046
Manganese	0.00038
Mercury	0.00026
Nickel	0.0021
Selenium	0.000024

Conversion Factors:	Assumed for natural gas	
1 lb/MMBtu =	1020	Btu/scf
1 ton =	2000	lb
1 year =	8760	hours

	Emission Factor lb/MMbtu	Source
TSP	0.0075	AP-42 Section 1.4 * 1020 Btu/scf
PM10	0.0075	AP-42 Section 1.4 * 1020 Btu/scf
PM2.5	0.0075	AP-42 Section 1.4 * 1020 Btu/scf
NOx	0.0490	AP-42 Section 1.4 * 1020 Btu/scf
SOx	0.0006	AP-42 Section 1.4 * 1020 Btu/scf
VOC	0.0054	AP-42 Section 1.4 * 1020 Btu/scf
CO	0.0824	AP-42 Section 1.4 * 1020 Btu/scf

HAPs		Size	72.6	MMBtu/hr
HAP Name	CAS #	lb/hr	Unc tpy	Lim tpy
Benzene	71-43-2	0.00015	0.00065	0.00065
Berylium	7440-41-7	0.00000	0.00000	0.00000
Cadmium	7440-43-7	0.00008	0.00034	0.00034
Chromium	7440-47-3	0.00010	0.00044	0.00044
Cobalt	7440-48-4	0.00001	0.00003	0.00003
Formaldehyde	50-00-0	0.00534	0.02338	0.02338
Hexane	110-54-3	0.12812	0.56116	0.56116
Lead	NA	0.00003	0.00014	0.00014
Manganese	7439-96-5	0.00003	0.00012	0.00012
Mercury	7439-97-6	0.00002	0.00008	0.00008
Naphthalene	91-20-3	0.00004	0.00019	0.00019
Nickel	7440-02-0	0.00015	0.00065	0.00065
Selenium	7782-49-2	0.00000	0.00001	0.00001
Toluene	108-88-3	0.00024	0.00106	0.00106
	Total HAP	0.13431	0.58826	0.58826

	TSP		PM10		PM2.5		NOx		SOx		VOC		CO	
	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	ton/yr	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy	lb/hr	Lim tpy
Boiler 2	0.54	2.37	0.54	2.37	0.54	2.37	3.56	15.59	0.04	0.19	0.39	1.71	5.98	26.19
Uncontrolled Boiler 2	0.54	2.37	0.54	2.37	0.54	2.37	3.56	15.59	0.04	0.19	0.39	1.71	5.98	26.19

POET Biorefining - Macon
Macon, Missouri
EP-FS-03 Cooling Tower Emission Calculations

Particulate emissions are estimated by analyzing the mass of condensed water that is released times the total mass of dissolved solids present in the water. The estimate is based on the assumption that all of the dissolved solids that are present in the water becomes airborne particulate matter upon evaporation of the release condensed water droplets. The released water droplets are referred to as the "drift loss" of the tower. The drift loss of the tower is usually referenced as a percent of the total water circulation rate of the tower.

PM to PM10 ratio:	70% % Drift PM	Reisman and Frisbie
PM to PM2.5 ratio	42% % Drift PM	Reisman and Frisbie

Conversions:

1 gallon =	3.78541	Liters
1 gram =	1000	mg
1 hour	60	minutes
1 pound =	453.6	grams
1 ton =	2000	pounds
1 year =	8760	hours
1 day =	24	hours

Assumptions:

Drift Loss = 0.005%

Water circulation flow =	18,500	gallons / minute
Water circulation flow =	70,030	liters / minute
Drift loss =	3.5	liters / minute
Total Dissolved Solids in cooling tower =	1500	milligrams / liter
Total Dissolved Solids in cooling tower =	1.5	grams / liter
PM-10 = Drift loss (l/min) x TDS (g/l)	5.3	grams/minute
g/min x 60 =	315.1	grams/hr

	lbs/hr	lb/day	g/s	TPY	g/s
Fugitive PM emissions=	0.69	16.7		3.04	
Fugitive PM10 emissions=	0.5	11.7	0.0613	2.1	0.0613
Fugitive PM2.5 emissions=	0.3	7.0	0.0368	1.3	0.0368

POET Biorefining - Macon
Macon, Missouri
FS005 Wet Cake - Fugitive Emissions

Process Description
Wetcake is the grain solids from the whole stillage centrifuges, before being conveyed to the DDGS dryers. If it is not dried to produce DDGS, wetcake is stored on an outdoor pad located adjacent to the process building. Wetcake contains a small amount of residual VOC and HAP which are emitted during storage, dependent on the boiling point of the compound. Wetcake is loaded into trucks using a frontend loader. Since the wetcake has a high moisture content, wetcake loadout is not a source of particulate matter emissions.

Modified wetcake is being produced as local users may prefer the product over DDGS. DDGS is produced by the two dryers operating in series. Modified wet cake has been dried by only one dryer, then syrup is added to bring the moisture to approximately 50%.

VOC and HAP Potential to Emit
Wetcake production storage and loadout is a source of VOC and HAP emissions (methanol) because the wetcake contains a small quantity of ethanol and HAPs. This source is not controlled. The emission factors for this process come from emissions testing at a similar facility. The operation of the dryers and DDGS cooler represent the "worst case" emission scenario and thus are presented in the potential to emit summary.

The emission factors for this process were based on a wet cake analysis completed in April 2015. All VOC and HAP in the wet cake are assumed to be emitted. This operating scenario will be limited based on production per calendar year. Wetcake will be produced when the dryers are off-line. Based on customer demand, a portion of the distillers grains production may be diverted from the dryers to the wetcake pad for off-site sale. Therefore, wetcake may be stored on the wetcake pad at the same time that the dryers and DDGS cooler are operating at full or partial capacity.

Wet Cake Acrolein, Acetaldehyde, and Formaldehyde Potential Emissions
There will be no acrolein, acetaldehyde, or formaldehyde present in the wet cake on the storage pad. The boiling point of acrolein is 127 °F, of acetaldehyde is 68 °F, and of formaldehyde is -2 °F. The majority of any acrolein, acetaldehyde, or formaldehyde present in the fermented beer would boil out in distillation. The ethanol vapor removed from the beer is condensed in a condenser at temperatures greater than 127 °F. Due to boiling points, acrolein, acetaldehyde, and formaldehyde vapor would not be condensed but rather sent to the distillation scrubber. Furthermore, there will be no acrolein, acetaldehyde, or formaldehyde present in the whole stillage, thin stillage, or wet cake (solids left after centrifuging). These materials are maintained at 175-180 °F while held in tanks and while centrifuged. If any acrolein, acetaldehyde, or formaldehyde did remain in the stillage from distillation, it would be emitted from the centrifuges as temperatures within the centrifuges remain greater than 127 °F.

Acrolein, acetaldehyde, or formaldehyde would only be emitted in the scrubber or centrifuges (RTO). Historical analytical results have supported this analysis as shown in the tables below.

Given:

Capacity =	60	tons/hr maximum wetcake production rate
Maximum Tons/Year =	312,500	tons/yr
Weight Cake Moisture Content =	64%	

Analytical Results, Sample Collected 4-17-2015 Ashton, IA

Ethanol		12.8	mg/kg of wet cake produced
Methanol		5.2	mg/kg of wet cake produced
Acetaldehyde	Non-detect @	4.99	mg/kg of wet cake produced
Formaldehyde	Non-detect @	4.99	mg/kg of wet cake produced
Acrolein	Non-detect @	0.02	mg/kg of wet cake produced
Acrolein Results, Sample Collected 3-16-2017 Macon, MO			
Acrolein	Non-detect @	0.048	mg/kg of wet cake produced

The March 2017 Acrolein test was completed on thin stillage. Per the testing in June 2016, wetcake is 36% dry solids by weight. PTE calculations assume wetcake is 64% thin stillage by weight.

Conversion Factors:

1 lb =	453592	mg
1 kg =	2.2	lb
1 ton =	2000	lb

The emission calculations assume that all VOC in the wet cake are emitted on-site. To account for process variables, a "safety factor" of 1.5 was applied to the detected compounds (ethanol and methanol).
Assumed Maximum Presence:

Detected compounds	150%	of level detected

Pollutant	Emission Factor	Emission Rate	
	lb/ton	lb/hr	Ton/yr
Ethanol	3.85E-02	2.31	6.01
Methanol	1.56E-02	0.94	2.44
Total VOC		**3.25**	**8.46**
Total HAP		**0.94**	**2.44**

POET Biorefining - Macon
Macon, Missouri
SV-006 Potential Emissions from the Fermenters, Distillation System, and DDGS Dryers.

Emissions from the RTO includes the emissions from the fermentation system, distillation system, DDGS dryers, and centrifuges. The fermentation system is a source of VOC and HAPs and are designated as SV004 when in bypass mode. The distillation system includes the beer wells, distillation columns, rectifiers, and molecular sieves and is routed through a scrubber and to SV005 when in bypass mode. The dryers have natural gas fired burners and are sources of TSP, PM10, NOx, SOx, CO, VOC and HAPs. The emission units are vented to a regenerative thermal oxidizer (RTO) and out SV006. The RTO will control VOC HAPs, and CO emissions. TSP emissions are controlled by the dryer multiclones.

The RTO has been designed to achieve a minimum of 98.0% reduction in VOC and 95.0% reduction in HAP emissions. Additionally, the RTO will achieve 90% reduction in TSP and CO emissions. The RTO control efficiencies are based on engineering estimates for TSP and CO, and performance tests for VOC and HAPs.

The DDGS dryers #1 and #2 have 68.5 MMBtu/hr natural gas fired burners. The dryers do not have the capacity to combust any other fuel. The dryers are connected in series, therefore, all of the DDGS from each production line is processed by the two dryers in series.

RTO VOC Control Efficiency =	98.00%
RTO HAP Control Efficiency =	95.00%
RTO NOx Control Efficiency =	0.00%
RTO CO Control Efficiency =	90.00%
RTO PM Control Efficiency =	90.00% in conjuction with dryer multiclones
RTO SO2 Control Efficiency =	0.00%
Proposed beerfeed=	633 gpm
Yearly operation limit=	8760 hours
EF safety factor =	1.4
EF safety factor (Acetaldehyde) =	3.2

Conversion: 2000 lbs = 1 ton

Natural Gas Combustion

Unit	Rated Capacity	
Dryers	137	MMBTU/Hr
RTO	18	MMBTU/Hr

	Emission Factor	
Pollutant	**lb/MMBTU**	**Source**
SO2	0.0006	AP-42 Section 1.4

AP-42 emission factors from Section 1.4 were converted to lb/MMBtu assuming a heating value of 1020 Btu/ft3 for natural gas.

Macon RTO Stack Test - 10/17/2017 **CO2 Plant In-Service**

Test Results	Run 1	Run 2	Run 3	Average
	lb/hr	lb/hr	lb/hr	lb/hr
PM[1]	3.35	2.31	3.06	2.91
PM10	3.35	2.31	3.06	2.91
PM2.5[2]	3.35	2.31	3.06	2.91
NOx	8.44	8.53	8.40	8.46
CO	0.93	0.89	0.88	0.90
VOC	0.57	0.57	0.69	0.61
Acetaldehyde[3]	0.04	0.04	0.04	0.04
Methanol	0.17	0.17	0.17	0.17
Formaldehyde	0.01	0.01	0.01	0.01
Acrolein[4]	0.05	0.05	0.05	0.05
Total HAPs	0.27	0.27	0.27	0.27
Process Rate	Run1	Run 2	Run 3	Average
	gpm	gpm	gpm	gpm
Beerfeed	570	570	570	570

Macon RTO Stack Test - 10/18/2017 **CO2 Plant Out of Service**

Test Results	Run 1	Run 2	Run 3	Average
	lb/hr	lb/hr	lb/hr	lb/hr
PM[1]	2.68	3.42	2.70	2.93
PM10	2.68	3.42	2.70	2.93
PM2.5[2]	2.68	3.42	2.70	2.93
NOx	8.73	8.61	8.62	8.65
CO	2.34	2.56	2.53	2.48
VOC	0.76	0.76	0.79	0.77
Acetaldehyde[3]	0.04	0.04	0.04	0.04
Methanol	0.14	0.14	0.18	0.15
Formaldehyde	0.01	0.01	0.01	0.01
Acrolein[4]	0.05	0.05	0.05	0.05
Total HAPs	0.24	0.24	0.26	0.25
Process Rate	Run1	Run 2	Run 3	Average
	gpm	gpm	gpm	gpm
Beerfeed	570	570	570	570

[1] Total PM, Filterable plus Condensable

[2] Total PM measured by USEPA Method 5 and 202. PM2.5 assumed to be equal to Total PM.

[3] All results were <detection limit for the test method. Lb/hr emission rate is assumed to = the detection limit.

[4] On 12/10/2018, Montrose Environmental Group, Inc. completed further data analysis on Acrolein for Macon's 2017 RTO Stack Test. The reprocessed data indicates < 0.1 ppmv for all 6 runs.

RTO Total Emissions

	Uncontrolled Potential to Emit		Controlled Potential to Emit			Permit Limits		Proposed Potential to Emit	
	lbs/hr	Unc. TPY	lbs/hr CO2 Plant ON	lbs/hr CO2 Plant OFF	TPY (Worst-case)	lbs/hr	TPY	lbs/hr	TPY
PM	45.21	198.04	4.52	4.56	19.99	4.83	N/A	4.83	21.16
PM10	45.21	198.04	4.52	4.56	19.99	4.83	N/A	4.83	21.16
PM2.5	45.21	198.04	4.52	4.56	19.99	4.33	N/A	4.33	18.97
NOx	13.15	57.62	13.15	13.46	58.96	11.96	N/A	13.46	58.96
CO	14.00	61.32	1.40	3.85	16.87	11.00	N/A	11.00	48.18
VOC	47.44	207.81	0.95	1.20	5.25	3.64	N/A	3.64	15.94
Acetaldehyde	2.84	12.46	0.14	0.14	0.62	N/A	N/A	0.14	0.62
Methanol	5.29	23.17	0.26	0.23	1.16	N/A	N/A	0.26	1.16
Formaldehyde	0.31	1.38	0.02	0.02	0.07	N/A	N/A	0.02	0.07
Acrolein	1.56	6.81	0.08	0.08	0.34	N/A	N/A	0.08	0.34
Total HAPs	10.00	43.80	0.50	0.46	2.19	N/A	N/A	0.50	2.19
SO2	0.09	0.40	0.09	0.09	0.40	N/A	N/A	0.09	0.40

POET Biorefining - Macon
Macon, Missouri
SV-006 HAP Emissions from Natural Gas Combustion (DDGS Dryers & RTO)

The facility will operate two DDG driers. Each drier will be fired on natural gas. There is no back-up fuel. The drier exhaust be directed to the RTO or TO/HRSGB at all times when the driers are operated.

Dryer Rated Capacity=	137	MMBTU/hr
RTO Rated Capacity =	18	
RTO HAP Control Non-metals =	97%	

Conversion Factors:

1 lb/MMBtu =	1020	Btu/scf
1 ton =	2000	lb
1 year =	8,760	hours

Emission Factors: AP-42, Section 1.4 Natural Gas Combustion, July 1998

Uncontrolled Potential to Emit							
	Emission Factor[1]	Dryers		RTO		Total	
HAP Pollutant	(lb/MMSCF)	lb/hr	Unc. TPY	lb/hr	Unc. TPY	lb/hr	Unc. TPY
Benzene	0.00210	2.82E-04	1.24E-03	3.71E-05	1.62E-04	3.19E-04	1.40E-03
Formaldehyde[2]	0.07500	-	-	-	-	-	-
Hexane	1.80000	2.42E-01	1.06E+00	3.18E-02	1.39E-01	2.74E-01	1.20E+00
Naphthalene	0.00061	8.19E-05	3.59E-04	1.08E-05	4.71E-05	9.27E-05	4.06E-04
Toluene	0.00340	4.57E-04	2.00E-03	6.00E-05	2.63E-04	5.17E-04	2.26E-03
Arsenic	0.00020	2.69E-05	1.18E-04	3.53E-06	1.55E-05	3.04E-05	1.33E-04
Berylium	0.00001	1.61E-06	7.06E-06	2.12E-07	9.28E-07	1.82E-06	7.99E-06
Cadmium	0.00110	1.48E-04	6.47E-04	1.94E-05	8.50E-05	1.67E-04	7.32E-04
Chromium	0.00140	1.88E-04	8.24E-04	2.47E-05	1.08E-04	2.13E-04	9.32E-04
Cobalt	0.00008	1.13E-05	4.94E-05	1.48E-06	6.49E-06	1.28E-05	5.59E-05
Lead	0.00050	6.72E-05	2.94E-04	8.82E-06	3.86E-05	7.60E-05	3.33E-04
Manganese	0.00038	5.10E-05	2.24E-04	6.71E-06	2.94E-05	5.77E-05	2.53E-04
Mercury	0.00026	3.49E-05	1.53E-04	4.59E-06	2.01E-05	3.95E-05	1.73E-04
Nickel	0.00210	2.82E-04	1.24E-03	3.71E-05	1.62E-04	3.19E-04	1.40E-03
Selenium	0.00003	3.36E-06	1.47E-05	4.41E-07	1.93E-06	3.80E-06	1.66E-05
Total		0.24	1.07	0.03	0.14	2.75E-01	1.21E+00

1 - Emission factors are from AP-42, 5th Edition, Section 1.4, 7/98

2 - Formaldehyde emissions were measured during the 2017 RTO performance test.

Controlled Potential to Emit						
	Dryers		RTO		Total	
HAP Pollutant	lb/hr	Limited TPY	lb/hr	Limited TPY	lb/hr	Limited TPY
Benzene	8.46E-06	3.71E-05	3.71E-05	1.62E-04	4.55E-05	1.99E-04
Formaldehyde	-	-	-	-	-	-
Hexane	1.25E-01	5.45E-01	3.18E-02	1.39E-01	1.58E-01	6.84E-01
Naphthalene	4.22E-05	1.85E-04	1.08E-05	4.71E-05	5.30E-05	2.32E-04
Toluene	2.35E-04	1.03E-03	6.00E-05	2.63E-04	2.95E-04	1.29E-03
Arsenic	2.69E-05	1.18E-04	3.53E-06	1.55E-05	3.04E-05	1.33E-04
Berylium	1.61E-06	7.06E-06	2.12E-07	9.28E-07	1.82E-06	7.99E-06
Cadmium	1.48E-04	6.47E-04	1.94E-05	8.50E-05	1.67E-04	7.32E-04
Chromium	1.88E-04	8.24E-04	2.47E-05	1.08E-04	2.13E-04	9.32E-04
Cobalt	1.13E-05	4.94E-05	1.48E-06	6.49E-06	1.28E-05	5.59E-05
Lead	6.72E-05	2.94E-04	8.82E-06	3.86E-05	7.60E-05	3.33E-04
Manganese	5.10E-05	2.24E-04	6.71E-06	2.94E-05	5.77E-05	2.53E-04
Mercury	3.49E-05	1.53E-04	4.59E-06	2.01E-05	3.95E-05	1.73E-04
Nickel	2.82E-04	1.24E-03	3.71E-05	1.62E-04	3.19E-04	1.40E-03
Selenium	3.36E-06	1.47E-05	4.41E-07	1.93E-06	3.80E-06	1.66E-05
Total	0.13	0.55	0.03	0.14	1.58E-01	6.90E-01

POET Biorefining - Macon
Macon, Missouri
EP-4 - Fermentation Process - RTO Bypass Stack

The RTO must occasionally be temporarily shut down for unscheduled maintenance or other operational reasons. In this event, the DDGS dryers will be shut down, however, the fermentation tanks, distillation systems, and centrifuges will continue to be operated in normal mode. This operating scenario will be limited to less than 500 hours per calendar year. The emissions from these sources will be vented to separate RTO bypass stacks. One stack will be for the fermentation scrubbers, one stack will be the distillation scrubber, four stacks will be for the whole stillage centrifuges and one stack will be for the corn oil centrifuges. The emissions for the distillation and fermentation stacks will be controlled by wet scrubbers. The VOC and HAP emission factors from the fermentation scrubber are based on the 2/4/14 performance test.

Scrubber VOC Control Efficiency =		98.00%
Scrubber HAP Control Efficiency =		50.00%
Proposed beerfeed=		633 gpm
Yearly operation limit=		500 hours
EF safety factor =		1.40
	1 ton =	2000 lb

2/5/2014 CO2 Fermentation Scrubber Stack Test Results

Test Results	Run 1	Average
Run 1 duration was from 9:12 - 17:15		
	lb/hr	lb/hr
VOC	21.12	21.12
Acetaldehyde	0.69	0.69
Acrolein[1]	0.0072	0.0072
Methanol	0.03	0.03
Formaldehyde	0.001	0.00

Process Rate	Run1	Average
	gpm	gpm
Beerfeed	530	530

[1]On 3/8/2019, Montrose Environmental Group, Inc. completed further data analysis on Acrolein for Macon's 2014 Fermentation Scrubber Stack Test. The reprocessed data indicates < 0.17 ppmv (0.0072 lb/hr).

Uncontrolled Potential to Emit		
	lb/hr	ton/yr
VOC	1075.0	268.8
Acetaldehyde	150.0	37.5
Acrolein	0.6	0.2
Methanol	2.5	0.6
Formaldehyde	0.1	0.0
Total HAP	153.2	38.3

Controlled Potential to Emit			
	lb/hr	ton/yr	
VOC	21.50	5.4	Permit Limit
Acetaldehyde	3.0	0.8	Permit Limit
Acrolein	0.012	0.003	
Methanol	0.05	0.01	
Formaldehyde	0.002	0.0004	
Total HAP	3.1	0.8	

POET Biorefining - Macon
Macon, Missouri
EP-05 Distillation Bypass

The RTO must occasionally be temporarily shut down for unscheduled maintenance or other operational reasons. In this event, the DDGS dryers will be shut down, however, the fermentation tanks, distillation systems, and centrifuges will continue to be operated in normal mode. This operating scenario will be limited to less than 500 hours per calendar year. The emissions from these sources will be vented to separate RTO bypass stacks. One stack will be for the fermentation scrubbers, one stack will be the distillation scrubber, four stacks will be for the whole stillage centrifuges and one stack will be for the corn oil centrifuges. The emissions for the distillation and fermentation stacks will be controlled by wet scrubbers. The VOC and HAP emission factors from the fermentation scrubber are based on the 2/4/14 performance test.

Scrubber VOC Control Efficiency =		98.00%
Scrubber HAP Control Efficiency =		50.00%
Proposed beerfeed=		633 gpm
Yearly operation limit=		500 hours
EF safety factor =		1.40
	1 ton =	2000 lb

3/11/2014 POET-Bingham Lake Distillation Scrubber Stack Test Results

Test Results	Run 1	Run 2	Run 3	Average	Emission factor
	lb/hr	lb/hr	lb/hr	lb/hr	lb/gal beerfeed
VOC	0.57	0.59	0.55	0.57	2.48E-05
Acetaldehyde	0.13	0.10	0.10	0.11	4.79E-06
Acrolein	0.01	0.01	0.01	0.01	4.35E-07
Methanol	0.002	0.002	0.002	0.002	8.70E-08
Formaldehyde	0.0004	0.0004	0.0004	0.0004	1.74E-08
Process Rate	Run1	Run2	Run3	Average	
	gpm	gpm	gpm	gpm	
Beerfeed	384.052	384.228	380.952	383.1	

3/1/2016 POET-Glenville Distillation Scrubber Stack Test Results

Test Results	Run 1	Run 2	Run 3	Average	Emission factor
	lb/hr	lb/hr	lb/hr	lb/hr	lb/gal beerfeed
VOC	0.323	0.318	0.311	0.32	9.98E-06
Acetaldehyde	0.01	0.01	0.01	0.01	3.15E-07
Acrolein	0.0003	0.0003	0.0003	0.00	9.44E-09
Methanol	0.001	0.001	0.001	0.001	3.15E-08
Formaldehyde	0.0002	0.0002	0.0003	0.0002	7.34E-09
Process Rate	Run1	Run2	Run3	Average	
	gpm	gpm	gpm	gpm	
Beerfeed	530.0	530.0	529.2	529.7	

Average Emission Factors:

	lb/gal beerfeed
VOC	1.74E-05
Acetaldehyde	2.55E-06
Acrolein	2.22E-07
Methanol	5.92E-08
Formaldehyde	1.24E-08

Uncontrolled Potential to Emit		
	lb/hr	ton/yr
VOC	46.3	11.6
Acetaldehyde	0.27	0.1
Acrolein	0.02	0.0
Methanol	0.006	0.0
Formaldehyde	0.001	0.0
Total HAP	0.3	0.1

Controlled Potential to Emit		
	lb/hr	ton/yr
VOC	0.93	0.23
Acetaldehyde	0.14	0.034
Acrolein	0.012	0.0030
Methanol	0.0032	0.0008
Formaldehyde	0.00066	0.00016
Total HAP	0.15	0.038

POET Biorefining - Macon
Macon, Missouri
EP-22, 23, 24, 25 Centrifuges - Bypass of RTO

Whole stillage is pumped to the centrifuges where the solids (wet cake) is separated from the liquid (thin stillage). A fraction of the residual VOC and HAP contained in the whole stillage is emitted from the centrifuges during the separation process. The centrifuges are normally vented to the RTO. Therefore, the emissions are uncontrolled only during RTO downtime.

Emission data from a performance test completed on April 18, 2017 for the centrifuge stacks at the POET plant in Mitchell, SD was used to calculate the potential to emit. The maximum performance test run result for each pollutant is increased linearly to correspond with the proposed centrifuge process rate and then a safety factor is added.

POET - Mitchell Stack Test Results April 18, 2017

Test Results	Run 1	Run 2	Run 3	Average	Maximum
	lb/hr	lb/hr	lb/hr	lb/hr	lb/hr
VOC	1.92	1.82	1.57	1.77	1.920
Formaldehyde	0.0020	0.0020	0.0020	0.002	0.002
Methanol	0.0040	0.0040	0.0040	0.004	0.004
Acetaldehyde	0.1620	0.1580	0.1230	0.148	0.162
Acrolein	0.0029	0.0029	0.0029	0.00294	0.003
Total HAP				**0.1566**	**0.1709**

Process rates associated with above stack test results

Process Rates =	Run 1	Run 2	Run 3	Average
	gpm	gpm	gpm	gpm
Centrifuges 1-4	831.68	831.43	821.48	828.2
Total				**828.20**

200 gallons liquid per minute per individual centrifuges
800 gallons liquid per minute through all centrifuges
12,000 gallons liquid per hour through all centrifuges
105,120,000 gallons liquid per year through all centrifuges
500 Limited RTO Bypass Condition hours per year
1.5 Safety Factor

	Uncontrolled Emission		Limited Emission Rate	
	lb/hr	tpy	lb/hr	tpy
Total VOC	**2.78**	**12.18**	**2.78**	**0.70**
Formaldehyde	0.0029	0.013	0.0029	0.0007
Methanol	0.0058	0.025	0.0058	0.0014
Acetaldehyde	0.235	1.028	0.235	0.059
Acrolein	0.0043	0.019	0.0043	0.00106
Total HAP	**0.25**	**1.08**	**0.25**	**0.062**

POET Biorefining - Macon
Macon, Missouri
EP-27 - DDG Fluid Bed Cooler

The following calculations are for the emissions of total suspended particulate (TSP), particulate matter less than 10 micron (PM10), and particulate matter less than 2.5 micron (PM2.5) from point sources at the proposed ethanol facility. For the purposes of these calculations, TSP, PM10, and PM2.5 are assumed to be equal.

The controlled potential TSP/PM10/PM2.5 emissions were calculated by multiplying the design grain loading from the fabric filter by the design flow rate for the source and converting to a pound per hour emission rate. The equation for the calculations is:

Ec lb/hr = G gr/dscf x 1 lb / 7000 gr x Q dscfm x 60 minutes / hour

Where:
E = Emission rate of TSP/PM10 in lb/hr
G = Outlet Grain Loading in grain per dry standard cubic foot of air
Q = Flow Rate in dry standard cubic feet per minute

The uncontrolled potential emission were calculated by multiplying the AP-42 emission factor by the annual throughput in tons/year. The equation for the uncontrolled emission rate is:

Eu ton/yr = Ef * Q

Where:
Eu = Uncontrolled Emission rate in ton/yr
Ef = AP-42 Emission Factor
Q = Annual Throughput

Qddgs = 200,000 tons/year

Conversion Factors:

1 ton =	2000	lb
1 lb =	7000	gr
1 year =	8760	hours
1 hour =	60	minutes

Assumptions:

Maximum controlled TSP/PM10 emission rate[1] = 0.004 gr/dscf
PM Control Efficiency[2] = 99%
Maximum controlled PM2.5 emission rate = 100% of PM10

Given:
Capacity = 22.83 ton/hr
All point source emissions from these emission units will be controlled by a fabric filter.
Exhaust Flow Rate = 26,500 dscfm

				Permit Limits	
	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Limited Uncontrolled[4]	Controlled[3]	Controlled[3]
	lbs/hr	Unc. TPY	TPY	lbs/hr	TPY
PM	90.86	397.95	397.95	0.91	3.98
PM10	90.86	397.95	397.95	0.91	3.98
PM2.5	90.86	397.95	397.95	0.91	3.98

[1] This is assumed loading. Stack tests have indicated that this is a conservative assumption
[2] Assumed control efficiency
[3] Controlled is based on maximum flow rate
[4] Limited is based on annual throughput

The DDGS cooler is also a source of VOC emissions because the DDGS retains a small quantity of ethanol after drying. This ethanol is emitted from the cooler stack. The cooler is not controlled. The concentration of VOC in the DDGS cooler exhaust ranges from 50 to 100 PPMV. The HAP and VOC emissions are based upon engineering estimates from similar facilities.

Proposed Beerfeed =	833.33 GPM
Hours of Operation=	8760 hrs/year
Safety Factor =	1.3

Macon, MO Stack Test - 10/23/2003

Test Results	Run 1	Run 2	Run 3	Average
	lb/hr	lb/hr	lb/hr	lb/hr
VOC	1.65	1.69	1.61	1.65
Acetaldehyde	0.07	0.10	0.14	0.10
Process Rate	Run1	Run 2	Run 3	Average
	gpm	gpm	gpm	gpm
Beerfeed	460	460	460	460

Laddonia, MO Stack Test - 5/27/2016

Test Results	Run 1	Run 2	Run 3	Average
	lb/hr	lb/hr	lb/hr	lb/hr
Methanol	0.003	0.004	0.004	0.004
Acrolein	0.01	0.01	0.01	0.01
Formaldehyde	0.004	0.004	0.004	0.004
Process Rate	Run1	Run 2	Run 3	Average
	gpm	gpm	gpm	gpm
Beerfeed	722	722	722	722

	Uncontrolled Potential to Emit		Controlled Potential to Emit	
	lb/hr	TPY	lb/hr	TPY
VOC	2.95	12.94	2.95	12.94
Acetaldehyde	0.18	0.81	0.18	0.81
Methanol	0.004	0.02	0.004	0.02
Acrolein	0.01	0.05	0.011	0.05
Formaldehyde	0.00	0.02	0.00	0.02
Total HAPs	0.21	0.90	0.21	0.90

POET Biorefining - Macon
Macon, Missouri
EP-28 - DDGS Silo

The following calculations are for the emissions of total suspended particulate (TSP), particulate matter less than 10 micron (PM10), and particulate matter less than 2.5 micron (PM2.5) from point sources at the proposed ethanol facility. For the purposes of these calculations, TSP, PM10, and PM2.5 are assumed to be equal.

The controlled potential TSP/PM10/PM2.5 emissions were calculated by multiplying the design grain loading from the fabric filter by the design flow rate for the source and converting to a pound per hour emission rate. The equation for the calculations is:

Ec lb/hr = G gr/dscf x 1 lb / 7000 gr x Q dscfm x 60 minutes / hour

Where:
E = Emission rate of TSP/PM10 in lb/hr
G = Outlet Grain Loading in grain per dry standard cubic foot of air
Q = Flow Rate in dry standard cubic feet per minute

The uncontrolled potential emission were calculated by multiplying the AP-42 emission factor by the annual throughput in tons/year. The equation for the uncontrolled emission rate is:

Eu ton/yr = Ef * Q

Where:
Q = Annual Throughput

Qddgs = 200,000 tons/year

Conversion Factors:

1 ton =	2000	lb
1 lb =	7000	gr
1 year =	8760	hours
1 hour =	60	minutes

Assumptions:

Maximum controlled TSP/PM10 emission rate[1] = 0.003 gr/dscf
Control Efficiency[2] = 99%
Maximum controlled PM2.5 emission rate = 100% of PM10

Given:
Capacity = 22.8 ton/hr
All point source emissions from these emission units will be controlled by a fabric filter.
Exhaust Flow Rate = 3,500 dscfm

	Uncontrolled	Uncontrolled (based on 8760	Limited Uncontrolled[4]	Controlled[3]	Controlled[3]
	lbs/hr	Unc. TPY	TPY	lbs/hr	TPY
PM	9.00	39.42	39.42	0.09	0.39
PM10	9.00	39.42	39.42	0.09	0.39
PM2.5	9.00	39.42	39.42	0.09	0.39

[1] This is assumed loading. Stack tests have indicated that this is a conservative assumption
[2] Assumed control efficiency
[3] Controlled is based on maximum flow rate
[4] Limited is based on annual throughput

DDGS handling is a source of VOC and HAP emissions. The following calculations are for the emissions of VOC and HAPs. The baghouse does not provide control of VOC or HAPs, and therefore the emissions are exhausted as uncontrolled.

POET - Gowrie Stack Test 11-9-17 - Grain Receiving/DDGS Loadout Baghouse

Test Results	Run 1	Run 2	Run 3	Average
	lb/hr	lb/hr	lb/hr	lb/hr
VOC	0.25	0.26	0.25	0.25
Acetaldehyde	0.023	0.024	0.024	0.024
Acrolein[1]	0.015	0.015	0.015	0.015
Methanol	0.008	0.008	0.008	0.008
Formaldehyde	0.003	0.003	0.003	0.003
Process Rate	Run1	Run 2	Run 3	Average
	ton/hr	ton/hr	ton/hr	ton/hr
DDGS Loadout	104	104	104	104

Emission Factors (DDGS Loadout - Captured Emissions)[2]	
	lb/ton
VOC	2.44E-03
Acetaldehyde	2.28E-04
Acrolein	1.44E-04
Methanol	7.69E-05
Formaldehyde	2.88E-05

Safety Factor = 3

Acrolein Adjustment Factor[3] = 0.5

	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Limited Uncontrolled[4]	Controlled	Controlled (based on 8760 hrs/yr)	Limited Controlled[4]
	lbs/hr	Unc. TPY	TPY	lbs/hr	lbs/hr	TPY
VOC	0.17	0.73	0.73	0.17	0.73	0.73
Acetaldehyde	0.016	0.068	0.068	0.016	0.068	0.068
Acrolein	0.0016	0.007	0.0072	0.0016	0.007	0.0072
Formaldehyde	0.005	0.023	0.023	0.005	0.023	0.023
Methanol	0.002	0.009	0.009	0.002	0.009	0.0087
Total HAPs	0.024	0.11	0.107	0.024	0.11	0.107

[1]On 1/13/2019, Montrose Environmental Group, Inc. completed further data analysis on Acrolein for Gowrie's 2017 DDGS Loadout Baghouse Stack Test. The reprocessed data indicates < 0.102 ppmv (0.015 lb/hr) for all 3 runs.

[2]Emission factors are determined based on stack tests completed at POET - Gowrie.

[3]Acrolein forms in the DDGS dryers from fat and fat breakdown products, with the majority being emitted through the RTO stack. A small amount may be present in between the voids of the DDGS solids, which is then emitted through the fluid bed cooler. Acrolein levels in DDGS handling emissions measured during performance tests have historically been less than the detection value. Therefore, the safety factor is assumed to be conservative.

[4]Limited is based on annual throughput.

Methodology

Emission Factor (lb/ton) = average stack test result (lb/hr) / average tested throughput (ton/hr)

Uncontrolled PTE (lbs/hr) = Emission factor (lb/ton) x proposed throughput (ton/hr) x safety factor

Uncontrolled PTE (tons/yr) = Uncontrolled PTE (lb/hr) x 8760 hours/yr / 2000 lbs/ton

Limited Uncontrolled PTE (tons/yr) = Emission factor (lb/ton) x annual throughput (tons/yr) / 2000 lbs * safety factor

Controlled PTE (lbs/hr) = uncontrolled PTE (lbs/hr)

Controlled PTE (tons/yr) = Uncontrolled PTE (tons/yr)

Limited Controlled PTE (tons/yr) = Limited uncontrolled PTE (tons/yr)

POET Biorefining - Macon
Macon, Missouri
EP-7 - DDGS Silo Bypass

The following calculations are for the emissions of total suspended particulate (TSP), particulate matter less than 10 micron (PM10), and particulate matter less than 2.5 micron (PM2.5) from point sources at the proposed ethanol facility. For the purposes of these calculations, TSP, PM10, and PM2.5 are assumed to be equal.

The controlled potential TSP/PM10/PM2.5 emissions were calculated by multiplying the design grain loading from the fabric filter by the design flow rate for the source and converting to a pound per hour emission rate. The equation for the calculations is:

Ec lb/hr = G gr/dscf x 1 lb / 7000 gr x Q dscfm x 60 minutes / hour

Where:
E = Emission rate of TSP/PM10 in lb/hr
G = Outlet Grain Loading in grain per dry standard cubic foot of air
Q = Flow Rate in dry standard cubic feet per minute

The uncontrolled potential emission were calculated by multiplying the AP-42 emission factor by the annual throughput in tons/year. The equation for the uncontrolled emission rate is:

Eu ton/yr = Ef * Q

Where:
Q = Annual Throughput

Qddgs = 200,000 tons/year
Assumes that the maximum percentage of annual production of DDGS sent to the floor is: 25%

Conversion Factors:

1 ton =	2000	lb
1 lb =	7000	gr
1 year =	8760	hours
1 hour =	60	minutes

Assumptions:
Maximum controlled TSP/PM10 emission rate[1] = 0.003 gr/dscf
Control Efficiency[2] = 99%
Maximum controlled PM2.5 emission rate = 100% of PM10

Given:
Capacity = 22.8 ton/hr
All point source emissions from these emission units will be controlled by a fabric filter.
Exhaust Flow Rate = 3,500 dscfm

	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Limited Uncontrolled[4]	Controlled[3]	Controlled[3]
	lbs/hr	Unc. TPY	TPY	lbs/hr	TPY
PM	9.00	39.42	39.42	0.09	0.39
PM10	9.00	39.42	39.42	0.09	0.39
PM2.5	9.00	39.42	39.42	0.09	0.39

[1] This is assumed loading. Stack tests have indicated that this is a conservative assumption
[2] Assumed control efficiency
[3] Controlled is based on maximum flow rate
[4] Limited is based on annual throughput

DDGS handling is a source of VOC and HAP emissions. The following calculations are for the emissions of VOC and HAPs. The baghouse does not provide control of VOC or HAPs, and therefore the emissions are exhausted as uncontrolled.

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Test Results	Run 1	Run 2	Run 3	Average
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Acetaldehyde	0.023	0.024	0.024	0.024
Acrolein[1]	0.015	0.015	0.015	0.015
Methanol	0.008	0.008	0.008	0.008
Formaldehyde	0.003	0.003	0.003	0.003
Process Rate	Run1	Run 2	Run 3	Average
	ton/hr	ton/hr	ton/hr	ton/hr
DDGS Loadout	104	104	104	104

Emission Factors (DDGS Loadout - Captured Emissions)[2]	
	lb/ton
VOC	2.44E-03
Acetaldehyde	2.28E-04
Acrolein	1.44E-04
Methanol	7.69E-05
Formaldehyde	2.88E-05

Safety Factor = 3

Acrolein Adjustment Factor[3] = 0.5

	Uncontrolled	Uncontrolled (based on 8760 hrs/yr)	Limited Uncontrolled[2]	Controlled	Controlled (based on 8760 hrs/yr)	Limited Controlled[4]
	lbs/hr	Unc. TPY	TPY	lbs/hr	lbs/hr	TPY
VOC	0.17	0.73	0.73	0.17	0.73	0.73
Acetaldehyde	0.02	0.068	0.07	0.016	0.068	0.068
Acrolein	0.00165	0.0072	0.0072	0.0016	0.0072	0.0072
Formaldehyde	0.002	0.0087	0.009	0.002	0.0087	0.0087
Methanol	0.01	0.023	0.023	0.01	0.023	0.023
Total HAPs	0.02	0.11	0.11	0.02	0.11	0.107

[1]On 1/13/2019, Montrose Environmental Group, Inc. completed further data analysis on Acrolein for Gowrie's 2017 DDGS Loadout Baghouse Stack Test. The reprocessed data indicates < 0.102 ppmv (0.015 lb/hr) for all 3 runs.

[2]Emission factors are determined based on stack tests completed at POET - Gowrie.

[3]Acrolein forms in the DDGS dryers from fat and fat breakdown products, with the majority being emitted through the RTO stack. A small amount may be present in between the voids of the DDGS solids, which is then emitted through the fluid bed cooler. Acrolein levels in DDGS handling emissions measured during performance tests have historically been less than the detection value. POET is working with stack testers to lower detection levels.

[4]Limited is based on annual throughput.

Methodology

Emission Factor (lb/ton) = average stack test result (lb/hr) / average tested throughput (ton/hr)

Uncontrolled PTE (lbs/hr) = Emission factor (lb/ton) x proposed throughput (ton/hr) x safety factor

Uncontrolled PTE (tons/yr) = Uncontrolled PTE (lb/hr) x 8760 hours/yr / 2000 lbs/ton

Limited Uncontrolled PTE (tons/yr) = Emission factor (lb/ton) x annual throughput (tons/yr) / 2000 lbs * safety factor

Controlled PTE (lbs/hr) = uncontrolled PTE (lbs/hr)

Controlled PTE (tons/yr) = Uncontrolled PTE (tons/yr)

Limited Controlled PTE (tons/yr) = Limited uncontrolled PTE (tons/yr)

POET Biorefining - Macon
Macon, Missouri

Storage Tanks (see TANKS 4.0 reports for lb/year VOC emissions)

		VOCs			
Unit Number	Tank Description	lb/year	lb/hr	Ton/year	Tank Number
EP-11	190 Proof Ethanol Storage Tank	743	0.08	0.37	T002
EP-12	Denaturant Storage Tank #1	3064	0.35	1.53	T001
EP-13	Denaturent Storage Tank #2	2504	0.29	1.25	T003
EP-14	Ethanol Storage Tank #1	231	0.03	0.12	T004
EP-15	Ethanol Storage Tank #2	231	0.03	0.12	T005
EP-29	Ethanol Storage Tank #3	476	0.05	0.24	T006

HAP	HAP Fraction[1]	
	Denaturant	Ethanol
Benzene	8.62E-04	
Ethylbenzene	5.06E-06	
Hexane	1.32E-02	
Toluene	2.96E-04	
Xylenes	3.19E-05	
Acetaldehyde		4.13E-03
Methanol		3.10E-04
Total	**0.01**	**0.0044**

Conversion Factors:
1 year = 8760 hours
1 ton = 2000 lb

	EP-11			EP-12			EP-13			EP-14			EP-15			EP-29		
HAP Name	Lbs/yr	Lbs/hr	Tons/yr	Lbs/yr	Lbs/hr	Tons/yr	Lbs/yr	Lbs/hr	Tons/yr	Lbs/yr	Lbs/hr	Tons/yr	Lbs/yr	Lbs/hr	Tons/yr	Lbs/yr	Lbs/hr	Tons/yr
Benzene				2.64	3.02E-04	1.32E-03	2.16	2.46E-04	1.08E-03									
Ethylbenzene				0.02	1.77E-06	7.75E-06	0.01	1.45E-06	6.33E-06									
Hexane				40.55	4.63E-03	2.03E-02	33.14	3.78E-03	1.66E-02									
Toluene				0.91	1.04E-04	4.54E-04	0.74	8.47E-05	3.71E-04									
Xylenes				0.10	1.11E-05	4.88E-05	0.08	9.11E-06	3.99E-05									
Acetaldehyde	3.07	3.50E-04	1.53E-03							0.95	1.09E-04	4.77E-04	0.95	1.09E-04	4.77E-04	1.97	2.24E-04	9.83E-04
Methanol	0.23	2.63E-05	1.15E-04							0.07	8.17E-06	3.58E-05	0.07	8.17E-06	3.58E-05	0.15	1.68E-05	7.38E-05
Total	3.30	0.00	0.002	44	0.0050	0.022	36.13	0.0041	0.018	1.03	0.00012	0.00051	1.026	0.00012	0.00051	2.11	0.0002	0.001

[1] HAP fractions for gasoline vapors used at plant

POET Biorefining - Macon
Macon, Missouri
Ethanol Loading Racks

Emission Factors: AP-42, Section 5.2, June 2008

Undenatured ethanol, denatured ethanol (95% to 98% ethanol) and E85 (70% to 85% ethanol) will be shipped by truck. Undenatured and denatured ethanol may also be shipped by rail. Railcars will be dedicated fleets, but the trucks may be used to carry gasoline prior to filling with ethanol. Both railcars and trucks will be filled by submerged loading process. Truck loadout operations will be controlled by a flare (CE18), which has a control efficiency of 98% for VOC and HAPs. Rail loadout operations will be uncontrolled. The rail loadout may also be used for truck loadout, or two trucks may be loaded at the same time. When the rail loadout is used for truck loadout, operations will be controlled by the flare.

Conversion Factors:

1 ton =	2000	lb
1 year =	8760	hours

According to AP-42, Chapter 5.2 - Transportation and Marketing of Petroleum Liquids (2008), the VOC emission factors for the truck and rail loading docks can be estimated from the following equation:

$$L = 12.46 \times (S \times P \times M) / T$$

where:
L = loading loss (lb/kgal)
S = a saturation factor (see AP-42, Table 5.2-1)
P = true vapor pressure of the liquid loaded (psia)
M = molecular weight of vapors
T = temperature of the bulk liquid loaded (degree R)

Previous Stored Liquid	S	P (psia)	M (lb/lb-mole)	T (degree R)	L (lb/kgal)
Gasoline (dedicated vapor balance)	1.0	5.58	[illegible]	518	8.65
Gasoline (clean cargo)	0.5	5.58	[illegible]	518	4.44
E-85 Ethanol (dedicated normal)	[illegible]	3.18	[illegible]	[illegible]	[illegible]
E-85 Ethanol (clean cargo)	0.5	3.18	56.02	518	1.48
Denatured Ethanol (dedicated normal)	0.6	0.85	49.86	518	0.60
Denatured Ethanol (clean cargo)	0.5	0.85	49.86	518	0.57
Undenatured Ethanol (dedicated normal)	0.6	0.75	46.07	518	0.50
Undenatured Ethanol (clean cargo)	0.5	[illegible]	[illegible]	518	0.43

Source Specific Emission Factors

		lb/kgal
The emission factor for loading denatured ethanol to rail which previously contained denatured ethanol = L (Denatured ethanol, normal) =	Denatured Ethanol to Rail	[illegible]
The emission factor for loading undenatured ethanol to rail which previously contained denatured ethanol = L (Denatured ethanol, dedicated vapor balance) - L (Denatured ethanol, clean cargo) + L (undenatured ethanol) =	Undenatured Ethanol to Rail	0.53
The emission factor for loading denatured ethanol to trucks which stored gasoline previously = L (gasoline, dedicated vapor balance) - L (gasoline, clean cargo) + L (denatured ethanol, clean cargo) =	Denatured Ethanol to Truck	5.0161
The emission factor for loading E-85 to trucks which previously contained denatured ethanol or E-85 = L (gasoline, dedicated vapor balance) - L (gasoline, clean cargo) + L (E-85, clean cargo) =	E-85 to Truck	5.90
The emission factor for loading undenatured ethanol to trucks which stored gasoline previously = L (gasoline, dedicated vapor balance) - L (gasoline, clean cargo) + L (undenatured ethanol, clean cargo) =	Undenatured Ethanol to Truck	4.88

1. Throughputs:

Worst Case Truck and Rail (MMgal/yr)	
Denatured Ethanol	57.00
E-85	1.00
Denatured Ethanol and E85 Loadout Combined	58.00
Undenatured Ethanol	64.30

Ethanol Loading Capacity

Truck 1 (gal/hr)	Truck 2/Rail (gal/hr)	Total (kgal/hr)
30,000	30,000	825.00

2. Hourly Potential to Emit (Annual Unlimited):

	Maximum Loading Capacity (kgal/hr)	Uncontrolled Emission Factor (lb/kgal)	Controlled Emission Factor (lb/kgal)	Emissions Uncontrolled (lb/hr)	Emissions Uncontrolled (tpy)	Control Efficiency %	Emissions Controlled (lb/hr)	Emissions Controlled (tpy)
Denatured ethanol loaded out via truck:	30	5.02	0.10	150.48	659.12	98%	3.01	13.18
Denatured ethanol loaded out via rail:	30	0.69	0.69	20.59	90.18	0%	20.59	90.18
Undenatured ethanol loaded out via truck:	30	4.86	0.10	145.82	638.68	98%	2.92	12.77
Undenatured ethanol loaded out via rail:	30	0.53	0.53	15.92	69.74	0%	15.92	69.74
E85 loaded out via truck:	30	5.90	0.12	177.13	775.83	98%	3.54	15.52
Worst case scenario =			0.69	327.61	1,435		38.83	[illegible]

Emissions Uncontrolled (lb/hr) = Throughput (kgal/hr) x Emission Factor (lb/kgal)
Emissions Controlled (lb/hr) = Emissions Uncontrolled (lb/hr) x (1 - Control Efficiency)

3. Limited Annual VOC Potential to Emit:

	Limited* Throughput (kgal/yr)	Uncontrolled Emission Factor (lb/kgal)	Limited* Emissions Uncontrolled (tpy)	Control Efficiency %	Limited* Emissions Controlled (tpy)
Denatured ethanol loaded out via truck:	58,000	5.02	145.45	98%	2.81
Denatured ethanol loaded out via rail:	58,000	0.69	19.22	0%	19.22
Undenatured ethanol loaded out via truck:	53,300	4.86	129.53	98%	2.59
Undenatured ethanol loaded out via rail:	53,300	0.53	14.14	0%	14.14
All E85 loaded out via truck:	1,000	5.90	2.95	98%	0.06
Worst case scenario =			145.45		[illegible]

* Total throughput is limited in order for HAP emission to be less than major source levels.
Emissions Uncontrolled (tpy) = Throughput (kgal/yr) x Emission Factor (lb/kgal) / 2000 lb/ton
Emissions Controlled (tpy) = Emissions Uncontrolled (tpy) x (1 - Control Efficiency)

Flare Control Efficiency, Truck =	98%
Flare Control Efficiency, Rail =	0%

4. Potential to Emit HAPs:

HAP	Gasoline HAP Fraction[1]	Denatured Ethanol HAP Fraction[2]	E85 HAP Fraction[3]	Uncontrolled: Denatured Ethanol to Rail (lb/kgal)	Uncontrolled: Denatured Ethanol to Truck (lb/kgal)	Uncontrolled: E85 to Truck (lb/kgal)	Controlled: Denatured Ethanol to Rail (lb/kgal)	Controlled: Denatured Ethanol to Truck (lb/kgal)	Controlled: E85 to Truck (lb/kgal)
Acetaldehyde	0.00E+00	[illegible]	[illegible]	[illegible]	0.00236	[illegible]	[illegible]	[illegible]	0.00004
Benzene	4.6[illegible]E-04	[illegible]	[illegible]	[illegible]	0.00397	0.00470	[illegible]	[illegible]	0.00010
Ethylbenzene	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0.00000
Hexane	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Toluene	[illegible]	[illegible]	[illegible]	0.00008	0.00134	[illegible]	[illegible]	[illegible]	[illegible]
Xylenes	[illegible]	[illegible]	[illegible]	0.00001	0.00015	[illegible]	[illegible]	[illegible]	0.00000
Methanol	[illegible]	[illegible]	[illegible]	0.00021	0.00018	[illegible]	[illegible]	[illegible]	0.00000
Total	[illegible]	[illegible]	[illegible]	0.01	0.07	[illegible]	0.00077	[illegible]	0.0012

[1] This is the highest HAP percentage for gasoline vapors as per the Flint Hills gasoline SDS, issue date of 12-03-2014. Assumed truck tanker hauled gasoline for prior load.
[2] Based on the average weight fraction of the denaturant used at the plant (Enterprise 2015 data) & weight fraction in 200 proof ethanol (POET analytical data Feb & March 2016).
No acrolein, acetaldehyde, or formaldehyde will be present in the undenatured ethanol. The boiling points of acrolein (127 °F), acetaldehyde (68 °F) and formaldehyde (-3 °F) are less than the temperatures sustained during the distillation process, so these compounds would only be emitted in the distillation scrubber or centrifuges (RTO).

Hourly Potential to Emit

HAP	Uncontrolled/Unflared: Denatured Ethanol to Rail (lb/hr)	Uncontrolled/Unflared: Denatured Ethanol to Truck (lb/hr)	Uncontrolled/Unflared: E85 to Truck (lb/hr)	Controlled/Flared: Denatured Ethanol to Rail (lb/hr)	Controlled/Flared: Denatured Ethanol to Truck (lb/hr)	Controlled/Flared: E85 to Truck (lb/hr)	Worst Case[1] (lb/hr)
Acetaldehyde	0.00	0.071	0.061	[illegible]	0.00142	0.00122	[illegible]
Benzene	0.02	0.12	0.14	[illegible]	0.00238	0.00282	[illegible]
Ethylbenzene	0.00	0.00	0.00	[illegible]	0.00001	0.00070	[illegible]
Hexane	2.27	1.83	3.35	[illegible]	0.03654	0.04495	0.116
Toluene	2.50	2.04	9.03	[illegible]	[illegible]	[illegible]	[illegible]
Xylenes	0.00	0.00	0.01	[illegible]	0.00008	[illegible]	[illegible]
Methanol	0.01	0.01	0.00	[illegible]	0.00011	[illegible]	[illegible]
Total	0.17	3.57	3.47	[illegible]	0.04	[illegible]	[illegible]

[1] Worst case is simultaneous loadout of E85 to rail and E85 to truck (Total HAP).

Limited Annual Potential to Emit

	Uncontrolled/Unflared: Denatured Ethanol to Rail	Uncontrolled/Unflared: Denatured Ethanol to Truck	Uncontrolled/Unflared: E85 to Truck	Controlled/Flared: Denatured Ethanol to Rail	Controlled/Flared: Denatured Ethanol to Truck	Controlled/Flared: E85 to Truck	Limited Potential to Emit After Control	Restricted PTE Before Control
	kgal/yr	kgal/yr	kgal/yr	kgal/yr	kgal/yr	kgal/yr	kgal/yr	
Worst Case Annual Scenario (Total HAPs)[1]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	
Worst Case Annual Scenario (Acetaldehyde)[2]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	
HAP	ton/yr	ton/yr	ton/yr	ton/yr	ton/yr	ton/yr	ton/yr	ton/yr
Acetaldehyde (Worst Case)	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Acetaldehyde	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Benzene	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Ethylbenzene	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Hexane	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Toluene	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Xylenes	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Methanol	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Total	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

[1] [illegible]
[2] [illegible]

Methodology
HAP emissions are based on worst-case emission scenario.
HAP emission factors are based on content of HAP in product (denaturant, gasoline, or ethanol), content of material in product (E85, denatured ethanol, or undenatured ethanol), and VOC emission factor calculations above.
PTE of HAP before Control (lb/hr) = Uncontrolled HAP Emission Factor (lb/kgal) x Loadout Rate (kgal/hr)
PTE of HAP before Control (tons/yr) = Worst Case PTE of HAP before Control (lb/hr) x 8,760 (hrs/yr) / 2,000 (lb/ton)
PTE of HAP after Control (lb/hr) = PTE of HAP before Control (lb/hr) x (1-Control Efficiency)
Limited PTE of HAP after Control (tons/yr) = Uncontrolled HAP Emission Factor (lb/kgal) x throughput (MMgal/yr) x (1-Control Efficiency) x 1,000 (kgal/MMgal) / 2,000 (lb/ton)

5. Flare Emissions from Combustion of VOCs from the Ethanol Loading Rack

Heat Input Capacity (MMBtu/hr)	Max Hourly Rate (1 Rail/Truck and 1 Truck) (kgal/hr)	Annual Production Limit (kgal/yr)
0.057	60.0	58,000

	Pollutant: CO**	Pollutant: NOx**
Emission Factor (lb/kgal)	0.084	0.0334
Potential to Emit in lbs/hr	5.04	2.00
Avg Limited Potential to Emit (lb/hr)	0.56	0.22
UNRESTRICTED Potential to Emit (TPY)	22.08	8.78
LIMITED Potential to Emit in tons/yr	2.44	0.97

** Emission factors for NO_x and CO are based on the information provided by the flare manufacturer ([illegible]).
PM, PM_{10}, $PM_{2.5}$, and SO_2 emission factors are negligible due to the [illegible] [illegible].
VOC emission calculations can be found above in loading rack calculations.

POET Biorefining - Macon
Macon, Missouri
FS007 Equipment Leaks - Fugitive Emissions

Equipment Leaks

Process Stream	Equipment Component Source	Product	Component Count	Emission Factor (lb/comp.-hr)	Uncontrolled VOC (lb/hr)	Uncontrolled VOC (ton/yr)	Subpart VVa Control Effectiveness	Controlled Rate (lb/hr)	TOC Weight (%)	Emitted Water (lb/hr)	Controlled VOC (lb/hr)	Controlled VOC (ton/yr)
Distillation (Ethanol Service)	Valves	Gas/Vapor	75	0.013134	0.99	4.31	92.00%	0.08	100.00%	0.000	0.079	0.345
	Valves	Light Liquid	175	0.008866	1.55	6.80	88.00%	0.19	100.00%	0.000	0.186	0.815
	Pump Seals	Light Liquid	6	0.04378	0.26	1.15	69.00%	0.08	100.00%	0.000	0.081	0.357
	Compressors	Gas/Vapor	0	0.5016	0.00	0.00		0.00	100.00%	0.000	0.000	0.000
	Relief Valves	Gas/Vapor	6	0.2288	1.37	6.01	92.00%	0.11	100.00%	0.000	0.110	0.481
	Sampling Connections	All	0	0.033	0.00	0.00		0.00	100.00%	0.000	0.000	0.000
	Open Ended Lines	All	0	0.00374	0.00	0.00		0.00	100.00%	0.000	0.000	0.000
	Connectors	All	777	0.004026	3.13	13.70	93.00%	0.22	100.00%	0.000	0.219	0.959
Fermentation (Beer Service)	Valves	Gas/Vapor	0	0.013134	0.00	0.00	92.00%	0.00	15.00%	0.000	0.000	0.000
	Valves	Light Liquid	0	0.008866	0.00	0.00	88.00%	0.00	100.00%	0.000	0.000	0.000
	Valves	Heavy Liquid	106	0.000506	0.05	0.23	88.00%	0.01	15.00%	0.005	0.001	0.004
	Pump Seals	Heavy Liquid	13	0.01804	0.23	1.03	69.00%	0.07	15.00%	0.062	0.011	0.048
	Compressors	Gas/Vapor	0	0.5016	0.00	0.00		0.00	15.00%	0.000	0.000	0.000
	Relief Valves	Gas/Vapor	25	0.2288	5.72	25.05	92.00%	0.46	15.00%	0.389	0.069	0.301
	Sampling Connections	All	13	0.033	0.43	1.88		0.43	15.00%	0.365	0.064	0.282
	Open Ended Lines	All	0	0.00374	0.00	0.00		0.00	15.00%	0.000	0.000	0.000
	Connectors	All	338	0.004026	1.36	5.96	93.00%	0.10	15.00%	0.081	0.014	0.063
	Totals				**15.10**	**66.13**		**1.74**		**0.902**	**0.834**	**3.655**

Tank Farm Equipment Fugitives

Process Stream	Equipment Component Source	Product	Component Count	Emission Factor (lb/comp.-hr)	Uncontrolled Rate (lb/hr)	Uncontrolled Rate (ton/yr)	Subpart VVa Control Effectiveness	Controlled Rate (lb/hr)	TOC Weight (%)	Emitted Water (lb/hr)	Controlled VOC (lb/hr)	Controlled VOC (ton/yr)
Denaturant Tanks	Valves	Gas/Vapor	0	0.013134	0.00	0.00	92.00%	0.00	100.00%	0.000	0.000	0.000
	Valves	Light Liquid	30	0.0089	0.27	1.17	88.00%	0.03	100.00%	0.000	0.032	0.140
	Pump Seals	Light Liquid	3	0.04378	0.13	0.58	69.00%	0.04	100.00%	0.000	0.041	0.178
	Compressors	Gas/Vapor	0	0.5016	0.00	0.00		0.00	100.00%	0.000	0.000	0.000
	Relief Valves	Gas/Vapor	2	0.2288	0.46	2.00	92.00%	0.04	100.00%	0.000	0.037	0.160
	Sampling Connections	All	4	0.033	0.13	0.58		0.13	100.00%	0.000	0.132	0.578
	Open Ended Lines	All	0	0.00374	0.00	0.00		0.00	100.00%	0.000	0.000	0.000
	Connectors	All	156	0.004026	0.63	2.75	93.00%	0.04	100.00%	0.000	0.044	0.193
	Totals (Tanks in Denaturant Service)				1.62	7.08		0.29		0.00	0.29	1.25
Non-denaturant tanks (190 and 200-Proof Tanks)	Valves	Gas/Vapor	0	0.013134	0.00	0.00	92.00%	0.00	100.00%	0.000	0.000	0.000
	Valves	Light Liquid	110	0.0089	0.98	4.29	88.00%	0.12	100.00%	0.000	0.117	0.515
	Pump Seals	Light Liquid	5	0.04378	0.22	0.96	69.00%	0.07	100.00%	0.000	0.068	0.297
	Compressors	Gas/Vapor	0	0.5016	0.00	0.00		0.00	100.00%	0.000	0.000	0.000
	Relief Valves	Gas/Vapor	5	0.2288	1.14	5.01	92.00%	0.09	100.00%	0.000	0.092	0.401
	Sampling Connections	All	5	0.033	0.17	0.72		0.17	100.00%	0.000	0.165	0.723
	Open Ended Lines	All	0	0.00374	0.00	0.00		0.00	100.00%	0.000	0.000	0.000
	Connectors	All	734	0.004026	2.96	12.94	93.00%	0.21	100.00%	0.000	0.207	0.906
	Totals (Tanks in Ethanol Service)				5.46	23.92		0.65		0.00	0.65	2.84
	Totals Tank Farm Equipment Fugitives				**7.08**	**31.00**		**0.93**		**0.00**	**0.93**	**4.09**

Total VOCs: Equipment Leaks and Tank Farm Equipment Fugitives **22.18** **97.13** **2.67** **0.90** **1.77** **7.75**

* Component count provided by source.

** Emission factors are from Protocol for Equipment leak Emission Estimates, EPA-453/R-95-017. Table 2-1 and Table 5-2

2. Fugitive HAP Emissions:

Fugitive HAP Emissions (tons/yr) = Controlled TOC (tons/yr) x HAP Fraction

Equipment Leaks (Distillation/Ethanol Service)

HAP	HAP Fraction	Fugitive HAP Emissions-Uncontroled (lbs/hr)	Fugitive HAP Emissions-Uncontroled (tons/yr)	Fugitive HAP Emissions-Controled (lbs/hr)	Fugitive HAP Emissions-Controled (tons/yr)
Acetaldehyde	4.13E-03	6.24E-02	2.73E-01	3.45E-03	1.51E-02
Methanol	3.10E-04	4.68E-03	2.05E-02	2.59E-04	1.13E-03
Totals		**6.70E-02**	**2.94E-01**	**3.70E-03**	**1.62E-02**

Equipment Leaks (Fermentation Service)

HAP	HAP Fraction	Fugitive HAP Emissions-Uncontroled (lbs/hr)	Fugitive HAP Emissions-Uncontroled (tons/yr)	Fugitive HAP Emissions-Controled (lbs/hr)	Fugitive HAP Emissions-Controled (tons/yr)
Acetaldehyde	3.21E-02	4.84E-01	2.12E+00	2.68E-02	1.17E-01
Methanol	7.82E-05	1.18E-03	5.17E-03	6.53E-05	2.86E-04
Formaldehyde	2.61E-04	3.94E-03	1.72E-02	2.18E-04	9.53E-04
Acrolein	1.04E-03	1.58E-02	6.90E-02	8.70E-04	3.81E-03
Totals		**5.05E-01**	**2.21E+00**	**2.79E-02**	**1.22E-01**

Tank Farm Equipment Fugitives in Ethanol Service

HAP	HAP Fraction	Fugitive HAP Emissions-Uncontroled (lbs/hr)	Fugitive HAP Emissions-Uncontroled (tons/yr)	Fugitive HAP Emissions-Controled (lbs/hr)	Fugitive HAP Emissions-Controled (tons/yr)
Acetaldehyde	4.13E-03	2.26E-02	3.72E-01	2.68E-03	1.17E-02
Methanol	3.10E-04	1.69E-03	2.79E-02	2.01E-04	8.81E-04

Tank Farm Equipment Fugitives in Denaturant Service

HAP	HAP Fraction	Fugitive HAP Emissions-Uncontroled (lbs/hr)	Fugitive HAP Emissions-Uncontroled (tons/yr)	Fugitive HAP Emissions-Controled (lbs/hr)	Fugitive HAP Emissions-Controled (tons/yr)
Benzene	8.62E-04	1.39E-03	6.10E-03	2.46E-04	1.08E-03
Ethylbenzene	5.06E-06	8.17E-06	3.58E-05	1.44E-06	6.32E-06
Hexane	1.32E-02	2.14E-02	9.37E-02	3.78E-03	1.65E-02
Toluene	2.96E-04	4.79E-04	2.10E-03	8.45E-05	3.70E-04
Xylenes	3.19E-05	5.15E-05	2.25E-04	9.09E-06	3.98E-05
Total HAPs in Tank Farm Equipment Fugitives		**0.048**	**0.502**	**0.007**	**0.031**

Total HAPs: Equipment Leaks and Tank Farm Equipment Fugitives	**0.620**	**3.009**	**0.039**	**0.169**

POET Biorefining - Macon
Macon, Missouri
Greenhouse Gas

Conversion Factors:

1 lb/MMBtu =	1020	Btu/scf
1 ton =	2000	lb
1 year =	8760	hours

Emission Factors: AP-42, Section 1.4 Natural Gas Combustion, July 1998

Combustion Emission Factors		
	Natural Gas	Diesel
	lb/mmcf	lb/HP-hr
CO2	120,000	1.15
CH4	2.3	0.00247
N2O	2.2	0.000021
Low Nox	0.64	-

Global Warming Potential Factor	
CO2	1
CH4	25
N2O	298

1. Green House Gas from Natural Gas Combustion

			CO2	CH4	N2O	CO2e
	MMBtu/hr	MMCF/yr*	tons/yr	tons/yr	tons/yr	tons/yr
Grain Dryer	31	266	15,974	0.31	0.57	16,153
Boiler #1	60.5	520	31,175	0.60	0.17	31,240
Boiler #2	72.6	624	37,410	0.72	0.20	37,488
Dryer #1	69	588	35,298	0.68	0.65	35,507
Dryer #2	69	588	35,298	0.68	0.65	35,507
RTO	18	155	9,275	0.18	0.05	9,294
Flare	0.057	0.487	29	0.001	0.0005	29
Potential Emission (tons/yr)	288.2	2,475	**164,460**	**3.2**	**2.3**	**165,219**

Methodology
All emission factors are based on normal firing.
MMBtu = 1,000,000 Btu
MMCF = 1,000,000 Cubic Feet of Gas
*Potential Throughput (MMCF) = Heat Input Capacity (MMBtu/hr) x 8,760 hrs/yr x 1 MMCF/1,020 MMBtu
The N2O Emission Factor for uncontrolled is 2.2. The N2O Emission Factor for low Nox burner is 0.64.
Emission Factors for Natural Gas Combustion are from AP 42, Table 1.4-2 SCC #1-02-006-02, 1-01-006-02, 1-03-006-02, and 1-03-006-03.
Emission Factors for Diesel Generator are from AP 42, Table 3.3-1 SCC #2-02-001-02, 2-03-001-01.
Greenhouse Warming Potentials (GWP) from Table A-1 of 40 CFR Part 98 Subpart A.
Emission (tons/yr) = Throughput (MMCF/yr) x Emission Factor (lb/MMCF)/2,000 lb/ton

CO2e (tons/yr) = CO2 Potential Emission ton/yr x CO2 GWP (1) + CH4 Potential Emission ton/yr x CH4 GWP (25) + N2O Potential Emission ton/yr x N2O GWP (298).

3. Green House Gas from Fermentation (Biogenic)

Fermentation Process

Given:

55,000,000 gallons of undenatured (200-proof) EtOH / year
46.06844 [g/mol] mole weight of EtOH
0.789 [g/cm^3] density of liquid EtOH
44.0095 [g/mol] mole weight of CO2

and:

$C_6H_{12}O_6$ + yeast = 2 CH_3CH_2OH + 2 CO_2
sugar + yeast = ethanol + carbon dioxide

Therefore:

$$\frac{55{,}000{,}000 \text{ gal 200-proof EtOH}}{\text{year}} \cdot \frac{0.789 \text{ g EtOH}}{1 \text{ cm}^3} \cdot \frac{3{,}785.41 \text{ cm}^3}{1 \text{ gal}}$$

$$= \frac{1.64\text{E}+11 \text{ g EtOH}}{\text{year}} \cdot \frac{1 \text{ mol EtOH}}{46.06844 \text{ g EtOH}}$$

$$= \frac{3{,}565{,}735{,}392 \text{ mol EtOH}}{\text{year}} \cdot \frac{2 \text{ mol } CO_2}{2 \text{ mol EtOH}}$$

$$= \frac{3{,}565{,}735{,}392 \text{ mol } CO_2}{\text{year}} \cdot \frac{44.0095 \text{ g } CO_2}{1 \text{ mol } CO_2} \cdot \frac{1 \text{ ton}}{907{,}184.74 \text{ g}}$$

= **172,982** tons CO_2 / year

Biogenic GHG= **172,982** CO2e Total in tons/yr

Total GHG Emissions (Combustion and Biogenic)			
CO2	**CH4**	**N2O**	**CO2e**
tons/yr	tons/yr	tons/yr	tons/yr
337,441.13	3.15	2.28	338,200.65

POET Biorefining - Macon, LLC
Macon, MO
Insignificant Activities

Corrosion Inhibitor Tank

The emissions from the 1,000 gal corrosion inhibitor storage tank are calculated using TANKS 4.0.9d softwa

Uncontrolled Emiss	lb/hr	ton/year
Total VOC	5.98E-03	2.62E-02
Methanol	5.96E-03	2.61E-02
Xylene	1.94E-05	8.50E-05
Ethylbenzene	4.57E-06	2.00E-05
Total HAP	5.98E-03	2.62E-02

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